Pontifícia Universidade Católica
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Lucas Martins Thimoteo

Explainable Ensemble Learning for Alzheimer's Disease Diagnosis

Dissertação de Mestrado

Dissertation presented to the Programa de Pós–graduação em Engenharia Elétrica, do Departamento de Engenharia Elétrica da PUC-Rio in partial fulfillment of the requirements for the degree of Mestre em Engenharia Elétrica.

Advisor : Prof. Marley Maria Bernardes Rebuzzi Vellasco
Co-advisor: Prof. Jorge Luís Machado Do Amaral

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Abstract

Thimoteo, Lucas Martins; Vellasco, Marley Maria Bernardes Rebuzzi (Advisor); Amaral, Jorge Luís Machado Do (Co-Advisor). **Explainable Ensemble Learning for Alzheimer's Disease Diagnosis**. Rio de Janeiro, 2022. 119p. Dissertação de Mestrado – Departamento de Engenharia Elétrica, Pontifícia Universidade Católica do Rio de Janeiro.

The Alzheimer's Disease is a neurodegenerative disease that degrades several brain functions and mostly affects elderly populations. With the general aging of the world population, treating aging related issues gains more and more importance. It is estimated that 40% of elderly population lives with some sort of Dementia. Up to this day, there is no cure for this disease. Hence, it is essential that more robust diagnostic tools are developed to expand and facilitate access to the diagnosis as soon as possible, in order to start necessary treatment. This dissertation aims to propose an Explainable Machine Learning solution for the Alzheimer's Disease diagnosis, by leveraging the use of multi-modality data, as well as explanations of the diagnosis, in order to offer a robust and trustworthy diagnosis. To achieve that, we employ a stacked ensemble learning strategy and make use of three data sources: Magnetic Resonance Scans, demographic information and cognitive tests. We show that when classifying patients with more advanced pathology against healthy ones, it is possible to achieve an area under the ROC curve of 0.936 by using Magnetic Resonance and demographic features. When classifying early pathology patients, area under the ROC curve is 0.917. We provide global explanations to assess the importance of features found by the ensemble learning classifier, as well as local explanations to provide specific diagnosis contribution of features. Moreover, we also explore visual explanations for the magnetic resonance, showing the most relevant parts of the brain in order to execute the partial Alzheimer's Disease diagnosis. Finally, we state that our proposed model imitates what specialists generally do to make a diagnosis: they look for evidence in more than on data source and they also ponder which evidence might have more relevance to specific situations.

Keywords

AI in Healthcare; Alzheimer's Disease Diagnosis; Interpretability; Explainability; Ensemble Learning.

Resumo

Thimoteo, Lucas Martins; Vellasco, Marley Maria Bernardes Rebuzzi; Amaral, Jorge Luís Machado Do . **Aprendizado de Comitê Explicável para Diagnóstico da Doença de Alzheimer**. Rio de Janeiro, 2022. 119p. Dissertação de Mestrado – Departamento de Engenharia Elétrica, Pontifícia Universidade Católica do Rio de Janeiro.

A Doença de Alzheimer é uma doença neurodegenerativa que degrada diversas funções cerebrais e afeta principalmente populações idosas. Com o envelhecimento geral da população mundial, o tratamento de questões relacionadas ao envelhecimento ganha cada vez mais importância. Estima-se que 40% da população idosa vive com algum tipo de Demência. Até hoje, não há cura para esta doença. Por isso, é fundamental que ferramentas diagnósticas mais robustas sejam desenvolvidas para ampliar e facilitar o acesso ao diagnóstico o mais rápido possível, a fim de iniciar o tratamento necessário. Esta dissertação tem como objetivo propor uma solução de Aprendizado de Máquina Explicável para o diagnóstico da Doença de Alzheimer, utilizando dados multimodais, bem como explicações do diagnóstico, a fim de oferecer um diagnóstico robusto e confiável. Para isso, empregamos uma estratégia de aprendizado de conjunto empilhado e utilizamos três fontes de dados: imagens de ressonância magnética, informações demográficas e testes cognitivos. Mostramos que ao classificar pacientes com patologia mais avançada contra saudáveis, é possível obter uma área sob a curva ROC de 0,936 usando Ressonância Magnética e características demográficas. Ao classificar pacientes com patologia precoce, a área sob a curva ROC é de 0,917. Fornecemos explicações globais para avaliar a importância das características encontradas pelo classificador de aprendizagem em conjunto, bem como explicações locais para fornecer contribuição de diagnóstico específica das características. Além disso, também exploramos explicações visuais para a ressonância magnética, mostrando as partes mais relevantes do cérebro para realizar o diagnóstico parcial da Doença de Alzheimer. Por fim, afirmamos que nosso modelo proposto imita o que especialistas geralmente fazem para executar um diagnóstico: buscam evidências em mais de uma fonte de dados e também ponderam quais evidências são mais relevantes para situações específicas.

Palavras-chave

IA na Saúde; Diagnóstico da Doença de Alzheimer; Interpretabilidade; Explicabilidade; Aprendizado de Comitê.

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List of Abreviations

Acc – *Accuracy*
AD – *Alzheimer's Disease*
ADL – *Activity Daily Living*
ADAS-Cog – *Alzheimer's Disease Assessment Scale-cognitive*
ADNI – *Alzheimer's Disease Neuroimaging Initiative*
AI – *Artificial Intelligence*
AIBL – *The Australian Imaging, Biomarkers . Lifestyle Flagship Study of Ageing*
AUC – *Area Under the Curve*
BCE – *Binary Cross Entropy*
BG – *Background Knowledge*
CAM – *Class Activation Mapping*
CDRSB – *Clinical Dementia Rating Scale Sum of Boxes*
CN – *Cognitive Normal*
CNN – *Convolutional Neural Networks*
CSF – *Cerebrospinal Fluid*
CT – *Computed Tomography*
DL – *Deep Learning*
DLi – *DeepLift*
DNN – *Deep Neural Networks*
EBM – *Explainable Boosting Machine*
FAQ – *Functional Assessment Questionnaire*
FDR – *Fisher Discriminate Ratio*
GAM – *Generalized Additive Models*
GAN – *Generative Adversarial Networks*
GC – *Guided Grad-CAM*
GRU – *Gated Recurrent Units*
LDA – *Linear Discriminant Analysis*
LIME – *Local Interpretable Model-Agnostic Explanations*
LR – *Logistic Regression*
LSTM – *Long Short Term Memory*
MCI – *Mild Cognitive Impairment*
ML – *Machine Learning*

MMSE – *Mini-mental State Examination*
MMML – *Multi Modal Machine Learning*
MMMT – *Multi Modal Multi Task*
MOCA – *Montreal Cognitive Assessment*
MRI – *Magnetic Resonance Image*
MSML – *Multi Source Machine Learning*
NC – *Normal Control*
OASIS – *Open Access Series of Imaging Studies*
PCA – *Principal Component Analysis*
PET – *Positron Emission Tomography*
Prec – *Precision*
Pred_Threshold – *Prediction Threshold*
RAVLT – *Rey Auditory Verbal Learning Test*
SAE – *Stacked Auto Encode*
SG – *Smoothgrad*
SHAP – *Shapley Additive Explanations*
SPECT – *Single-photon Emission Computed Tomography*
SVM – *Support Vector Machine*
TRABSCOR – *Time to Complete Part B of the Trail Making Test*
WAS-RC – *Weissler Intelligence Scale*
XAI – *Explainable Artificial Intelligence*

1
Introduction

With the fast spread of computers over the second half of the XX century, we heavily rely on computational systems for several areas of our lives. Since the early 1960's [1], computer science has been applied to the medical area. The first appearance of computer-aided processes was only to store data and make simple calculations. However, as machines became more powerful, more complex tasks started to be delegated to computers. Not only that, but medicine itself made huge advances with the support of computer science. Today, common medical and healthcare procedures are computer-aided, such as most medical imaging, like Magnetic Resonance Images (MRI), Computed Tomography (CT) and Ultrasonography, and also blood exam variables calculations.

During the 1980s, there was a substantial rise of Artificial Intelligence (AI) being applied to healthcare and medicine, employing methods such as fuzzy systems and Bayesian models [2]. Today, the use of AI in the medical domain is extensive. It is possible to define two branches of applications: the use of AI focused on a single patient; and the use of AI for healthcare systems and processes [3]. While the use of AI for healthcare systems focuses on medical facilities´ general processes, such as managing resources for patient allocation or predicting patient readmission, single patient-related AI applications focus on analyzing one's personal traits.

In general, a patient's life cycle in a medical facility can be defined by the following steps: Initial trial, diagnosis, treatment and prognosis, monitoring and finally managing. AI can be applied during the entire patient life cycle. To our understanding, the three most sensitive areas are clinical trials, diagnosis, and prognosis [4].

Applications of AI for a patients initial trial can be related to traumas [5] and fractures identification [6], as well as hemorrhage detection [7]. Basically, these applications focus on the classification or detection tasks during the patients' first contact with a medical facility, emergency room included. These applications should be fast and accurate and provide at least human-level performance. That way, clinicians and other professionals can rely on their results and focus on other tasks. However, as pointed out in [8], plenty of clinical trials AI applications are still under human-level performance, or when they are superior, there is a long processing time, making them unlikely to be used.

Diagnosis and prognosis of patients are probably the most famous and most common areas of AI applications. Here, AI systems are entrusted with the task of telling if a patient has a medical condition (diagnosis support); for example, chronic pulmonary diseases [9]. We can cite other common applications such as breast cancer diagnosis [10], diabetic retinopathy diagnosis [11] and also Alzheimer's Disease diagnosis [12]. For the prognosis tasks, AI should predict an accurate outcome about a patient´s situation, for example, the evolution of breast cancer [13], so that doctors can execute the proper treatment. Although these types of applications are not required to have a fast processing time, it is also crucial that they are well calibrated to avoid miss classification and wrong diagnoses. In a binary classification scenario, false negative events could cause patients the fake impression of a healthy situation and consequently delay or even deny them the possibility of treatment in time. On the other hand, false positive can cause implicate in unnecessary and sometimes harmful medications or medical procedures.

Another critical aspect to consider when developing AI systems for healthcare is the level of understanding of the system´s outcome. In other words, it is vital to make sense of how the model 'thinks' by providing explanations about the model output. We argue that model explainability has the same level of importance as the model´s predictive power, at least in the healthcare domain. Otherwise, how could a doctor trust an AI breast cancer diagnosis if they have no clue about the most critical aspects for the model? Providing clear explanations about results can guide professionals in their subsequent actions. Lundberg presents this excellent work [14] on explaining predictions for preventing hypoxemia of patients during surgeries. Besides predicting the hypoxemia risk for the next minutes, the system also provides which variables lead to the current outcome. This can guide the professionals to act before the event occurs, decreasing the chances of the obit.

Moreover, while most Machine Learning (ML) and Deep Learning (DL) works focus on processing one type of information or data source, that goes against the common practice in the medical domain. Healthcare professionals generally use the context of several types of information in order to decide a final diagnosis. In [15], the authors propose a multi-modality architecture that takes into consideration medical imaging and clinical data to detect pulmonary embolism. However, having applications that ingest and process different sources of data poses an additional challenge for two reasons: it is naturally more difficult to obtain diverse sources of data about the same subject due to time and resource reasons; DL and ML strategies that can deal with multi-modality are more complex to develop.

Furthermore, besides room for improvement on the aforementioned topics about AI in Healthcare, it is also essential to target efforts to prepare for the next generational Healthcare issues, such as the general aging of the population. After all, with advances in technology and medicine, besides providing food, water, and healthcare services to more people worldwide, humans could considerably extend life expectancy. According to this report of the World Health Organization [16], in 2030, 1 in 6 people will have over 60 years old. With the overall improvement of quality of life, global population aging brings new challenges such as hearing loss, pulmonary diseases, diabetes, depression, and Dementia [17]. A report from 2018 indicates that over 50 million people worldwide are living with Dementia [18]. These numbers tend to rise, mainly due to the increase in life expectancy over the following decades.

Alzheimer's Disease (AD) is a degenerative disease that affects several brain functions [19], and it is the most common cause of Dementia in people over 65 years old. It is characterized by the massive amounts of the proteins amyloid-beta and tau in the brain, which culminates in obstructing cognitive functions [20]. One of the most common symptoms is progressive memory decline over recent events. There may also be early signs, such as not recognizing familiar faces, disorientation of time and place, significant changes in mood and behavior, and withdrawal from work or social activities. Although not yet available for primary medical settings, in some cases, it is possible to diagnose these conditions during a pre-dementia stage, generally called Mild Cognitive Impairment (MCI). To correctly diagnose this stage, it is necessary laboratory-measured biomarkers and might be the best opportunity to treat the disease [20].

Up to this point, there is no scientific consensus about the actual causes of this disease, and it is confirmed that diagnosis can only be executed post-mortem by analyzing brain cells and abnormal plaque deposits [21]. However, by collecting useful biomarkers, such as magnetic resonance scans, computed tomography, and genetic testing, it is possible to obtain a working diagnosis of the disease[22].

Over the last two decades, there have been several computer-aided techniques to assist the Alzheimer's diagnosis [12]. Early approaches focused heavily on kernel methods, such as Support Vector Machines, to draw decision boundaries between healthy subjects and AD subjects. Recent works proposed deep learning based approaches, that learn features directly from neuroimages, having comparable or superior performance to more traditional methods [23]. However, we point out that the vast majority of the works developed so far, although showing excellent predictive performance, are not concerned about providing explanations of their model´s predictions, leading to a black-box diagnosis. Moreover, we see very few applications that consider processing different data sources together in order to execute a more complete and robust AD diagnosis, similar to what real doctors and other healthcare professionals would do [24, 25]. Hence, it is of great importance the development of robust and explainable techniques to provide a trustworthy diagnosis to give people a chance to seek treatment as soon as possible.

1.1 Objectives

The main objective of this work is to propose an accurate and explainable machine learning based methodology for the Alzheimer's Disease diagnosis, considering different data sources in order to achieve a robust and clear diagnostic prediction.

Secondary objectives are:

- Propose an ensemble architecture that combines the following data sources: MRI scans, patient demographics information, psychological/cognitive tests;
- Evaluate the classification performance of the ensemble and of each data source for the diagnosis of early and advanced Alzheimer cases.
- Discuss the importance of each data sources for the Alzheimer's Disease Diagnosis through explanation methods;
- Discuss the efficacy and clarity of image explainability on MRI scans;
- Compare the classification performance and explainability of each separate learning module against the ensemble learning models;
- Discuss overall ensemble behavior by analysing global explanations;
- Discuss biased predictions, fairness and models behavior under specific circumstances by analysing local explanations;

1.2 Contributions

The main contributions of this work seek to fill two research gaps on computer-aided AD diagnosis: build an explainable and multi-modality artificial intelligence tool. We achieve this by providing one final architecture that takes into consideration demographic and neuroimaging data and explains the contribution of each data source and data feature to the final diagnosis. In other words, it is possible to know which variables, brain-related or demographics, contributed more to the diagnosis, as well as to assess specific brain imaging explanations that indicate the most relevant areas of the brain for a particular diagnosis.

To achieve that, we initially built four separate learning baselines experiments using the following data:

1. 2D Sagittal view of a 3D MRI scan processed by a Deep Convolutional Neural Networks (CNN);

2. 2D Coronal view of a 3D MRI scan processed by a Deep Convolutional Neural Networks;

3. 2D Axial view of a 3D MRI scan processed by a Deep Convolutional Neural Networks;

4. Demographics and Cognitive Tests processed by a traditional Machine Learning model;

Then, we tried several stacked/layered ensemble architectures by merging data as follows:

– Prediction probabilities (or scores) of Deep Convolutional Neural Networks (CNN) that process 2D slices of a 3D MRI scan;
– Combination of CNNs scores directly with demographic features;
– Combination of CNNs scores directly with cognitive tests and demographics features;
– Combination of CNN scores with a traditional ML model score that has the demographics and cognitive tests as input features.

Based on the results obtained, we argue that since the final model has very few false-negative cases, our approach can safely reject most healthy patients, sparing them more thorough exams. Moreover, our approach serves as a non-invasive passive cooperation screening tool to obtain an accurate biomarker of Alzheimer's Disease. Overall, we state that our architecture can potentially speed up Alzheimer's Disease diagnosis by providing accurate and trustworthy predictions.

1.3 Dissertation Structure

This dissertation is divided into six additional chapters:

- *Chapter 2 - Artificial Intelligence for Alzheimer's Disease Diagnosis*: Discusses what we know about Alzheimer's Disease pathology, the state of the art for machine learning based diagnostic tools and also points out research gaps and possibilities.
- *Chapter 3 - Explanation and Learning Methods*: Presents some important concepts about machine learning interpretability and explainability and also discusses the theoretical background of the explanation methods that are used during the experiments.
- *Chapter 4 - Proposed Architecture*: Lays out the proposed architecture of our work, explaining data sources that are used, data processing and model training pipelines, as well as the experimental arrangement to test each architecture.
- *Chapter 5 - Experiments and Results*: Presents the results obtained by the architectures previously discussed, comparing experiments classification performance and discussing advantages and limitations of each approach.
- *Chapter 6 - Explanations of Predictions*: Debates the interpretation and explanation of the obtained models during the experiments. This chapter focuses on both global explanations for assessing overall models behavior and also local explanations to explore specific diagnoses executed by the models, such as true positives and false positives.
- *Chapter 7 - Conclusion*: Discusses the advantages of the newly proposed architecture, its innovative aspects, and limitations. Also, based on achieved performance results and explainability clarity, we point out potential future works that can further improve this work.

2
Artificial Intelligence for Alzheimer's Disease Diagnosis

As briefly introduced before, Alzheimer's Disease (AD) is a neurodegenerative disease that causes loss of cognitive brain function. Up to this day, it is still not fully understood the pathogenesis of AD. The scientific community believes that it is caused by the abnormal build-up of proteins in and around brain cells. However, it is not known what causes this process to begin. Also, scientists know that this process begins many years or even decades before any symptom appears [19]. A recent study shows new evidence about the first appearance of tau seeds - the accumulation of tau proteins - that could indicate Alzheimer's Disease [26]. Authors suggest that, instead of the prior belief of the spreading of protein accumulation across the brain, there is a local replication of this anomalous behavior in neocortical areas of the brain.

The first signs of AD could be showing some cognitive impairment, like struggling with language or the loss of some short-term memory, which are generally confused with signs of normal aging [27]. At this stage, a patient can be classified as having Mild Cognitive Impairment (MCI). It is estimated that roughly 44% of patients with MCI convert to AD within a few years [22]. In the literature, these converter patients are generally called progressive MCI (pMCI) or converter MCI (C-MCI) [24]. On the other hand, MCI patients that do not convert are called stable MCI (sMCI) or non-converter MCI (NC-MCI). It is essential to mention that the MCI group of patients is quite heterogeneous, meaning that patients do not have a clear manifestation pattern of symptoms. For this reason, this group is the most difficult to execute a precise diagnosis [28]. That said, the earlier and more accurate the diagnosis is, the chances of effective treatment are higher, providing patients with a better quality of life.

Although the confirmed diagnosis can only be obtained after the patients death, a working diagnosis of Alzheimer's can be carried out with several biomarkers, such as cerebrospinal fluid (CSF), electroencephalogram, neuropsychological examination, neuroimaging and genetic detection. The clinical neuropsychological examinations usually include the mini-mental state examination (MMSE), the Clinical Dementia Rating Sum of Boxes (CDRSB), the Weissler intelligence scale (WAS-RC), the activity of daily living (ADL), and the Alzheimer's disease assessment scale-cognitive subscale (ADAS-Cog) [29].

Neuropsychological/cognitive exams, such as MMSE and CDRSB serve as auxiliary exams to help clinicians determine the severity of dementia.

The most common neuroimaging techniques for AD diagnosis are Magnetic Resonance Images (MRI), Positron Emission Computed Tomography (PET), and Single-photon Emission Computed Tomography (SPECT). With the advance of these techniques, they have become the most reliable and most intuitive form of diagnosis. Structural MRI (sMRI) in high resolution can display soft brain tissues, distinguishing white matter from gray and hence offering an understanding about the brain structural integrity. PET and SPECT images can give an understanding of brain metabolic rates. Therefore, these images can complement each other[23].

Since 2005, Machine Learning techniques have been applied to aid the AD diagnosis, making use of several different biomarker features, with a remarkable increase over the past ten years [12]. That is probably due to the fact of the launch of publicly available data on the internet, such as the ADNI 1 (Alzheimer's Disease Neuroimaging Initiative), in 2004 [30] and the OASIS 1 (Open Access Series of Imaging Studies) [31] in 2007. More recently, Deep Learning techniques have been applied, leveraging the ability to learn features directly from images and making a more precise diagnosis [25, 32]. However, we point out that very few studies focus on providing explanations for the diagnosis with explanation methods, as will be discussed later.

2.1 Traditional Machine Learning for AD Diagnosis

Due to technology and data availability limitations, earlier ML approaches for AD diagnosis focused only on classifying subjects on Cognitive Normal (CN) or Alzheimer's Disease (AD), which is the most straightforward binary classification setup that can be done for the AD diagnosis problem [12]. Simpler strategies, such as Support Vector Machines (SVM) or ensemble of trees (Random Forests, Extra Trees, etc.), could achieve results over 90% of accuracy on a balanced set [33, 34].

Traditional computer vision methods have been used to calculate volume and/or shape characteristics of the hippocampus and cerebral cortex regions of the brain [35, 36], or whole brain region segmentation methods to identify changes in gray matter shape and volume [37]. Thus, most of these methods require the support of a domain specialist. Other computer methods were also applied in order to reduce and/or select the dimensionality of raw data, such as the Principal Component Analysis (PCA), the Linear Discriminant Analysis (LDA), and Fisher Discriminate Ratio (FDR) to select useful voxels as features

[38, 39].

It is also important to mention that most of the aforementioned works have lower performance (generally less than 80%) on identifying MCI subjects from healthy control, also called Normal Control (NC), and AD subjects. This suggests that ML techniques still struggle to model the MCI group with available data. Moreover, as shown in [12] and [23], plenty of works employ these algorithms on tiny datasets, with less than 50 total subjects, making their results less reliable, at least from a machine learning algorithmic generalization perspective.

Therefore, it is important to analyze scenarios where not only data from more subjects are accessible, but also more sources of data, for example, genomics and neuropsychological exams for the same subjects are available. Consequently, the collection of more data opens space for applying more complex methods, such as Deep Learning (DL) based approaches.

2.2
Deep Learning for AD Diagnosis

As said before, traditional ML techniques heavily rely on some domain knowledge to execute a robust feature extraction process over neuroimages. As opposed to this, the most common DL approaches propose the use of Convolutional Neural Networks (CNN) as an end-to-end feature learning process to classify subjects on AD, MCI, or NC (normal control, not to be confused with non-converter) [40]. Since the CADDementia challenge [41], which evaluated the performance of models with sMRI data, several DL-based works have been proposed. Data used for DL-based methods are generally 2D or 3D PET and/or MRI scans from AD, MCI, and NC groups.

Recent studies show excellent results for high resolution 3D MRI, such as the ones proposed by [42] and [43]. While Wegmayr and colleagues propose a straightforward 3D CNN strategy to process the entire data from end to end, Hosseini-Al and colleagues propose a Domain Adaptation method. First, they train a Stacked Auto Encoder (SAE) to extract features from 3D MRI scans. After the unsupervised training of the SAE, they train a fully connected network while fine-tuning the feature extractor. The authors justify this choice, over a simpler 3D CNN architecture, due to the scarcity of data and the ability of the SAE to extract local and global features.

The work from [44] uses slices of 3D sMRI scans to model the AD diagnosis problem as an unsupervised anomaly detection by reconstruction loss. Authors achieve interesting results for MCI vs. AD classification, an area under the curve (AUC) of 0.78 and, as expected, better results for AD vs. NC classification, having an AUC of 0.92.

Moreover, some works propose using Generative Adversarial Networks (GAN) for data augmentation on supervised tasks and for 3D slices reconstruction as an unsupervised task. In [45], authors were able to improve results up to 10% by using GANs to generate synthetic PET 2D scans. They pose that their work is still limited to having separate networks generating each class: AD, CN or MCI. It would be interesting to have a multiclass GAN that can generate data for all three classes to increase the domain's generalization.

More recently, strategies have been proposed to classify subjects and provide explanations about the diagnosis. In [32], the authors show a 3D CNN framework capable of providing visual explanations as heatmaps overlaying on the image. They first modify the original Class Activation Mapping (CAM) [46] for some VGG and Resnet architectures. Then, they also propose a 3D Grad-CAM [47] approach for the same architectures. They conclude that Grad-CAM-based methods are easier to deploy, given there is no need to modify the last convolutional layer, and they offer better results. However, they point out that explanations might be too blurry when upsampled from the last convolutional layer to the original image resolution, due to the extreme image compression rate.

In [25], the authors propose an agnostic explanation framework, called the Swap Test, for the AD vs. NC problem. Also, they compared it with a simpler approach called the Occlusion Test. The Swap Test is also based on an occlusion/perturbation strategy and replaces patches with similar patches from images from another subject. Quantitative and qualitative results show that the Swap Test can provide more consistent explanations when compared to the Occlusion Test. Also, by analyzing some images provided in the paper, it is possible to see high importance assigned to the brain´s hippocampal region for true positive AD.

Lastly, it must be noted that most of the works mentioned in this section make use of earlier proposed deep architectures such as AlexNet [48] and VGG-16 [49]. Not surprisingly, the neuroimages for AD are much harder to acquire when compared to data acquisition for other tasks, such as face recognition or simple object identification. Given the difficulty of acquiring data for the AD problem, a semi-supervised learning approach could be applied, using unannotated MRI or PET samples from subjects with similar characteristics to AD subjects (age, gender, demographic aspects, among others). Also, Domain Adaptation techniques can be applied to leverage similar tasks that could ingest the same type of data (MRI, PET, SPECT, etc). In [50], authors show that even when a small number of unlabeled samples is used for a semi-supervised learning algorithm, there is a considerable improvement when compared to a baseline supervised learning only. In [51], authors show some interesting results by aggregating data from ADNI, CADDementia Challenge and AIBL (The Australian Imaging, Biomarkers &. Lifestyle Flagship Study of Ageing) for a domain adaptation strategy. However, these works do not explore Deep Learning based architectures, only simpler ML algorithms, such as SVM and ElasticNet.

2.3
Multi-Modal Machine Learning for AD Diagnosis

The concept of multi-modality learning comes from how we humans learn from the surrounding environment. Generally, we intrinsically use more than one sense to execute a given task, such as vision, smell, hearing, and motor functions. When humans are employed on any learning task, they can learn by simultaneously hearing, seeing, writing, and moving. Therefore, we can say these senses are complementary to each other to give us humans a richer experience of the environment. This concept can be applied to machine learning, being called Multi-Modal Machine Learning (MMML), sometimes Multi Source Machine Learning (MSML), or even Ensemble Learning [52]. In this case, a given ML or DL model would learn representations from different data sources and data types, such as tabular data, images, signals, etc.

For the AD diagnosis, MMML strategies should process data from domains such as MRI scans, PET scans, cerebrospinal fluid (CSF) features, demographics data, and longitudinal (temporal) structured data. It is important to say that, with the increase of publicly available data, such as the publication of ADNI 2 in 2011 [30], ADNI 3 in 2016 [53], and the TADPOLE Challenge [54] launched in 2017, the community is being pushed towards multi-modality approaches.

An interesting approach is to aggregate both MRI and PET scans in order to obtain higher performance, as presented in [55]. The authors propose a deep 3D CNN architecture that takes 3D MRI scans and 3D PET scans for binary classification between the groups AD versus NC, sMCI versus pMCI and pMCI versus NC. A total of 5 experiments were made, two single modality experiments and three multi-modality experiments. The first multi-modality experiment was a simple concatenation of 3D MRI and 3D PET over the channel direction, considering three dimensions for the spatial localization of voxels and a fourth dimension containing color/channel information. The second was the use of a Siamese convolutional network that learns features from PET and MRI simultaneously with shared weights over the convolutional layers. The last experiment had two separate convolutional branches for 3D MRI and 3D PETs, concatenated before they are used as inputs in a fully connected network. All multi-modality experiments showed better results than the single modality ones. The siamese network architecture found the highest accuracy and sensitivity, while the separate branches scheme showed the highest specificity and AUC. Worst results were found for the alternate training between MRI and PET. The authors point out that for the sMCI vs pMCI case, better results were found when training the network with AD versus NC samples. In other words, a higher classification performance between stable and progressive MCIs - early pathology stages - is obtained when training with features from normal subjects against subjects with more advanced pathology of AD. They argue that this might be due to the fact that sMCI and pMCI features are closer in space, being a hard task to learn. That implies it might be useful to aggregate more sources of data, in order to learn more meaningful features for the sMCI and pMCI case.

In [29], authors use 2D MRI and 2D PET on two separate CNNs and use their output as an auxiliary diagnosis for a final diagnosis. The final diagnosis is made by a so-called Comprehensive Diagnosis that assigns weights to network predictions and also to clinical result diagnosis. The authors affirm that the clinical neuropsychological criteria can be directly converted into AD groups by setting thresholds for the mini-mental state examination (MMSE) and clinical dementia rating (CDR) scores. The weight assigning strategy is done with a correlation analysis, where the output of the PET and MRI networks are compared. If they are consistent, that is, if they predict the same class, correlation and, consequently, weights are higher for the outputs of the networks and lower for the MMSE and CDR scores. In case the predictions diverge and have lower correlation, MMSE and CDR scores criteria receive higher weights. The proposed method shows a higher AUC.

Furthermore, besides applying multi-modality for classifying the patient's actual stage, it can be beneficial to aggregate data in the time domain to know how a patient's condition could progress through time. The work from [24] deals with this problem by proposing a Gated Recurrent Unit (GRU) based multi-modal framework that learns temporal representations from the following domains: cognitive performance, MRI, cerebrospinal fluid, and demographic information. Their training is separated into two steps. First, each GRU is trained over a single modality. Then, their output is concatenated and fed into an L1 regularized logistic regression. This setup makes it possible to deal with the scarcity of overlapping data, i.e., patients that contain data from all domains, since they are training each modality separately. However, for that same reason, it is not possible to find meaningful multi-modality features, because each GRU is optimized only for a single domain. Although they present superior results compared to single modality training and multi-modality without temporal data, it would be interesting to fine-tune the GRU during the aggregation step to learn multi-modality features.

In [56], authors propose a multi-modal multi task (MMMT) Deep Learning framework that predicts a patient's AD group, as well as four psychological exam results that can help diagnose Alzheimer's Disease. They integrate five heterogeneous data sources containing longitudinal information - temporal information - with demographic information, personality, and physical traits such as weight and height, which they refer to as background knowledge (BG). The temporal data are from the following domains: MRI extracted features, PET extracted features, Neuropathology data, cognitive scores, and neuropsychological exams. The presented architecture comprises a set of 1D convolutions followed by a BiLSTM cell block fed by each domain. Their outputs are concatenated and processed by a series of fully connected layers and then are concatenated again along with the preprocessed BG data. There are five separate outputs for the architecture: the classification of the patient (AD vs. MCI vs. NC), the regression of mini-mental state examination score (MMSE), the regression of clinical dementia rating sum of boxes (CDRSB), the regression of functional assessment questionnaire score (FAQ) and the regression of Alzheimer's diseases assessment scale (ADAS-Cog). The MMSE is an 11-question measure that tests five areas of cognitive function: orientation, registration, attention and calculation, recall, and language [57]. The CDRSB is a score obtained through an interview protocol that assesses the patient's cognitive functions in six different areas: memory, orientation, judgment and problem solving, community affairs, home and hobbies, and personal care [58]. This score can also be used to stage levels of dementia in patients. The FAQ evaluates the degree of independence for the performance of ten activities of daily living [59]. Finally, the ADAS-Cog evaluates the severity of cognitive and non-cognitive behavioral dysfunctions characteristic of a subject with Alzheimer's Disease [60]. All the regressions output single score values that are equivalent to the final result of each psychological exam.

Also, they make variations in the main architecture by concatenating the CNN and BiLSTM output instead of stacking them. They show that the best results - higher classification accuracy and lower regression error - are achieved with all modalities along with BG data. Also, they analyze the influence of increasing the temporal window, i.e., feeding the model with data containing more time steps. They considered each time step as 6 months and found that most of the time the model degrades with larger time steps (over 3-time steps). Lastly, the authors state that, up to late 2019, their work is the only one to consider the MMMT problem along with temporal data and that this approach is closer to what a real doctor would do. Indeed, for real medical scenarios, the doctor would execute a handful of exams over time, accompanying the patient's evolution and making a periodical diagnosis.

2.4 Research Gaps and Future Directions for the Application of Artificial Intelligence to Alzheimer's Disease Diagnosis

The first and most explicit line of future work is to improve classification metrics for Mild Cognitive Impairment (MCI) groups. As mentioned before, this is a heterogeneous group, and roughly 40% convert to Alzheimer´s Disease group. Hence, it is also important to identify potential converters (pMCI) from stable subjects (sMCI) to give proper early treatment. It is at least curious to notice that, as shown in [55], training a network for the binary task AD versus NC and testing on sMCI versus pMCI leads to better results than training it with sMCI vs. pMCI. This suggests that AD versus NC may provide better features to distinguish between healthy and non healthy brain structures. That raises the question: Since MCI is a heterogeneous group, what type of data could better represent them? Maybe genomics data could be integrated into existing methods to help identify subjects from this group. Besides aggregating more sources of data, we also encourage future research on understanding what differences and similarities from MCI to other groups. A joint effort of computational intelligence experts and domain knowledge experts could point the right direction on finding more robust patterns for the MCI group.

Also, while all works mentioned in this review focused on classifying subjects within the scope of Alzheimer's Disease progression (AD vs. MCI vs. NC), no work proposed the classification of AD against other forms of dementia. We state that before classifying the progression of AD in a given patient, they should be carried for a trial to identify other potential forms of dementia besides only AD [23]. That way, incorrect diagnosis and prognosis can be avoided, improving patients' chances of survival and quality of life. Also, we understand that developing this type of work might be more challenging since it would require a curated database that incorporates several forms of the disease with several data sources.

About Deep Learning architectures, as far as we are concerned, there is still no literature on Semi-Supervised Learning, Transfer Learning, or Domain Adaptation applied to AD diagnosis. By using these approaches, we believe one could achieve better results, for example, by using images to learn shape, volumetric, and functional generic brain features, without explicit supervision, for AD groups. Then, it would be possible to fine-tune their architecture for a specific AD feature learning process. This opens space to explore more powerful deep architectures and state-of-the-art learning algorithms while simultaneously pushing the boundaries for generalizing the AD diagnosis problem for a more diverse population, even on the scarcity of data availability. Moreover, leveraging unannotated data from other domains to enhance performance for a specific task is closer to real-world scenarios. For example, a hospital from a given locality with insufficient data about AD subjects could execute these strategies to aggregate data from other hospitals or other localities.

Also, there is a notable gap in the literature about providing explanations methods for predictions of ML and DL models. One could argue that some traditional ML models employed for AD diagnosis are already interpretable, such as a Logistic Regression, an SVM with Linear Kernel, or a Decision Tree. However, for any DL-based method, it is impossible to understand and to interpret predictions just by analyzing its weights. One could ask questions such as: Why is the given subject being classified with AD? Was it a high level of brain structure related features? Was it due to brain activity features? Was the neuropsychological exam that provided the most valuable information? Or is it a deformation on the brain that could be spotted on the MRI scan? Without explanations methods, such as SHAP values [61], or interpretable models [62], there is no way to know how predictions are made. Thus, how can a model be trusted if there is no understanding on how it reasons? Therefore, explainability and interpretability of results play an important role, and it should be explored in future works. We can say that the following works [25, 63, 64, 65] are the first ones to introduce an explainability analysis for DL methods to the AD diagnosis, besides only being preoccupied with achieving higher scores and beating existing methods. Therefore, since these works date only from 2019 and 2020, we can safely argue that there is much ground to explore when discussing interpretability for this task.

Additionally, it is of utmost importance for the community to keep developing MMML techniques that can leverage heterogeneous data sources for AD diagnosis and any medical diagnosis. Very few strategies in the literature have worked on learning features from heterogeneous data in its natural form. Most works apply feature extraction and/or feature selection techniques to execute a data fusion step before the actual training step. Even though learning from heterogeneous data poses a natural challenge, applying more complex algorithms, such as Deep Learning, could find hidden patterns when combining these data and, consequently, improve model performance.

Our work aims to fill some of the research gaps shown so far, proposing an explainable AI tool that is able to execute and explain the Alzheimer's Disease diagnosis, using different data sources. We leverage from Deep Learning models to process brain scans and also interpretable machine learning models to serve as a final classifier, pondering DL models outputs as well as data from demographics and cognitive tests.

3
Explanation and Learning Methods

This chapter aims to present the key concepts of explainability, a brief taxonomy about explanainability, and discuss the explanations methods that are used on the proposed architecture. We propose the use of two explainable models for the ensemble architecture: a simple Logistic Regression model and a state-of-the-art nonlinear generalized additive model, called Explainable Boosting Machine. For the MRI scan explanations we propose the use of two back propagation approaches: Guided Grad-CAM, which is a gradient-based approach, and DeepLift, which compares activations of neurons based on reference values. All these models are described in the following sections.

3.1
General Definition of Explainability

First and foremost, before diving into the specificity of the methods mentioned above, let us discuss our current understanding about Artificial Intelligence (AI) explainability under the light of the present literature. First of all, there is no formal mathematical definition for what is considered explainability. In fact, many authors use the terms interpretability and explainability interchangeably [66, 67].

In some works, the word *interpretability* is more closely related to understanding the behavior of a model/system in a macro way, meaning to understand why it behaves the way it does [62, 68]. As Miller states in [69], "*Interpretability is the degree to which a human can understand the cause of a decision*". In the context of Machine Learning, it means to understand how the model makes a prediction based on a group of features. Adadi and Berrada say that *explainability* is closely related to interpretability: "*Interpretable systems are explainable if their operations can be understood by humans*". Moreover, the term Explainable Artificial Intelligence (XAI) represents the general effort of the community to provide reliability and transparency on the use of AI [70]. The word *explanation* is generally tied to understanding a specific event and its potential cause, rather than understanding the nature of the entire system itself [62]. In the context of ML, explanations try to show the causes behind a single prediction or group of predictions solely based on the input features.

However, in our work, the term *explanation* will be used to refer to both the overall/macro and local/specific behavior of a model.

The taxonomy concerning XAI is not unique and can be divided under several topics [71]. The main taxonomy criterion separates methods into intrinsically interpretable models or explanation methods, followed by their specificity and locality, as shown in Figure 3.1.

Intrinsically interpretable models can be interpreted after they are trained. The main advantage of these models is their natural transparency [72]. Moreover, since these models are transparent and interpretable, so are their predictions. Some examples of interpretable models are linear regression algorithms, rule-based models, fuzzy inference systems, and also decision trees [62, 73]. **Explanation methods**, on the other hand, are generally used on top of trained models to extract some level of clarity about their predictions. This information can be extracted directly from the model structure or through the analysis of data instances [66]. The main advantage of this approach is the ability to evaluate trained models and treat them as a complete black box in plenty of cases. We will address both interpretable models and explanation methods as just explanations for simplicity.

The **specificity** of explanations can be separated into two categories: model specific explanations and model agnostic explanations. The first one is related to methods that require access to a model's inner structure, such as the feature importance calculated by the Random Forest based on its sub decision trees split over each feature, or the Grad-CAM, which needs access to the last convolutional layer of a neural network in order to provide visual explanations. Model specific explanations can take advantage of using model structures without making any obscure or simplistic assumptions that might lead to false explanations. However, sometimes it is not possible to access a model inner structure, which leads us to the use of agnostic explanations, such as the LIME or SHAP methods, that use a group of linear models to explain predictions based on how the output of a model responds to perturbations in the input [74, 75].

Lastly, the **locality** aspect of an explanation can be divided into: local explanations and global explanations. As briefly commented across this work introduction, global explanations seek to evaluate the overall behavior of a model at a macro level. Given a set of features and their domain distribution, the global explanation indicates which ones have a higher degree of relevance for the model decision. On the other hand, local explanations analyze single predictions or a small set of predictions in order to assess feature importance. In this case, they show importance degrees relative to a specific input sample.

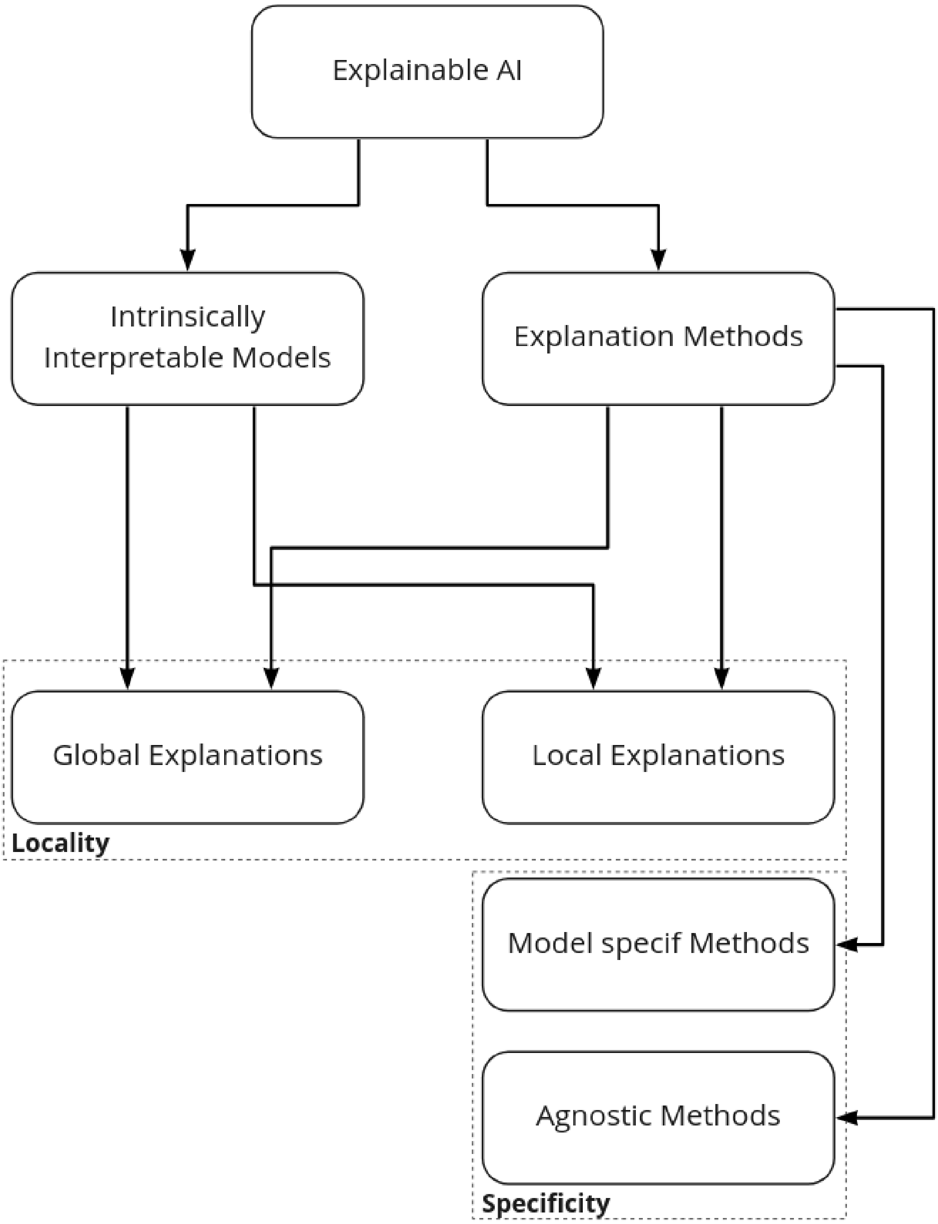

Figure 3.1: Explainable Artificial Intelligence Taxonomy

Nonetheless, it is tough to obtain true global explainability for any ML model due to the difficulty or even impossibility of collecting data on every existing distribution, which directly impacts the model´s operative boundaries. Hence, if a data sample is large enough to represent most of the expected data during model functioning, we can say it is possible to obtain a contained global explanation for that model. In practice, global explanations are calculated based on a training or test set for the model [61].

For more details about XAI taxonomy and further discussions, we recommend the reader to check the following literature: [62, 76].

3.2 Explainable Machine Learning Models

Moving on to the explainable ML models used in this work, we picked two intrinsically interpretable models for the simplicity of evaluation and analysis, one linear and the other nonlinear. After all, the main goal of using explanations in ML applications is to enable humans to make sense of what the models are 'thinking'.

3.2.1 Logistic Regression

Although one might criticize the use of a simple Logistic Regression (LR) [77], we must state that this model is still being heavily used in the community, showing comparable performance to more powerful methods. Just to cite an example, the work presented in [78] shows pretty good performance of a LR when compared to tree and kernel-based models for a Covid-19 diagnosis on patients with suspected symptoms attending to a hospital.

Therefore, the LR can offer a direct global explanation based on its adjusted weights for the dataset at hand. Particularly, a LR model provides single feature importance, not considering any form of feature interaction. The decision boundary of the model is given by:

$$p(t) = \frac{1}{1 + e^{-t}} \tag{3-1}$$

$$t = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n \tag{3-2}$$

where p(t) varies from 0 to 1, having a hard threshold value, in order to transform continuous predictions into categorical predictions. A general rule of the thumb is if p(t) $>$ 0.5, p(t) = 1 and if p(t) $<$ 0.5, pt(t) = 0. That way we can easily execute binary classifications.

When looking at Equation 3-2, assuming that all features are normalized for the same scale, if for a given feature x_1 there is a β_1 weight, for another feature x_2 there is a weight β_2 and β_1 is ten times greater than β_2, it is reasonable to say that the feature x_1 is considerably more important to the decision of the model. That is, x_1 will have ten times more impact than x_2. One limitation of this model is that local explanations cannot be directly assessed, as the model adjusts the same feature importance for an entire data distribution, rather than subsets of data.

3.2.2 Explainable Boosting Machine

The Explainable Boosting Machine (EBM) is a state-of-the-art tabular ML model proposed by the AI Research staff of Microsoft [79]. The mathematics behind this model relies on the alliance of boosting techniques with generalized additive models (GAM) while maintaining intrinsic interpretability. The EBM builds a generalized additive model with pairwise interactions by executing a round-robin training procedure on one feature at a time, using a really small learning rate. Its general equation is given by:

$$g(E[y]) = \beta_0 + \Sigma f_j(x_j) + \Sigma f_{ij}(x_i, x_j) \tag{3-3}$$

where $g(E[y])$ is the link function, f_j and f_{ij} are shape functions. The link function is a general representation that adapts the EBM to different tasks, such as regression or classification, and could be an identity function, logistic function or softmax function. The shape functions are defined by splines or a set of boosted regression trees and their purpose is to add nonlinearity to achieve higher accuracy.

The EBM brings one major improvement when compared to the original GAM [80]: the ability to capture feature interaction importance, that is the compound effect of two features in a model. In real-world scenarios, feature interactions are many times constraints of the problem, i.e., a change made in one feature impacts the value of other features. This constraint is often not considered in many explainability algorithms since most of them rely on local linear approximations. Therefore, we state that the ability to capture the pairwise interactions contributes to enhancing the model's accuracy while keeping it explainable.

Another advantage of this model in comparison to the LR, or other explanation methods, is its ability to provide global and local explanations. Due to its nonlinear nature, the importance of features may vary depending on the input domain. Therefore, local explanations are represented by the value of each additive term - feature, or pair of features - in the decision function. Global explanation is achieved by measuring the standard deviation of each shape function, to assess the overall contribution of each term for the final score.

For more details about the mathematical framework of shape functions construction and computational training procedure, please check [81, 82].

3.3 Explanations of Deep Learning

Finally, since we intend to process magnetic resonance images to execute the AD diagnosis, we shall consider using Deep Learning models. Due to the building blocks nature of stacking neuron layers followed by a nonlinear activation function, even neural networks containing only two hidden layers are impossible to interpret. Hence, explanations have to be carried through external algorithms.

Explanations for Deep Neural Networks (DNN) can be achieved by feature visualizations, pattern generation analysis and neuron activation. The most common explanations for computer vision tasks are feature importance visualizations, which consist of finding which parts or patterns of an image contributed more to a particular outcome [83].

While there is a myriad of deep explanations in the literature [84] and it is of our interest to explore MRI explanations, we will attain our efforts to two visual explanation methods: Guided Grad-CAM [47] and DeepLift [85]. Additionally, we will also use one visual explanation enhancement technique called SmoothGrad [86]. These approaches are based on backpropagation calculations and need to access DNNs inner structure in order to calculate the final visual explanation.

We justify the choice of these algorithms due to their wide use in the literature [25, 87] and its visual explanations that can provide general areas highlights as well as pattern highlights on important areas of an image. We seek to investigate whether or not gradient-based approaches (GuidedGraCAM) can be outperformed by alternative backpropagation calculations (DeepLift).

3.3.1 Guided Grad-CAM

Guided Grad-CAM stands for Guided Gradient Class Activation Mapping and is actually the combination of the Grad-CAM and the Guided Backpropagation methods [47]. The Grad-CAM calculation steps are as follows:

1. Gradient calculation from the output up to the last convolutional layer of the DNN, right before the first fully connected layer:

$$grad_{ij}^{k} = \frac{\delta y^{c}}{\delta Conv_{ij}^{k}} \tag{3-4}$$

where c is the predicted class, k is the number of feature maps in the last convolutional layer and i,j are feature map pixel indices. $grad_{ij}^{k}$ is the gradient value at each pixel of each feature map.

2. Global average pooling of the gradients, transforming them in a vector of scalar weights (w_k):

$$\boldsymbol{w_k} = \frac{1}{i.j}\sum_{i}\sum_{j}\frac{\delta y^{c}}{\delta Conv_{ij}^{k}} \tag{3-5}$$

3. Sum of products between the weights vector and each feature map on the last convolutional layer (C_k), resulting into a matrix of the same size as the feature maps:

$$\boldsymbol{C_{(m,n)}} = \sum_{k}\boldsymbol{w_k.C_k} \tag{3-6}$$

In other words, this is a weighted sum of the features maps over the filters direction (k). The size of the C matrix is (m,n).

4. ReLU activation to eliminate negative values:

$$\boldsymbol{R_{(m,n)}} = ReLU(\boldsymbol{C_{(m,n)}}) \tag{3-7}$$

5. Upscale of the resulting $R_{(m,n)}$ matrix to the size of the input image (u,v):

$$\boldsymbol{L_{(u,v)}^{Grad-CAM}} = upscale(\boldsymbol{R_{(m,n)}}) \tag{3-8}$$

The Guided Backpropagation is actually the gradient calculation through all the network layers, from the output to all the way up to the input image, with one key modification: each layer gradient passes by a ReLU activation function to eliminate negative contributions.

The final Guided Grad-CAM is obtained by the element-wise multiplication of the Guided Backpropagation and Grad-CAM results. This multiplication is possible due to the final upscale operation of the Grad-CAM, that transforms the explanation obtained to the size of the image. The general equation is given by:

$$\boldsymbol{L_{(u,v)}^{GuidedGrad-CAM}} = \boldsymbol{L_{(u,v)}^{Grad-CAM}} * \boldsymbol{L_{(u,v)}^{GuidedBackprop}} \tag{3-9}$$

3.3.2 DeepLift

DeepLift stands for Deep Learning Important Features, and it was proposed in [85]. Although this is also a backpropagation approach, its main essence differs from the Guided Grad-CAM, for it does not directly calculate the gradient. Instead, it uses a set of rules and strategies that consider the activation differences from a reference, the propagation of these differences through the network, and a separate calculation for negative and positive contributions. As stated by the authors, DeepLift can deal with gradient saturation and thresholding problems that the Guided Grad-CAM and other gradient based approaches cannot. This is achieved by calculating the difference from reference values on neurons, instead of calculating the gradient. Another key aspect to consider when using the DeepLift is the choice of a reference value that, as shown in the original work, have a great impact on how explanations are generated.

To present the general idea of the DeepLift formulation, let $\boldsymbol{x}$ be a feature vector of size i, $\boldsymbol{x} = [x_1, x_2, ..., x_i]$; $\boldsymbol{h}$ be an intermediate vector of neurons (hidden layer) of size j, $\boldsymbol{h} = [h_1, h_2, ..., h_j]$; $\boldsymbol{y}$ be the output layer. Let us consider the output as being only a single neuron for simplicity. The main steps necessary to calculate the contribution of each feature of $\boldsymbol{x}$ with respect to the output y are:

1. Make a normal forward pass with the given input $\boldsymbol{x}$ to calculate the output value. The neural network is denoted by the generic function f:

$$y = f(\boldsymbol{x}) \tag{3-10}$$

2. Pick a reference value ($\boldsymbol{x^0}$) and make a forward pass to calculate the output value (y^0) with respect to the reference:

$$y^0 = f(\boldsymbol{x^0}) \tag{3-11}$$

3. Calculate the difference from reference in every neuron, through the entire network, all the way from the input to the output:

$$\Delta y = y - y^0 \tag{3-12}$$

$$\Delta \boldsymbol{h} = \boldsymbol{h} - \boldsymbol{h^0} \tag{3-13}$$

$$\Delta \boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x^0} \tag{3-14}$$

4. Calculate multipliers for neurons, based on the ratio between the current layer and the previous layer. This step is similar to a partial derivative that is calculated towards all possible directions/variables. For brevity, we show part of the multipliers between y and $\boldsymbol{h}$:

$$\boldsymbol{m_{\Delta h \Delta y}} = \frac{\Delta y}{\Delta \boldsymbol{h}} \tag{3-15}$$

$$\boldsymbol{m_{\Delta h \Delta y}} = [m_{\Delta h_1 \Delta y}, m_{\Delta h_2 \Delta y}, ..., m_{\Delta h_j \Delta y}] \quad (3\text{-}16)$$

$$\boldsymbol{m_{\Delta h \Delta y}} = [\frac{\Delta y}{\Delta h_1}, \frac{\Delta y}{\Delta h_2}, ..., \frac{\Delta y}{\Delta h_j}] \quad (3\text{-}17)$$

5. We propagate the multipliers from the output layer to the input layer, by applying a chain rule for multipliers to build a vector of same size as the input dimension:

$$\boldsymbol{m_{\Delta x \Delta y}} = \sum_{\boldsymbol{j}} \boldsymbol{m_{\Delta x \Delta h_j} m_{\Delta h_j \Delta y}} \quad (3\text{-}18)$$

$$\boldsymbol{m_{\Delta x \Delta y}} = [m_{\Delta x_1 \Delta y}, m_{\Delta x_2 \Delta y}, ..., m_{\Delta x_i \Delta y}] \quad (3\text{-}19)$$

where:

$$m_{\Delta x_1 \Delta y} = \sum_j m_{\Delta x_1 \Delta h_j} \boldsymbol{m_{\Delta h_j \Delta y}} \quad (3\text{-}20)$$

$$m_{\Delta x_2 \Delta y} = \sum_j m_{\Delta x_2 \Delta h_j} \boldsymbol{m_{\Delta h_j \Delta y}} \quad (3\text{-}21)$$

$$... \quad (3\text{-}22)$$

$$m_{\Delta x_i \Delta y} = \sum_j m_{\Delta x_i \Delta h_j} \boldsymbol{m_{\Delta h_j \Delta y}} \quad (3\text{-}23)$$

6. Calculate the final contributions by executing the scalar product between the multipliers vector and the difference from input;

$$\boldsymbol{C_{\Delta x \Delta y}} = \boldsymbol{m_{\Delta x \Delta y}} \cdot \Delta \boldsymbol{x} \quad (3\text{-}24)$$

Therefore, the final contributions with respect to $\boldsymbol{x}$ are the actual explanations for a given data instance.

Additionally, DeepLift has a set of rules to deal with negative and positive contributions, as well as cancelling effects caused by zero values during Δ calculations. These rules deal with it by assuming that every difference from reference is a sum of positive and negative parts: $\Delta y = \Delta y^+ + \Delta y^-$. This separation is propagated backwards over the entire network and a RevealCancel rule or Rescale rule can be applied. The RevealCancel rule sets the average impact of Δy^+ to the average impact of Δx^+ before and after considering the influence of Δx^- during multipliers chain rule calculation. The Rescale rule is a simplistic assumption that negative and positive Δ have the same impact over the multipliers. Hence, when a given $\Delta = 0$, they assign a 0.5 value for the multiplier. We will not go over details of these implementations and we point to the Appendices of the original paper [85].

Moreover, since the DeepLift operates in a manner differently from plain gradient calculations, it is possible to obtain both positive and negative contributes of the input, relative to the output. We must state that so far no work in literature actually explored this additional feature of providing negative and positive contributions since it is naturally a harder interpretation to make. For our image classification problem, we can leverage from this feature and consider positive contributions relevant areas of the brain and negative contributions areas that are either confusing to the model or that are really similar between healthy and pathological patients.

3.3.3 SmoothGrad

Lastly, the SmoothGrad (SG) technique was proposed to be used as either a stand-alone explanation method or explanation improvement, affirming that it is possible to obtain less noisy explanations by adding Gaussian noise to the gradient calculation [86]. The original paper does not provide a strong mathematical framework to support this affirmation. However, through several empirical experiments and findings, the authors show that it is possible to obtain visually smoothed and more coherent explanations. The SmoothGrad general formulation is a simple average of n gradient based explanations, having as input an image X, added with some Gaussian noise $N(0, \sigma^2)$, and it is given by:

$$M_{smooth}(X) = \frac{1}{n}\sum_{1}^{n} M_{grad}(X + N(0, \sigma^2)) \tag{3-25}$$

where the term denoted by M_{grad} stands for a single gradient based explanation with Gaussian noise.

They state that the averaging combined with small amounts of noise can bring a smoothing effect and eliminate potential outliers. Another essential consideration made by the authors is that it is important to cap outlier values due to the fact that many gradient based methods end up resulting in a few pixels with a much larger gradient value when compared to their neighbors pixels [88]. For that, they suggest capping extreme values at a 99% percentile in order to obtain more coherent explanation maps.

Other authors tried to prove the SG intuition proposal analytically and showed that the visually smoothing factor is actually caused by a series of higher order partial derivatives [89]. However, they do not show any practical example or application of their theoretical findings.

Nevertheless, in later works, we already see the SG being used along with other techniques to obtain enhanced explanations [87, 90]. Therefore, due to its straightforward application, the SmoothGrad can safely be combined with any other explanation method by simply averaging a number of N noisy samples.

4
Proposed Architecture

This chapter presents the proposed models and the necessary data pipeline for Alzheimer's Disease diagnosis. As mentioned earlier during this work, we propose an explainable AI solution that is able to process data from different sources, in order to provide a robust and clear diagnosis.

We chose three main data sources: MRI scans, Demographics and Cognitive Tests. The Magnetic Resonance Image Scans were chosen due to their high availability and usage across several works to diagnose AD. Demographic information - age, gender, race, educational level, etc - about patients are generally collected for registration purposes and cognitive tests can indicate several levels of cognitive impairment as well as AD stages, by analysing a myriad of behavioral, sensory and memory functions.

Figure 4.1 shows the general architecture, containing the fundamental building blocks to obtain an explainable and multi-modality classifier for the Alzheimer's Disease diagnosis. It is important to mention that demographics data and cognitive tests data are in the tabular format, while the MRI scans are in the 3D image format.

The first necessary steps are **raw data acquisition** and **data preprocessing**, to get appropriate data to feed the models. Also, there is a need of data alignment between different data sources, represented by the dashed line that links the two preprocessing boxes. Particularly, it is indispensable to collect data from patients that have all data sources available. We will go over details about this strategy later on.

After data is preprocessed, **separate learning classifiers** are trained on their respective domain (source). Then, the final ensemble classifier is trained with the prediction scores of each separate learning classifier. Lastly, after training the ensemble classifier, it is possible to generate predictions and explanation of predictions in order to get a robust and clear diagnosis.

Each block of this general architecture is discussed in details ahead.

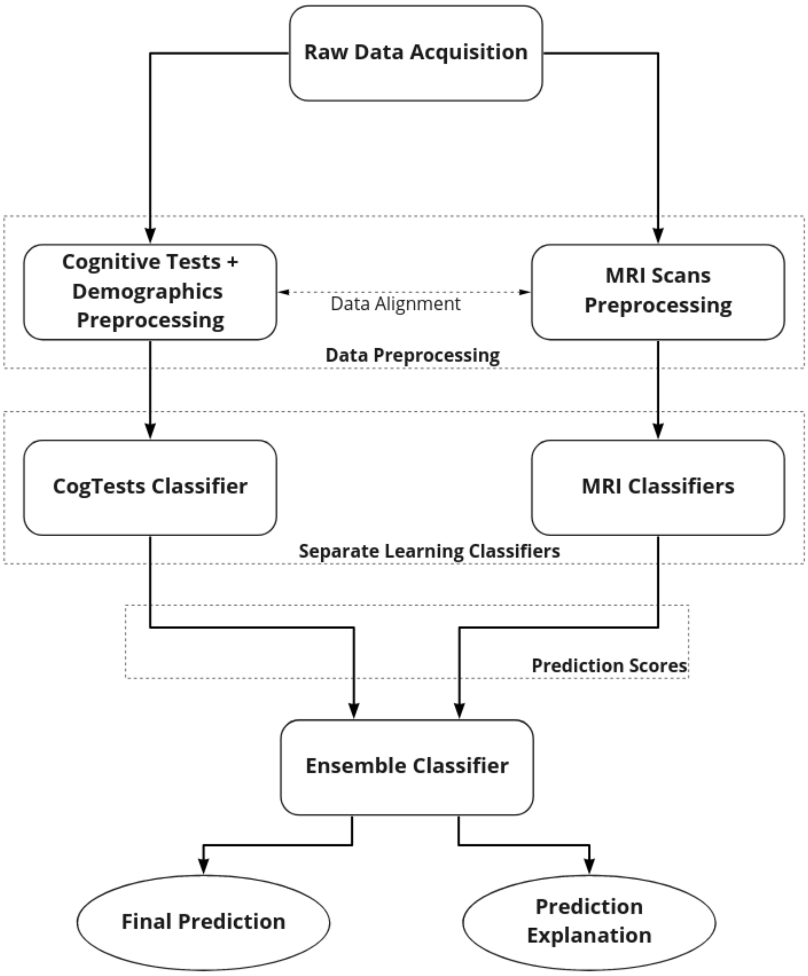

Figure 4.1: General Pipeline Architecture

4.1 Raw Data Acquisition and Data Preprocessing

For brevity, the group of demographics data and cognitive tests will be referred as CogTests from now on, except when explicitly stated.

Following the data acquisition, it is necessary to perform some cleaning and standardization steps to make the data adequate to be processed by the models. Since MRI scans are in 3D image format, and demographics and cognitive tests are in tabular format, these two data types must be preprocessed separately. Figure 4.2 shows all the steps to obtain the processed data, where the left side corresponds to the CogTests preprocessing and the right side to the 3D MRI scans preprocessing.

The CogTests preprocessing pipeline followed the standard tabular data preprocessing pipeline. We opted to remove data samples with missing values, remove highly correlated features - any feature in a pair of features that has over 0.9 linear correlation -, and use the one-hot encode for categorical features,

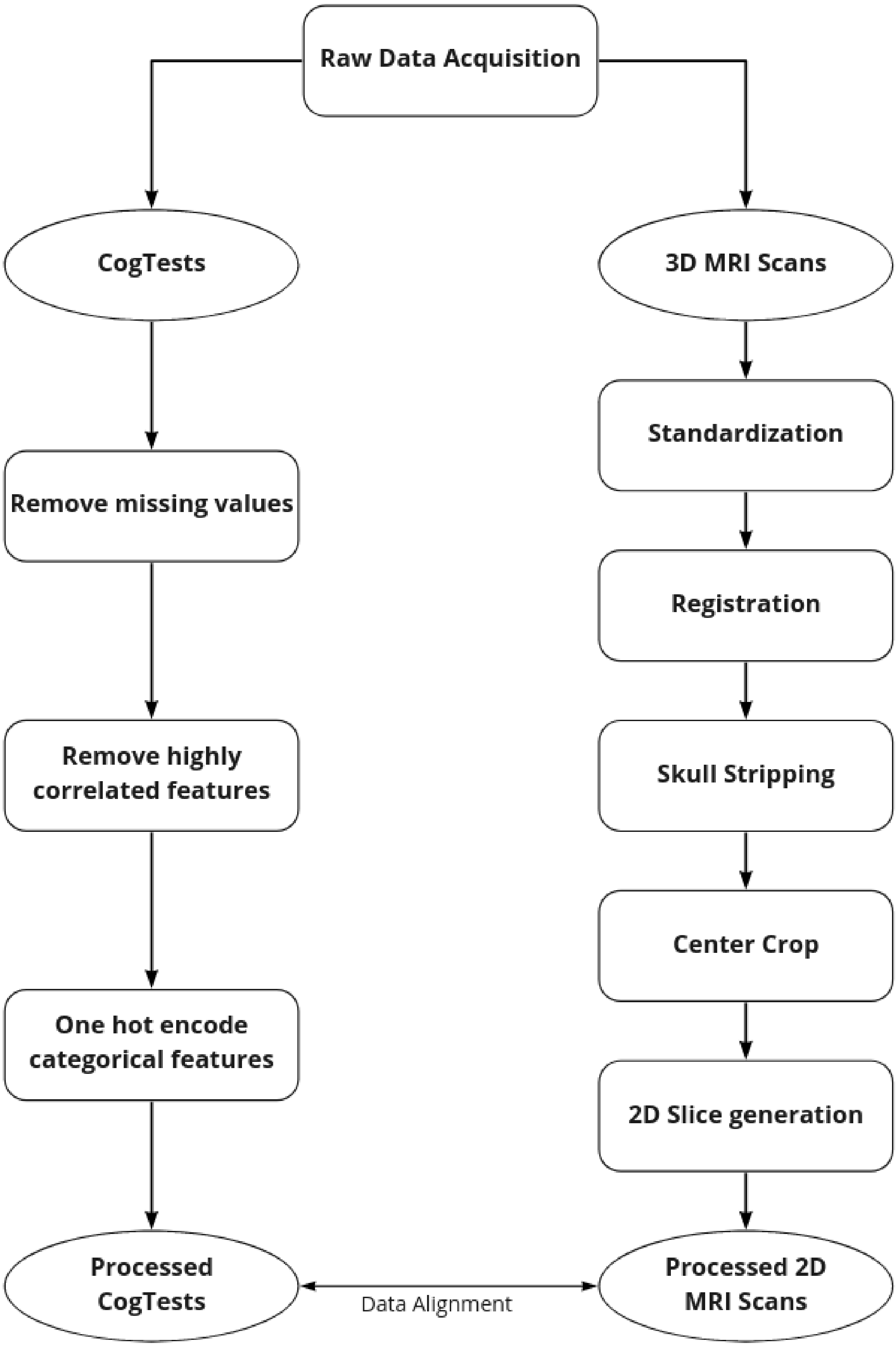

Figure 4.2: Data Preprocessing Pipeline - Left side branch: CogTests preprocessing; right side branch: MRI scans preprocessing

that is, binarize each category into one column containing only 0 or 1.

Since the MRI preprocessing steps is less obvious to most readers, and also quite more complex, we will go over each block in more details. This MRI preprocessing pipeline was inspired on previous works that used 3D and 2D MRI scans for AD diagnosis and obtained a high classification performance [63, 64].

4.1.1 MRI Standardization

The MRI Standardization process is the rescale of voxels' (3d pixels) intensity to a reference range. This step is necessary to establish an initial base of comparison between all images and all patients. It is known that even for the same patient and equipment, different voxel intensities might occur [91]. Moreover, due to potential outlier voxels with really high or low intensity, it is also beneficial to clip extreme values prior to the standardization. Therefore, the we set a clipping value on the 99.8% and 0.02% percentiles prior to rescaling. This follows a procedure already seen in previous works, such as [25].

The voxel intensity rescale generally uses a brain Atlas, which will be discussed briefly in the next section, as the reference. Thus, the maximum and minimum values of a brain Atlas will be used to rescale all images that are being processed on the MRI data pipeline.

4.1.2 MRI Registration

During the magnetic resonance scan, a patient may make subtle movements, or lean their head in a way that is not aligned with the scanner, generating brain scans that might be tilted or shifted to the sides. The objective of an MRI Registration is to spatially align different brain images, so the brain anatomical regions occupy roughly the same places. There are several registration methods [92], using linear and non-linear transformations. This process also makes use of a brain Atlas, also called registration Atlas, that servers as a fixed reference for the moving brain images that are being adjusted.

A brain Atlas is a type of brain template, containing a set of well known annotated regions. The purpose of an Atlas is to map brain areas of a new patient to already known regions. Using an Atlas can help on tasks such as brain segmentation, brain injury detection, brain degeneration detection, etc [93]. For the Alzheimer's Disease problem, one the most common Atlas is the MNI-ICBM-152 model, which was first introduced to the community in [94] and is an Atlas to specifically study older adults. Furthermore, it is important to say that the choice of a registration Atlas is not universal, meaning that the one Atlas might not serve a certain type of population or cohort. As Lee and colleagues point out in their work [95], they found significant variations between the MNI-ICBM-152 model and their proposed Atlas based on a Korean elderly population.

Additionally, regardless of brain Atlas model of choice, the same Atlas/template must be used throughout the entire MRI preprocessing pipeline, to keep consistency over patient data.

Therefore, since we want to preserve the brain morphology in order to assess the Alzheimer's Disease pathology, we chose to apply the so-called Affine registration, as it does not cause any deformation to the base structure of the brain [96]. This registration procedure executes a sequence of rotations, translations and rescaling, to match the moving brain image with the registration Atlas, by measuring some sort of similarity metric, such as the pixel correlation ratio [97].

4.1.3 Skull Stripping

After having the MRIs standardized and registered, we need to extract the regions of interest and remove irrelevant ones, that is, extract only brain related matter and remove any other non brain tissues and the skull. This procedure is called brain extraction or skull stripping and it is used across several neuroimaging diagnostic tasks, including Alzheimer's Disease diagnosis [98, 99].

One could argue that we could let the deep learning models extract relevant features automatically to make the AD classification. While that implication is valid, previous works have already shown improved performance when extracting only brain related matter for AD related tasks [63, 100]. Moreover, by removing irrelevant parts of the image, we automatically reduce computational overhead.

4.1.4 Center Crop

Subsequently, we crop the 3D MRI scans on a cubic volume of size 100x100x100 voxels. The cropping must be made in the center of the 3D image, where it contains the most relevant parts that are affected by the brain deterioration caused by the the development of the Alzheimer's Disease. This will reduce even further the computational cost of the experiments.

4.1.5
2D Slices Generation

We slice the final 3D MRI into several 2D images. The slicing is done over the three main brain orientations: Sagittal, Coronal and Axial. As will be shown soon, each orientation slice will feed a specific Deep Convolutional Neural Network (CNN). An additional benefit of this step is another computational cost reduction, since 3D MRIs would require 3D convolutions, which are computationally heavier when compared to 2D convolutions.

4.1.6
MRI and CogTests Data Alignment

After preprocessing CogTests and MRIs, there is one additional step to correctly setup up the ensemble experiments. Since we are dealing with three different data sources (MRI, cognitive tests and demographics) that are collected separately, it might happen for some patients to have incomplete or missing data. The data alignment is nothing more than keep records of which data source is available for a given patient.

In practice, this means mapping patients with only MRI or CogTests available, as well as patients with all data sources available. Since we area treating each learning module separately, it is possible to use patients containing only MRI data to train the MRI classifiers, and patients containing only CogTests data to train the CogTest classifier. However,as will be shown further ahead, the ensemble architectures that use both data types can only be built using patients with all data sources and data features available.

4.2
Separate Learning Classifiers

The main purpose of each separate learning classifier - MRI and CogTests classifiers - is to serve as an intermediate building block for the ensemble architectures, by generating prediction scores using only one data source. Additionally, these classifiers can be used as baseline experiments to measure the classification performance of the ensemble architectures.

Therefore we have proposed the following separate learning classifiers:

1. Deep Convolutional Neural Network processing 2D MRI sagittal slices;

2. Deep Convolutional Neural Network processing 2D MRI coronal slices;

3. Deep Convolutional Neural Network processing 2D MRI axial slices;

4. Traditional Machine Learning model processing demographics and cognitive tests tabular data.

We note that the CogTests classifier will not appear in all proposed ensemble architectures, as will be discussed shortly.

4.3 Ensemble Learning Architectures

We propose a total of four different ensemble architectures that are gradual variations of each other. The final ensemble classifier for these architectures is an explainable machine learning model that can receive the prediction scores of each intermediate classifier, as well as tabular features. The explanations of the predictions are made with respect to the input features.

The first architecture is just a simple combination of all MRI CNNs and it is shown in Figure 4.3. The input for each CNN is the respective 2D orientation slice extracted from the preprocessed MRI. The inputs for the ensemble classifier are prediction scores ranging from 0 to 1, and the higher the score, the higher the chances of having the pathology. For this simpler architecture, although using only one data source, we gather 3 different points of view from the brain in order to provide a more robust prediction and explanation.

The second and third architecture are shown in Figure 4.4. The only difference between them is that while the second architecture only uses demographics along with CNNs, the third adds the CDRSB score, which is one of the main cognitive tests for dementia profiling. Therefore, the inputs for this ensemble classifier are prediction scores ranging from 0 to 1, categorical and numerical features. These architectures provide prediction and explanation based on processed information about brain structure - CNN predictions - demographics information and a cognitive function information through the CDRSB score.

The last architecture is presented in Figure 4.5. This case is the most complex and most complete one, where one can assess the diagnosis through brain structure processed diagnosis along with neuropsychological exams and demographics processed diagnosis. Here, the CogTests model can use as input features the consolidated or partial scores from neuropsychological exams as well as demographic features.

The first and second ensemble architectures can be seen as screening steps to assess the Alzheimer's Disease, since they require less information and also passive cooperation of the patient. Third and fourth architectures make use of cognitive tests results, which may take more effort to collect, since it is unlikely

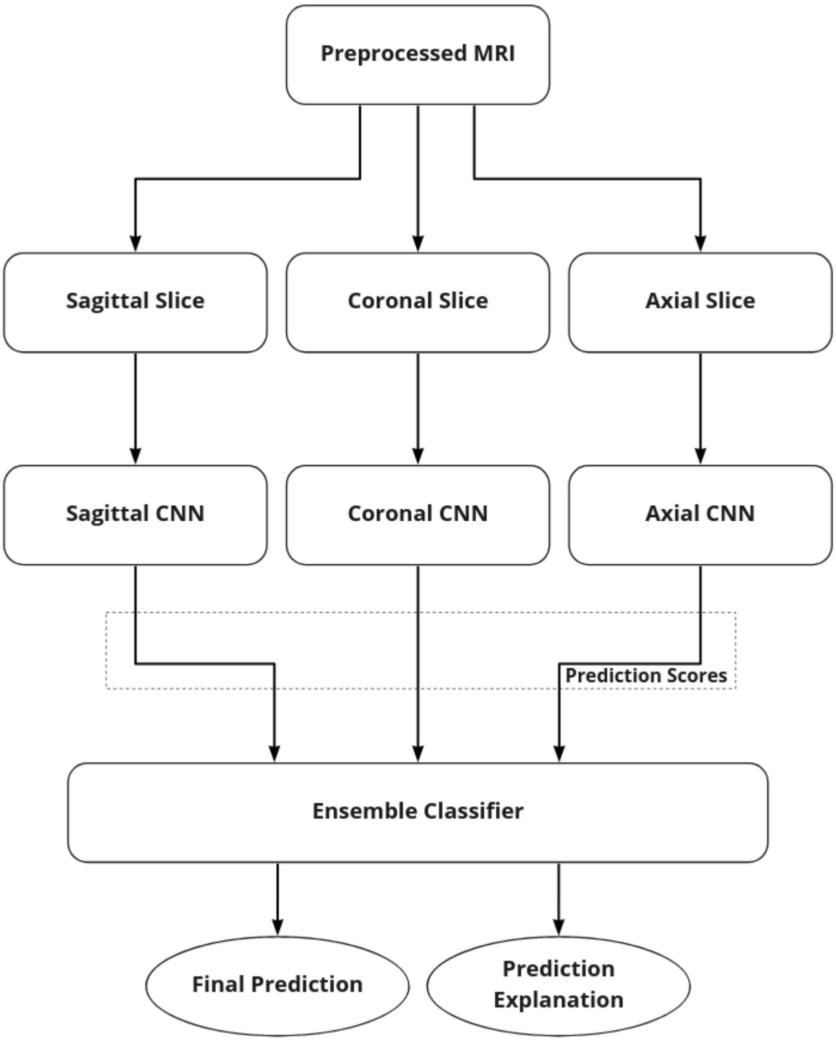

Figure 4.3: Ensemble of 2D MRI CNNs

a patient would take all cognitive tests at one single visit. Furthermore, it is important to mention that all data necessary to feed any of the architectures are collected through non invasive procedures.

4.4 Experimental Setup

All proposed architectures were thought to be tested on binary classification, that is, classifying if a patient has or not the pathology. Therefore, it is important to present two experimental setups that can test the proposed architectures and pipelines.

The first experiment is the most traditional one, that seeks to classify a patient as a Cognitive Normal (CN) or Alzheimer's Disease diagnosis (AD). This experiment will be addressed as ADxCN case.

The second experiment seeks to classify a patient between Cognitive Normal (CN) or Mild Cognitive Impairment (MCI), and it will be addressed

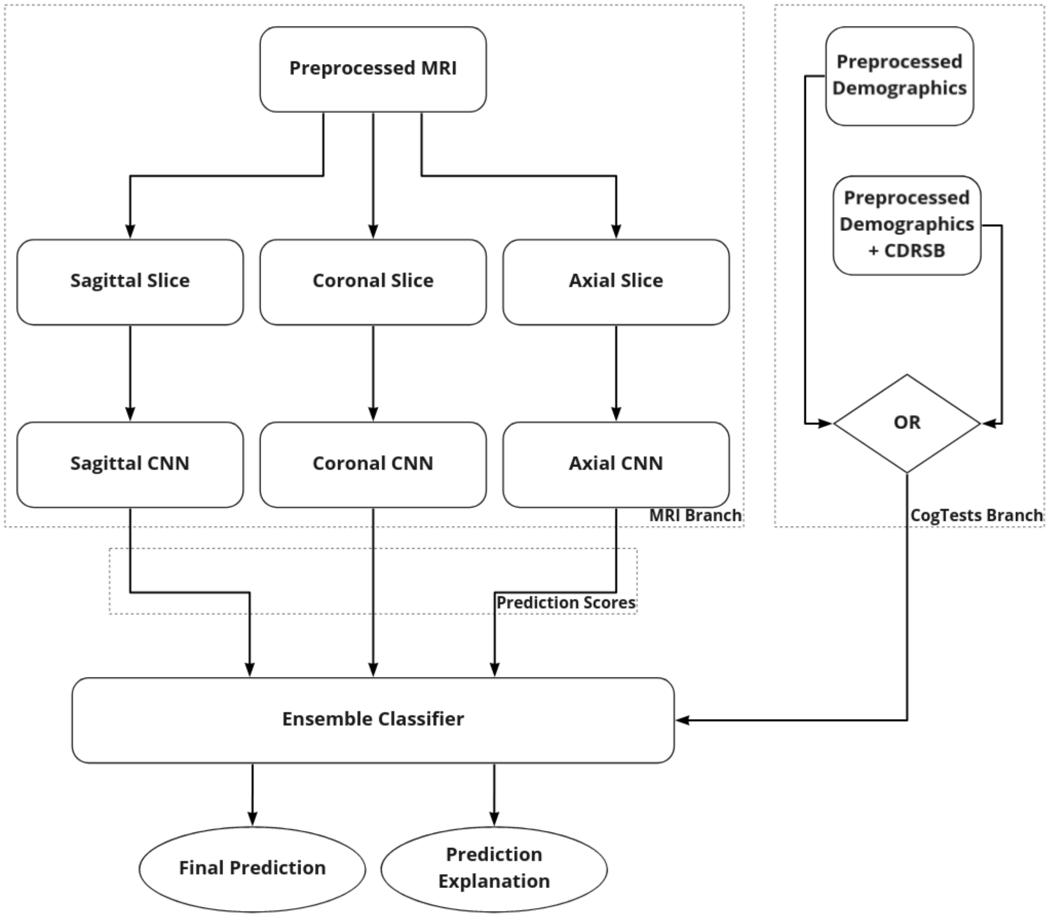

Figure 4.4: Ensembles of 2D MRI CNNs, Demographics and CDRSB

as MCIxCN. As discussed before, previous works show that classifying AD patients using structural information is easier when compared to MCI patients, for patients in more advanced stages of the disease manifest brain changes more often.

The next chapter describes the implementation details of the pipeline and architectures, and the experimental results obtained for the classification performance. Afterwards, Chapter 6 discusses in more details the explanations of predictions made by the experimental models.

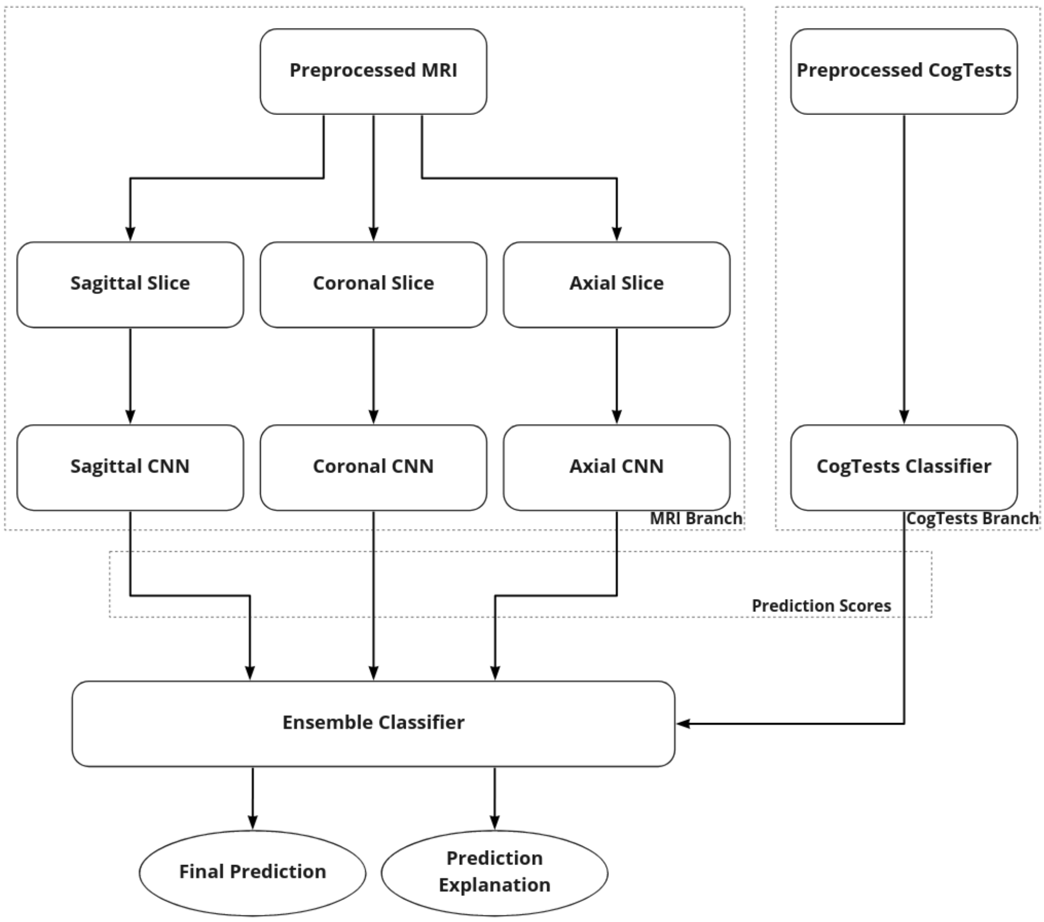

Figure 4.5: Ensembles of 2D MRI CNNs and CogTests Model

5
Experiments and Results

Before addressing the experimental results, we present the available computational setup and assess the preprocessed data. Afterward, we present the **Separate Learning Experiments**, analyzing the performance of each architecture containing only one source of data as an input. Next, the **Ensemble Learning Experiments**, contains more than one data source as input, such as cognitive tests, demographics, and prediction scores from other models. Finally, we discuss the obtained results for the test set by analyzing the Receiving Operating Characteristics Curve and a table showing important metrics, such as F1, AUC, and Accuracy.

5.1
Computational Setup

All the experiments and data processing were carried out on a Google Colab Pro environment, with a virtual machine with the following setup: Debian Linux operational system, 25Gb RAM, 2.4GHz quad-core processing unit, 200Gb SSD, GPU Nvidia K80, Python 3.8. Deep learning and machine learning packages used were PyTorch 1.10, Scikit-learn 1.21, and Interpret.ML 0.2.7. MRI processing packages used were ANTsPy 0.2 and DeepBrain 0.1. All the code and notebooks used can be found at this URL: `https://github.com/lucasthim/mmml-alzheimer-diagnosis`

5.2
Preprocessed Data Assessment

All data used in this work was acquired at from the Alzheimer's Disease Neuroimaging Initiative (ADNI) [30]. This initiative started in 2004 in the United States and is composed of several clinical trials that follows patients throughout years in order to assess the diseases progression. Different types of data are collected at multiple research centers from large cohort of subjects, including clinical, genetic, MRI, PET and biospecimen data.

The data sources used for the experiments are:

– Tabular data consisting of cognitive tests and demographic data of patients.

- MRI data in the form of 2D images.
- Tabular data of prediction scores from trained models that are used in the ensemble experiments.

We initially downloaded approximately 5000 MRIs, ranging from 1.5T to 3T field intensity and containing T1 or T2 weights. Then, after a few initial experiments, we decided to narrow down the dataset to samples that had an initial MRI correction treatment such as Bias field correction [101] and Nonlinear Gradient unwarping [102]. This decision led to a total of 3089 3D MRIs available.

During the MRI preprocessing pipeline, some images were removed due to inconsistencies. Although the automated skull stripping, executed with the DeepBrain package, has a high performance, some exceptions and bugs might occur, such as the algorithm deleting entire sections of the brain. Since we did not have an expert to manually strip the skulls, we opted to just remove these images. A safety check was implemented in order to filter out images that failed the skull stripping and registration process. This check was a sum of all voxel values. In case this sum is equal to zero, the image gets rejected, just to exclude critical failures of the automated skull stripping process. Although other values rather than zero could have been chosen, during early manual checks we realized that the algorithm either output an excellent stripped brain or a blank image. Only 7 images were excluded during this process, resulting in 3082 3D MRIS available for the experiments.

The ADNI repository provides a consolidated file with roughly 15000 samples of patient visits containing over 100 features such as psychological tests, brain extracted features, and demographics.We did not execute any missing data imputation for the CogTests (cognitive tests + demographics) tabular data and kept only patients with all features available. As a result, we obtained a total of 4991 data samples for the experiments.

We selected all demographic information available: years of education, gender, race, ethnicity, marital status and age. We one hot encoded the races and ethnicity, and for the marital status, we only kept a binary flag indicating if the patient is widowed or not. We argue that being a widow is a big challenge for anyone, specially for elderly people, who have not uncommonly lived long years with their partners.

There were over 50 features related to partial or final results concerning the cognitive tests available. We looked for the most common cognitive tests in the literature and selected the following ones:

– CDRSB - Clinical Dementia Rating Scale–Sum of Boxes: obtained through an interview protocol that assesses the patient's cognitive functions in six different areas: memory, orientation, judgment and problem solving, community affairs, home and hobbies, and personal care [58]. This score can also be used to stage levels of dementia in patients.

– MMSE - Mini Mental State Examination: an 11-question measure that tests five areas of cognitive function: orientation, registration, attention and calculation, recall, and language [57].

– ADAS-Cog - Alzheimer's Disease Assessment Scale–Cognitive: evaluates the severity of cognitive and non-cognitive behavioral dysfunctions characteristic of a subject with Alzheimer's Disease [60]. The available features for this exam were: ADAS11, ADAS13 (an extension of the ADAS11), and ADASQ4 (a partial result of the ADAS11).

– FAQ - Functional Assessment Questionnaire: evaluates the degree of independence for the performance of ten activities of daily living [59].

– MOCA - Montreal Cognitive Assessment: 30-question test that evaluates short-term memory, executive functions, attention, language abilities, and abstraction. This test helps to predict dementia for MCI subjects, as well as decide if an in-depth diagnosis is needed to assess Alzheimer's Disease [103].

– TMT - Trail Making Test: Evaluates attention mechanisms of a subject through two sub-tests of connecting numbers and letters [104]. The feature available for this test is the Time to Complete Part B of the Trail Making Test (TRABSCOR).

– RAVLT - Rey Auditory Verbal Learning Test: evaluates episodic declarative memory of a subject. This test provides scores for assessing immediate memory, new verbal learning, susceptibility to interference (proactive and retroactive), retention of information after a period, and memory recognition [105]. Features available for this test are: RAVLT_immediate (evaluates the immediate memory), RAVLT_learning (evaluates the verbal learning), RAVLT_perc_forgetting (evaluates the percentage of forgetting after a certain period of time).

These features are relevant because they represent in general the capacity of a patient to execute tasks such as: recalling recent events; recognizing loved ones; naming common objects; executing house chores such as cooking and cleaning; develop mathematical calculations. The feature distribution for the tabular data is shown in Tables 5.1 and 5.2.

Feature	Mean	Std	Min	Max
AGE	72.45	6.96	55.0	91.4
YEARS_EDUCATION	16.26	2.63	6.0	20.0
CDRSB	1.54	2.09	0.0	17.0
ADAS11	9.81	6.58	0.0	52.0
ADAS13	15.13	9.56	0.0	67.0
ADASQ4	4.45	2.93	0.0	10.0
MMSE	27.50	2.89	6.0	30.0
RAVLT_immediate	38.00	13.12	2.0	75.0
RAVLT_learning	4.64	2.75	-5.0	14.0
RAVLT_forgetting	4.31	2.76	-28.0	15.0
RAVLT_perc_forgetting	54.21	37.20	-400.0	100.0
TRABSCOR	108.44	69.52	0.0	510.0
FAQ	3.72	6.31	0.0	30.0
MOCA	23.56	4.18	0.0	30.0

Table 5.1: CogTests Numerical Features Distribution

Feature	yes	no
MALE	2655	2336
WIDOWED	531	4460
RACE_WHITE	4601	390
RACE_BLACK	199	4792
RACE_ASIAN	86	4905
HISPANIC	192	4799

Table 5.2: CogTests Categorical Features Distribution

We keep the validation and test sets exactly the same across all separate and ensemble experiments to compare them directly. That way, it is possible to compare tunning phases with the validation set and the final results with the test sets. On the other hand, we loosen this restriction for the training set, in order to gather more available data for each data source.

Therefore, we can take advantage of patients who executed only some parts of the exams. For example, while some of the patients in the ADNI repository only went through magnetic resonance procedures, other patients only took demographics or might have done all the exams. As may be expected, only patients who went through all the exams are considered in the **Ensemble Learning Experiments**.

Moreover, since ADNI is a study that tracks patients throughout the years and each data sample represents a patient's visit to the research center, we noticed that it is possible to have more than one data sample per patient. In order to avoid data leakage, we carefully split the experiments datasets - train, validation, test - by patients instead of by data sample. Thus, when training, validating and testing models, we are using completely different patients, assuring that models performance are not artificially inflated by induced patient based features.

Table 5.3, 5.4 and 5.5 show the dataset distribution for the MRI data, Cognitive Tests data and Ensemble data, respectively. Reminding that the ensemble dataset is composed by the full intersection of patient data samples from MRI and Cognitive Tests, leading to 870 data samples in training, 428 in validation, and 429 in test sets.

Class	**Train**	**Validation**	**Test**
AD	626	73	65
CN	1793	273	283
MCI	663	82	81
AD + CN	1722	346	348
MCI + CN	1733	355	364

Table 5.3: MRIs Class Distribution

Class	**Train**	**Validation**	**Test**
AD	636	73	65
CN	1286	273	283
MCI	2212	82	81
AD + CN	1922	346	348
MCI + CN	3498	355	364

Table 5.4: Cognitive Tests Class Distribution

Class	**Train**	**Validation**	**Test**
AD	155	73	65
CN	554	273	283
MCI	161	82	81
AD + CN	709	346	348
MCI + CN	715	355	364

Table 5.5: Ensemble Class Distribution

5.3
Separate Learning Experiments

5.3.1
MRI Deep Learning Training

As previously stated, although we had access to the 3D MRIs, we opted for using only 2D views for a matter of interpretability and also computational resource management. Since the current literature is unclear about specific regions or coordinates of the 3D MRIs that can be used for deep learning concerning the AD diagnosis problem, there is also a need of finding which 2D views can yield good results.

For that reason, the MRI Deep Learning training is divided into two phases: first, find a 2D view that can bring the highest results; and second, tune a model using the chosen 2D MRI views. For the sake of simplicity, from now on we will refer to any 2D view as a 'slice' of the 3D MRI, having a particular orientation, such as coronal, axial or sagittal, as shown on the example in Figure 5.1.

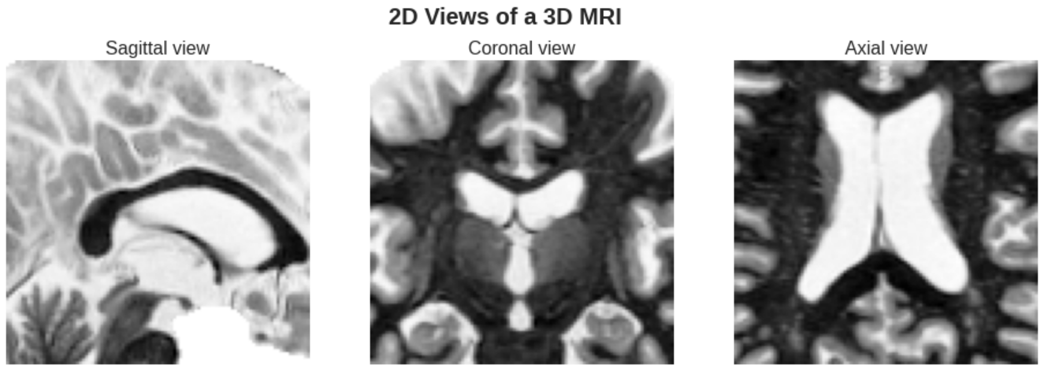

Figure 5.1: 2D views of a 3D MRI

5.3.1.1
2D MRI Slice Search

Since the 3D MRIs have a dimension of 100x100x100, by slicing each 3D image in every possible manner, we have 100 slices per orientation, totalizing 300 potential slices of dimension 100x100x1. To execute the 2D slice search, we will train and validate one deep learning model for each one of the potentatial slices. We opted to use simpler deep learning models for this search, such as the VGG11 and VGG13. As previously discussed, the VGGs still prove to be powerful for image classification tasks, and the VGG11 and VGG13 are their simplest architectures. Also, all the VGGs are being used with batch normalization layers.

For this experiment stage, a simpler DL model spared time and computational resources since it is more important to measure the overall performance for each 2D MRI slice rather than tunning a DL model to reach its full potential. This step can be seen as a feature selection process executed in more traditional machine learning experiments. **The slices are chosen based on the highest Area Under the Receiver Operating Characteristic Curve (AUC) of the validation set due to the class imbalance already discussed earlier.**

For the **ADxCN slice search**, we took as a starting point the results shown in [25], where Nigri executed a similar MRI preprocessing step and also ran a 2D slice search across all orientations, which he refers to as MRI planes. Therefore, instead of swiping across all slices, we only ran experiments around the areas that provided the best results in Nigri's work. Following his convention, each orientation has a slice index that ranges from 0 to 99. The initial search results are presented in Figure 5.2. We executed approximately 60 experiments.

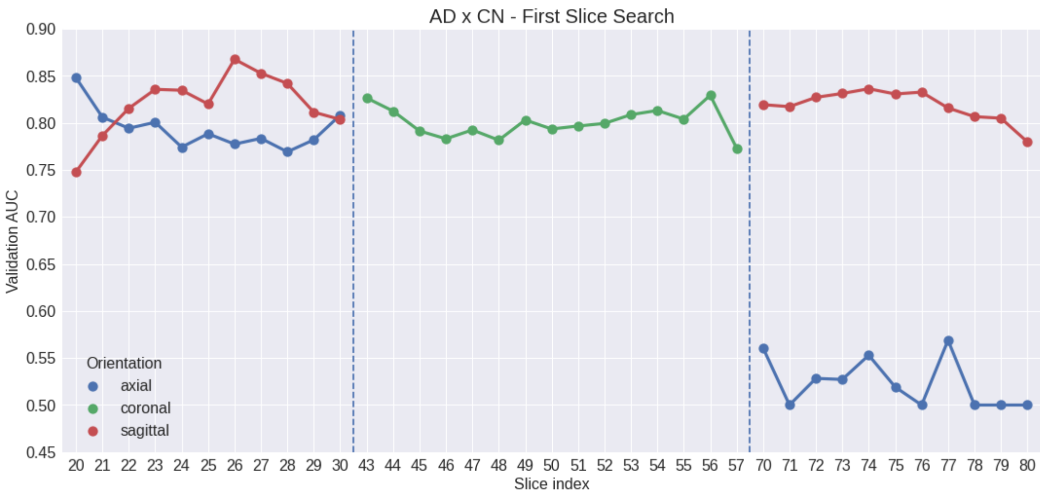

Figure 5.2: First Slice Search for ADxCN case

We notice that the area around 70 to 80 index for the Axial orientation shows really poor performance, in accordance with Nigri's results. We state that since we did not have direct access to the preprocessed data used in his work, as well as more clarification about his coordinate choices, we had to execute an Axial search around 20 to 30 and 70 to 80 to double check the real top performing area.

For the sake of reproducibility of this work, our voxel (3D pixels) coordinate system is as follows:

– **Sagittal orientation**: from 0 to 99 index is equivalent to the left to the right side of the head.

- **Coronal orientation**: from 0 to 99 index is equivalent to posterior (rear part of the head) to anterior (front of the head).
- **Axial orientation**: from 0 to 99 index is equivalent to inferior (below the chin) to superior (top of the head).

Afterwards, we filtered the top results for each orientation and performed an additional search with a slightly more powerful model, the VGG13, yielding the results shown in Figure 5.3. For the sagittal orientation, we decided to pick the top points around the 26 and 74 slices, since these areas provide really good results. For the coronal orientation, we picked top points around the 43 and 56 slices. Lastly, for the axial orientation, we picked top performing spots near the 21 and 29 and ignored the slices above 70. The intuition here was to explore areas near peaks shown in Figure 5.2, this is why we see a 'break' on the indices of the slices in Figure 5.3.

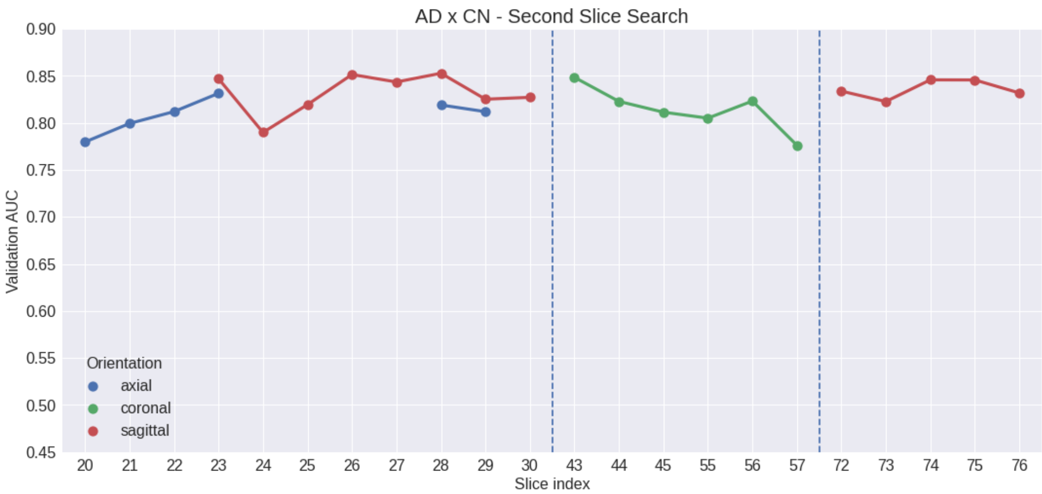

Figure 5.3: Second Slice Search for ADxCN case

Finally, the slices chosen for the ADxCN experiment were the following:

- Coronal: 43
- Axial: 23
- Sagittal: 26

The Coronal 43 and Axial 23 were chosen based on the VGG13 results. Since the Sagittal 26 and 28 had a practical tie, we chose the one that also performed better in the first search.

For the **MCIxCN slice search** we chose to search across every slice for every orientation, since we had no prior reference of similar work. This process, although extensive, seems reasonable due to the fact that early stage of AD or other forms of dementia might show differently through the patients when compared to more advanced cases of AD. That suggests that this difference might also manifest in different parts of the brain being affected. For this search, we only used the VGG11 model and even indexed slices. Results are shown in Figure 5.4. Here we executed 150 experiments.

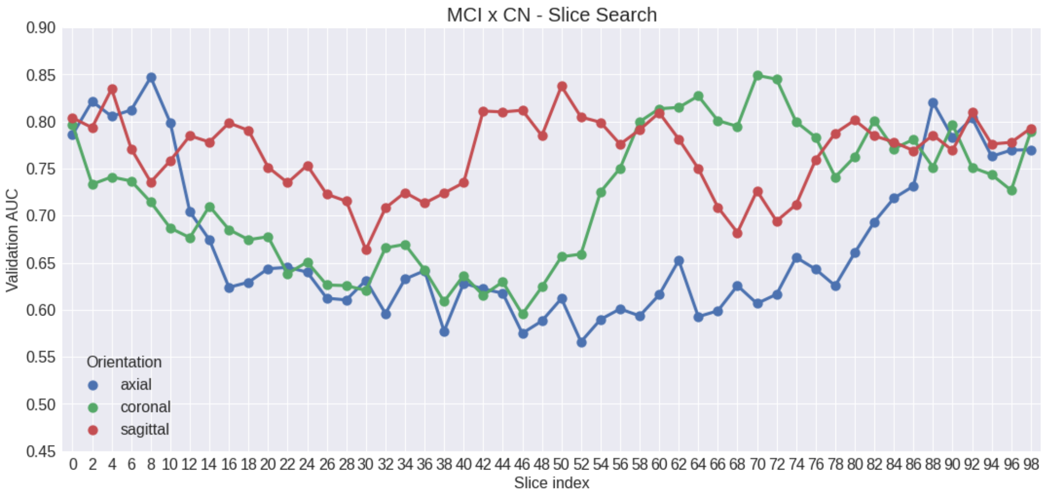

Figure 5.4: Slice Search for MCIxCN case

Finally, the slices chosen for the MCIxCN experiment were the following:

- Coronal: 70
- Axial: 8
- Sagittal: 50

We see very distinct performances across brain areas when comparing it to the ADxCN experiments. This implies that even for healthy patients (CN), the model considers different brain regions as relevant or more discriminative when the classification tasks change from ADxCN to MCIxCN.

From now on, the experiments using the chosen slices will be referred as: AXIAL_23, CORONAL_43, SAGITTAL_26, AXIAL_8, CORONAL_70 and SAGITTAL_50.

5.3.1.2
Final Models Evaluation

After choosing the top-performing slices, we ran more experiments to search for the best combination of some hyperparameters in order to obtain higher performance. We started testing the VGG16, ResNet34 and VGG19 models. However, after a few runs, the VGG19 showed superior performance for both training and validation on all slices. Hence, we opted to keep only the VGG19 for the rest of our experiments. We executed a grid search across the hyperparameters shown in Table 5.6 for both ADxCN and MCIxCN experiments. The early stopping metric was the validation AUC.

Hyperparameter	Experimented Values
learning_rate	0.001, 0.0001, 0.00005, 0.00001, 0.000005, 0.000001
batch_size	8,16,32,64
optimizer	Adam,SGD
loss	Binary Cross Entropy, Focal Loss
max_epochs	100,150,200
early_stopping_patience	10,20,50

Table 5.6: Experimented Hyperparameters

We tested two losses because using the BCE (Binary Cross Entropy) on the MCIxCN led the model to overfit, most likely due to memorizing the simplest features of CN patients. When using the Focal Loss, the model focuses better on differentiating hard cases by giving less attention to higher prediction probabilities. Easy cases where the model is really sure of the prediction contribute less to the loss than hard cases where the model is not so sure. By definition, MCI patients are "closer" to CN patients when compared to AD ones. Also, we noticed that most MCIxCN experiments required lower learning rates when compared to the ADxCN experiments and when using the same learning rates, the training for the MCIxCN became extremely unstable.

For the **ADxCN experiment**, the parameters that showed best results in validation set were: learning_rate=0.0001, batch_size=16, optimizer=Adam, loss=Binary Cross Entropy, max_epochs=100, early_stopping=10. For the **MCIxCN experiment**, the parameters that showed best results in validation set were: learning_rate=0.000001, batch_size=16, optimizer=Adam, loss=Focal Loss, max_epochs=150, early_stopping=20.

After having all models tuned and validated, we present the ROC Curve for all three ADxCN models on the test set Figure 5.5 for comparison: AXIAL_23, CORONAL_43 and SAGITTAL_26. The AXIAL_23 model seems slightly superior to the others. However, when running a DeLong AUC comparison test [106], we could not reject the null hypothesis for the p-value for all comparisons is higher than 0.39.

Table 5.7 summarizes some important metrics. The decision threshold was selected based on the closest point to the highest sensitivity and lowest false positive rate on the ROC curve. That is, the point closest to the vertex (0,1) of the graph. **It is important to note that the threshold was selected using the validation set and not the test set;** otherwise, this would be considered cheating, since by definition we do not have access to any information about the test set. Interestingly, all selected thresholds are closer to 0, suggesting that even weak activations, or lower values of important features, might indicate a positive outcome. We notice that all models suffer from a low Precision, that is being caused by some false positive cases.

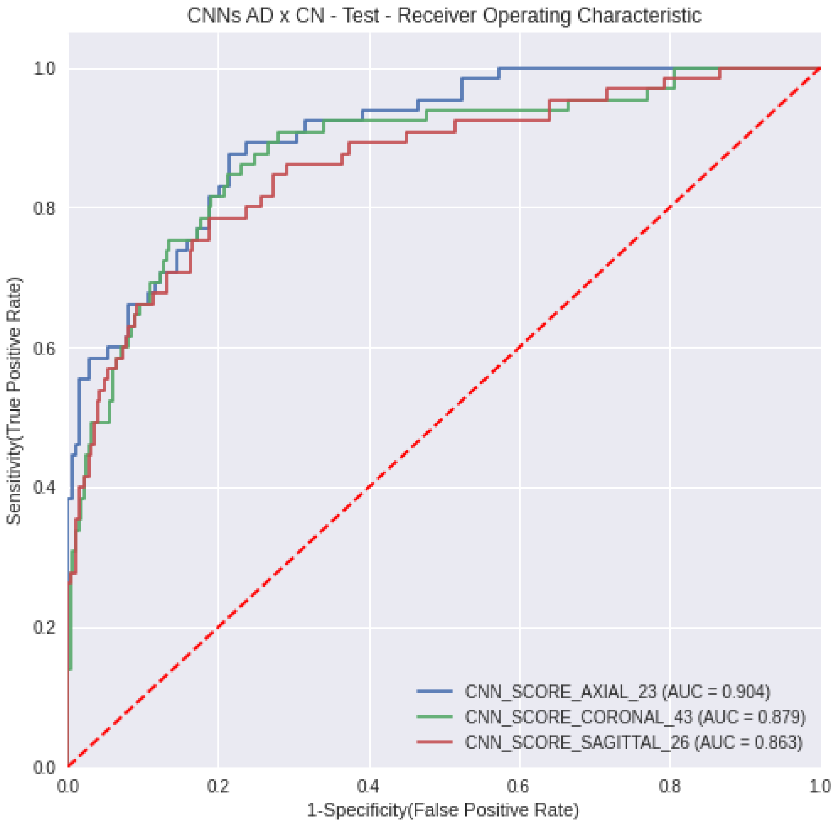

Figure 5.5: CNNs ROC Curves for ADxCN case

Model	AUC	F1	Acc	Prec	Recall	Pred_Threshold
AXIAL_23	0.904	0.595	0.805	0.485	0.769	0.004
CORONAL_43	0.879	0.609	0.819	0.510	0.754	0.164
SAGITTAL_26	0.863	0.580	0.787	0.459	0.785	0.002

Table 5.7: CNNs Performance on ADxCN Test Set

Figure 5.6 shows the result ROC curve for all three models on the MCIxCN test set. Here, the same pattern of the ADxCN follows: Really high values of AUCs and poor performance on the Precision of all three slices, reflecting on a lower F1, as shown in Table 5.8.

In general, we see a higher number of false-positive cases where CN patients ended up being identified as MCI or AD. This shows that although achieving a really good performance, the MRIs have some limitations. Also, it is possible to notice that the classification performance of the CNxMCI is superior to the CNxAD case. This might have happened due to the randomness of selecting patients from the original database. However, on the Ensemble sections, it will be shown that it is possible to achieve superior classification performance for the ADxCN.

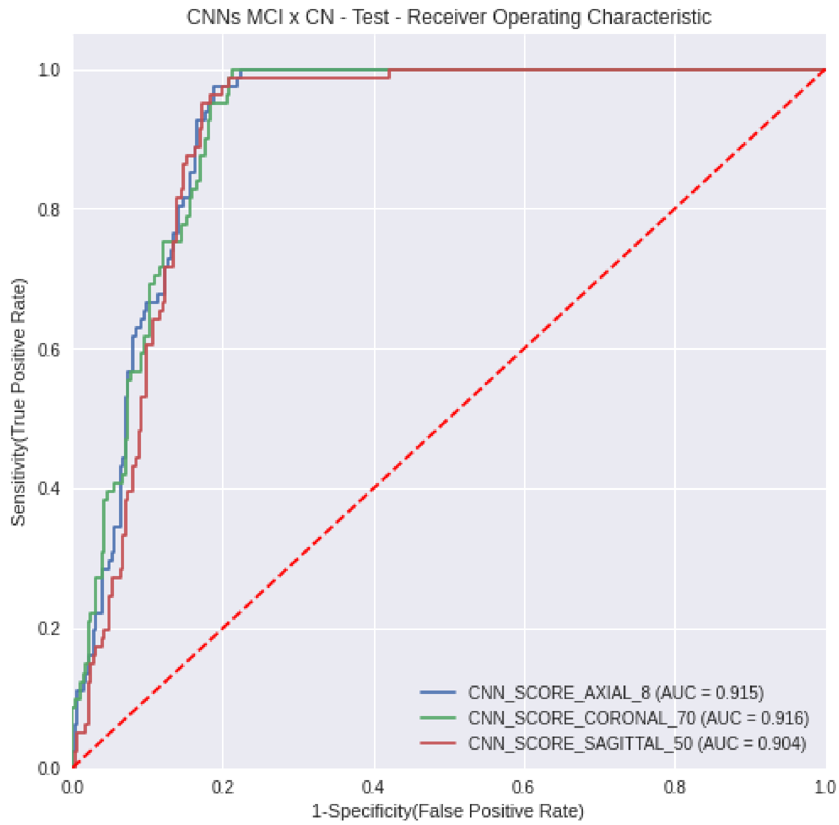

Figure 5.6: CNNs ROC Curves for MCIxCN case

Model	AUC	F1	Acc	Prec	Recall	Pred_Threshold
AXIAL_8	0.915	0.708	0.843	0.605	0.852	0.322
CORONAL_70	0.916	0.688	0.835	0.595	0.815	0.466
SAGITTAL_50	0.904	0.732	0.843	0.591	0.963	0.265

Table 5.8: CNNs Performance on MCIxCN Test Set

5.3.2
Cognitive Tests Training

For these experiments we used two models: Explainable Boosting Machine (EBM) and Logistic Regression (LR). As discussed before, the EBM model offers great overall performance for classification tasks and is also intrinsically interpretable [79]. The LR model is a well known linear model and besides being simple, still shows great performance on classification tasks on healthcare, such as shown in [78].

We chose not to tune any hyperparameter of these models since we achieved from the beginning great results for training and validation sets. Results for the test set can be visualized in Figure 5.7 and Table 5.9 for the ADxCN experiment and Figure 5.8 and Table 5.10 for the MCIxCN experiment. We see excellent results for all cases and this is no surprise, since most of the cognitive tests used as input for the models are used by experts to determine early and advanced stages of the AD/MCI. Also, we still notice a decrease on the Precision on the MCIxCN case, similar to the CNNs behavior.

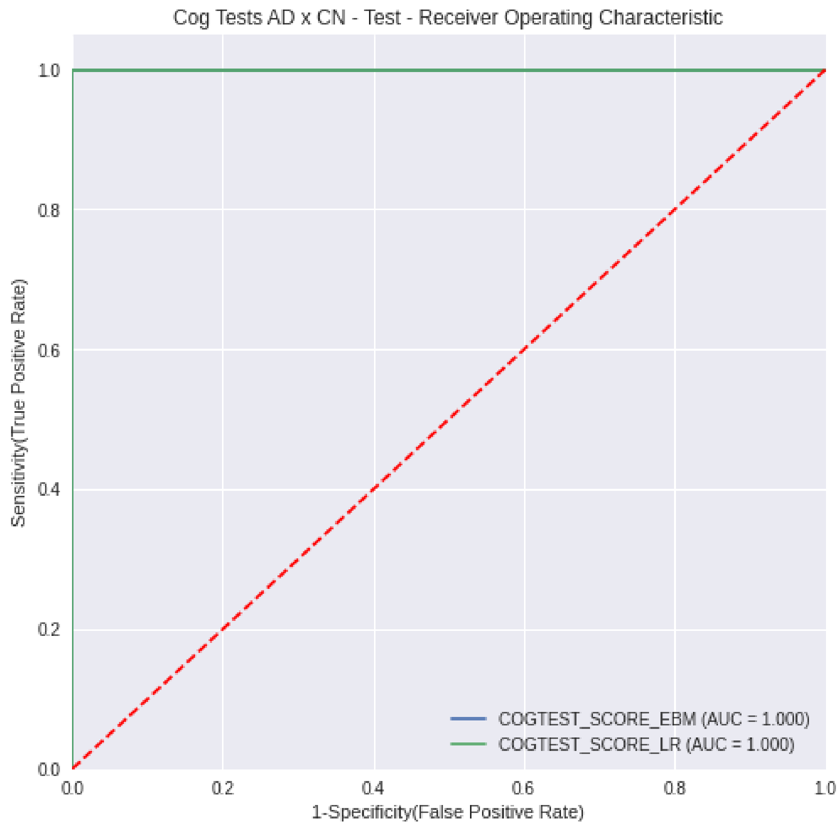

Figure 5.7: Cog Tests models ROC Curves for ADxCN case

Model	AUC	F1	Acc	Prec	Recall	Pred_Threshold
EBM	1.0	0.992	0.997	1.0	0.985	0.809
LR	1.0	0.984	0.994	1.0	0.969	0.620

Table 5.9: Cognitive Tests Models Performance on ADxCN Test Set

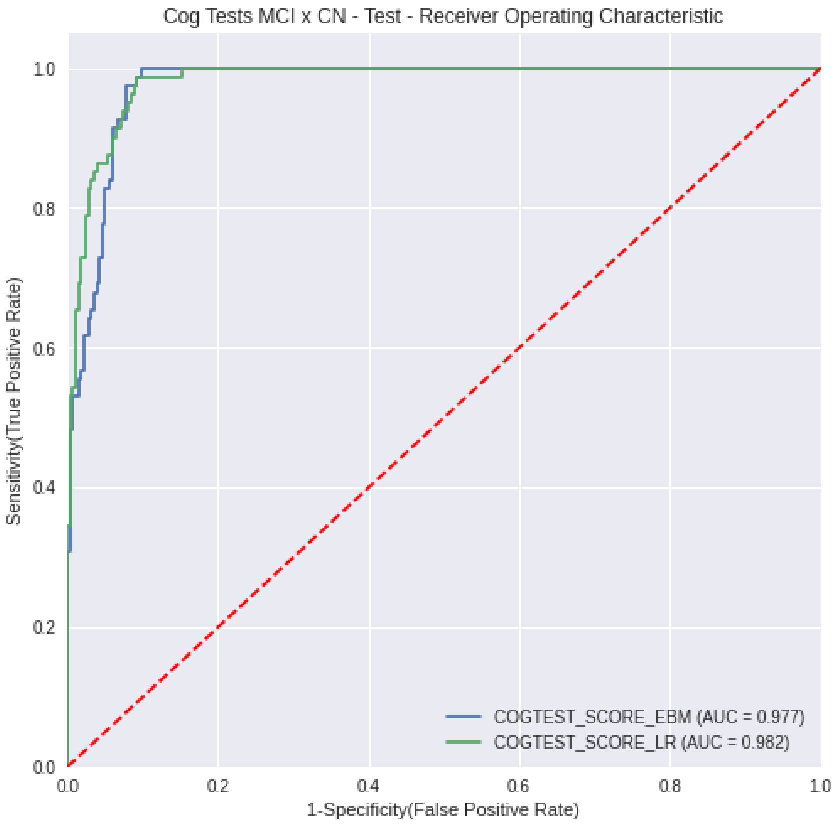

Figure 5.8: Cog Tests models ROC Curves for MCIxCN case

Model	AUC	F1	Acc	Prec	Recall	Pred_Threshold
EBM	0.977	0.857	0.931	0.798	0.926	0.785
LR	0.982	0.854	0.929	0.784	0.938	0.677

Table 5.10: Cog Tests Models Performance on MCIxCN Test Set

5.4 Ensemble Learning Experiments

The ensemble experiments will be presented according to the architectures discussed in the previous chapter. It is important to note that the prediction scores of the CNNs and cognitive tests models were collected during the separate learning experiments phases. Therefore, we can say that each learning module was trained separately.

All ensemble experiments used the EBM and LR models. ROC curves for the test set are presented separately for each experiment type to compare performance between the EBM and LR. Lastly, all experiments are compared against each other by selecting the best model on validation set for each case.

5.4.1 Ensemble of MRI CNNs

The first ensemble experiment consisted on training a model with the prediction scores of the 3 CNNs trained on one slice. The intuition behind this ensemble is to increase the diagnosis performance by leveraging relevant information extracted from three different views of the brain. Figures 5.9 and 5.10 show the performance for the ADxCN and MCIxCN, respectively. We use as baseline comparison the AXIAL_23 model for the ADxCN and the CORONAL_70 model for the MCIxCN, which were the best separate learning model.

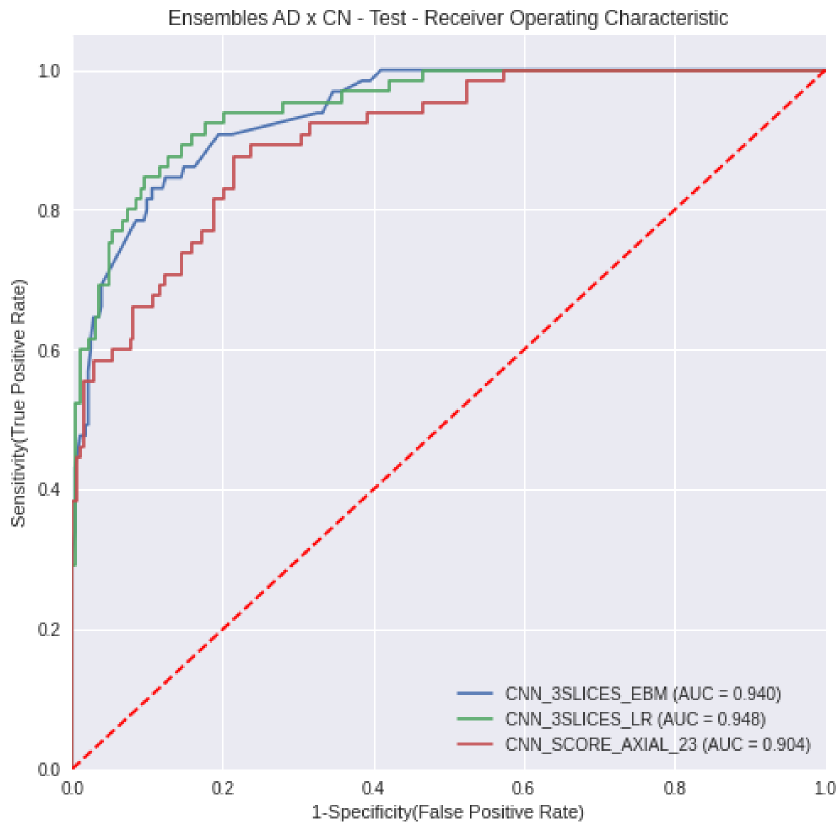

Figure 5.9: Ensemble of MRI CNNs ROC Curves for ADxCN case

For the ADxCN experiment, it is clear that both ensembles (EBM and LR) are superior to the baseline AXIAL_23 model. The DeLong test result presents a p-value less than 5%, for this case, refuting the null hypothesis and showing that these models are statistically different. On the other hand, for the MCIxCN case we see that there is no apparent increase in performance when compared to the CORONAL_70 model. While AD patients indicate more clearly changes in the brain, the same might not hold true to MCI patients. For this reason, an ensemble of different MRI slices does not improve the performance for the MCIxCN classification task.

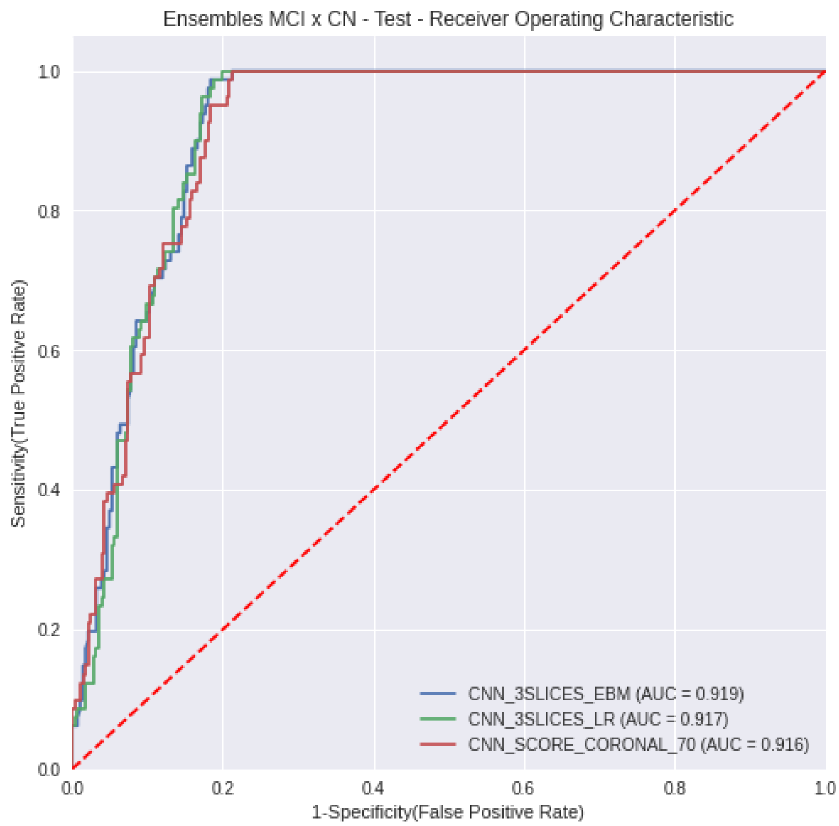

Figure 5.10: Ensemble of MRI CNNs ROC Curves for MCIxCN case

5.4.2
Ensemble of MRI CNNs and Demographics

The second ensemble consisted of the prediction scores of the 3 CNNs trained on one slice, plus demographic features of the patient, such as age, gender, years of education and widow indication. This ensemble follows the same idea of the previous one, with the added benefit of considering demographic information about the patient. Figures 5.11 and 5.12 show the performance for the ADxCN and MCIxCN, respectively. We also use as baseline comparison the AXIAL_23 model for the ADxCN and the CORONAL_70 model for the MCIxCN.

Following the same pattern of the first ensemble, we see an improvement in performance for the ADxCN case and a similar performance for the MCIxCN. Moreover, contrary to what was expected, the presence of demographic features ended up decreasing the performance of this ensemble, when compared to the first one.

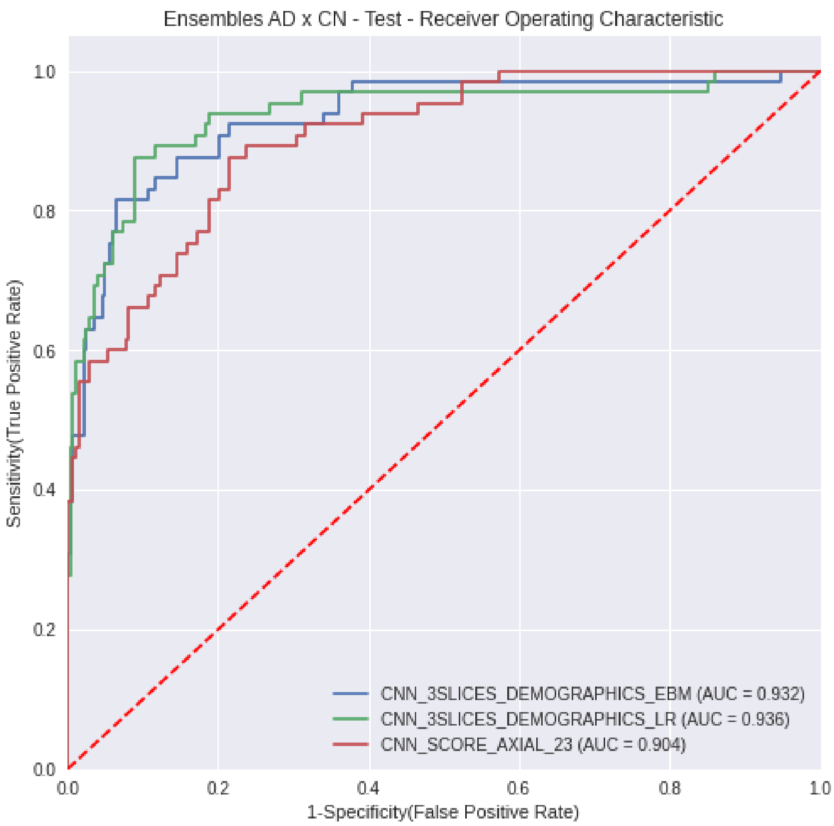

Figure 5.11: Ensemble of MRI CNNs + Demographics ROC Curves for ADxCN case

5.4.3
Ensemble of MRI CNNs, Demographics and CDRSB

Since it is known that the CDRSB test plays an important role to the AD and MCI diagnosis, the third experiment consists of CNNs predictions scores, the same demographics of the previous experiment and the CDRSB score. Figures 5.13 and 5.14 show the performance for the ADxCN and MCIxCN, respectively. Here, we also use as baseline comparison the AXIAL_23 model for the ADxCN and the CORONAL_70 model for the MCIxCN. Additionally, we compare the experiments with the Cognitive Tests model, which servers as an upper bound, since it has achieved perfect performance for the ADxCN and near perfect performance for the MCIxCN case.

For both ADxCN and MCIxCN we see a significant improvement on the performance. That can be explained by the presence of the CDRSB score.

5.4.4
Ensemble of MRI CNNs and Cog Tests Model

As the last ensemble experiment, we chose to combine all separate learning model scores to train a new ensemble, that is, the 3 CNNs models

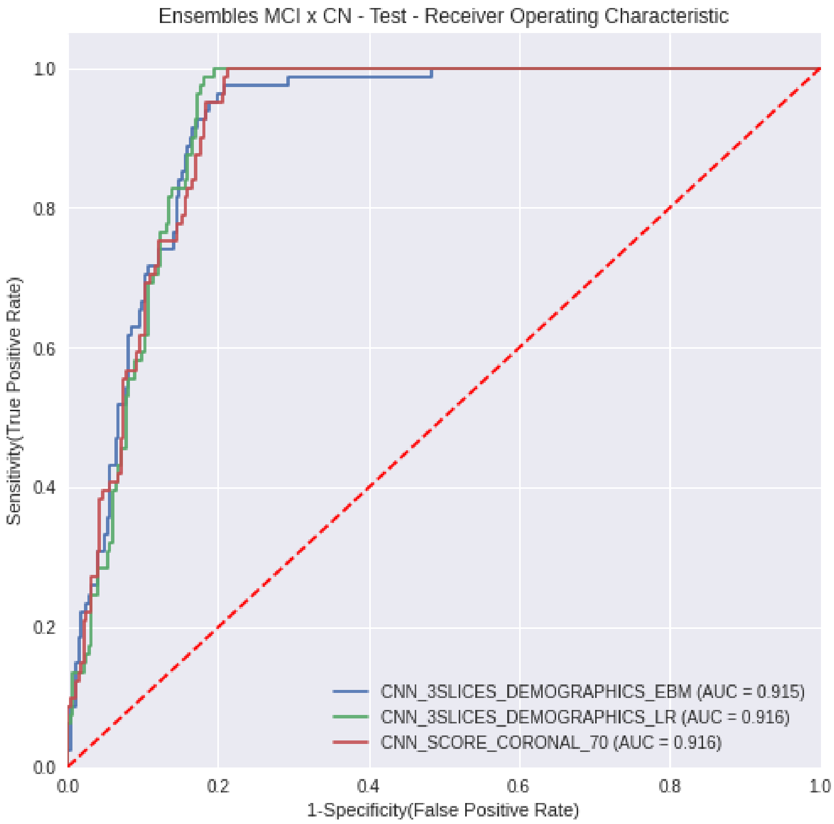

Figure 5.12: Ensemble of MRI CNNs + Demographics ROC Curves for MCIxCN case

plus the cognitive tests model. Figures 5.15 and 5.16 show the performance for the ADxCN and MCIxCN, respectively. We also present a comparison with the best single slice CNNs and the cognitive test model.

As expected, by having the prediction score provided by the cognitive test model the performance is nearly perfect for both cases.

5.5 Comparing All Experiments

So far we have seen that all ensemble experiments show excellent results, being the ones using the CDRSB metric the top performing. On most ensemble experiments, we see very similar performance between the LR and EBM models, except on the third ensemble (CNNs+Demographics+CDRSB) for the MCIxCN, on which the LR outperformed the EBM. Let us do a final comparison, between all experiments, by picking the best model for each category.

For the ADxCN case, Figure 5.17 presents all ROC curves. As stated previously, it is clear that using information about all three Coronal, Axial and Sagittal views of the MRI bring an interesting performance. We see a

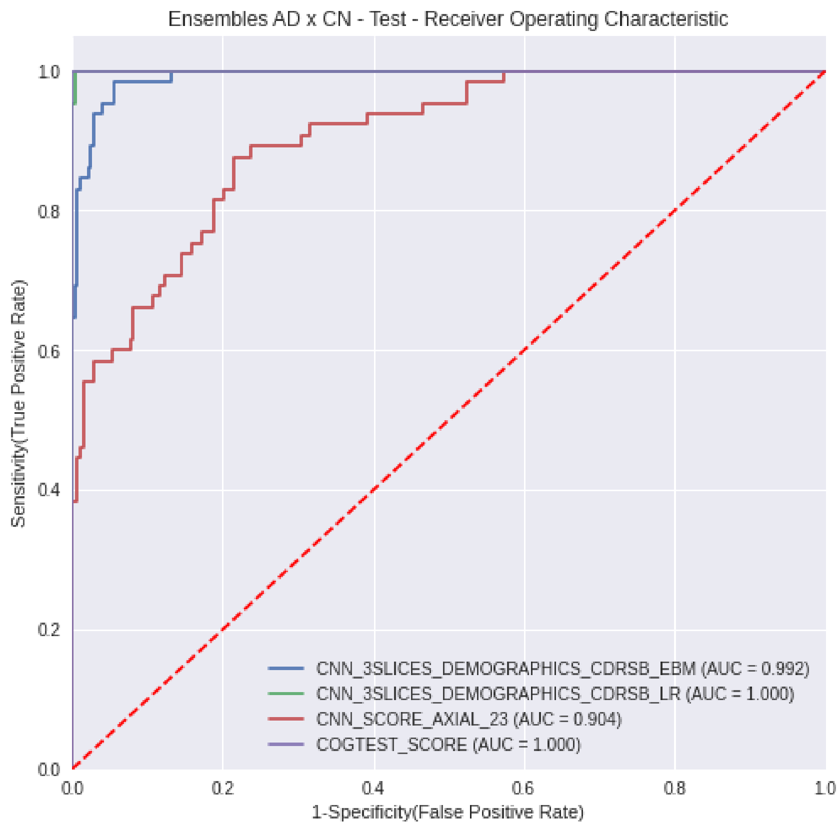

Figure 5.13: Ensemble of MRI CNNs + Demographics + CDRSB ROC Curves for ADxCN case

second jump in performance when we take into consideration the CDRSB as a feature. Also, since the CDRSB has showed to be a feature with extreme predictive power, we chose to compare it against all experiments as a bonus experiment. Therefore, we performed a test using only the CDRSB score to predict the diagnosis of a patient. Not surprisingly, using this feature alone can already achieve great performance on predicting the Alzheimer's Disease. However, specialist hardly rely on predicting a patient outcome base only on one feature alone.

Table 5.11 presents several performance metrics for all models. Due to space limitations we changed the name of the ensembles to the following: ENSEMBLE_1 refers to the first ensemble, using only the 3 CNNs models; ENSEMBLE_2 to the second ensemble, using the 3CNNs plus demographics; ENSEMBLE_3 to the third ensemble, using the 3 CNNs models plus demographics and CDRSB; ENSEMBLE_4 the fourth ensemble, using the 3 CNNs models and the cognitive tests model. Figure 5.18 makes a visual comparison of the AUC. We see that all experiments performed above 0.85 AUC, and all ensemble experiments performed above 0.93 AUC.

For the MCIxCN case, Figure 5.19 shows the ROC curve for all ex-

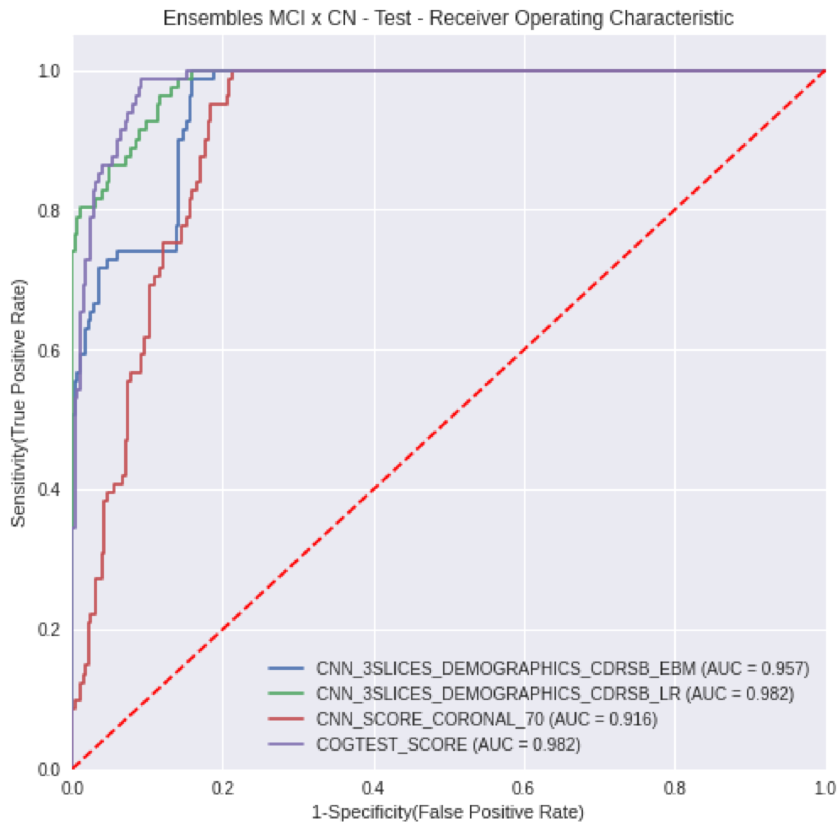

Figure 5.14: Ensemble of MRI CNNs + Demographics + CDRSB ROC Curves for MCIxCN case

Model	AUC	F1	Acc	Prec	Recall	Pred_Threshold
SAGITTAL_26	0.863	0.580	0.787	0.459	0.785	0.002
CORONAL_43	0.879	0.609	0.819	0.510	0.754	0.164
AXIAL_23	0.904	0.595	0.805	0.485	0.769	0.004
COGTEST	1.000	0.984	0.994	1.000	0.969	0.620
ENSEMBLE_1	0.948	0.713	0.871	0.609	0.862	0.077
ENSEMBLE_2	0.936	0.632	0.799	0.480	0.923	0.00005
ENSEMBLE_3	1.000	0.970	0.989	0.942	1.000	0.067
ENSEMBLE_4	0.995	0.926	0.971	0.887	0.969	0.005
CDRSB	0.999	0.962	0.986	0.955	0.969	1.500

Table 5.11: All Experiments Performance on ADxCN Test Set. ENSEMBLE_1 refers to the first ensemble, using only the 3 CNNs models; ENSEMBLE_2 to the second ensemble, using the 3CNNs plus demographics; ENSEMBLE_3 to the third ensemble, using the 3 CNNs models plus demographics and CDRSB; ENSEMBLE_4 the fourth ensemble, using the 3 CNNs models and the cognitive tests model.

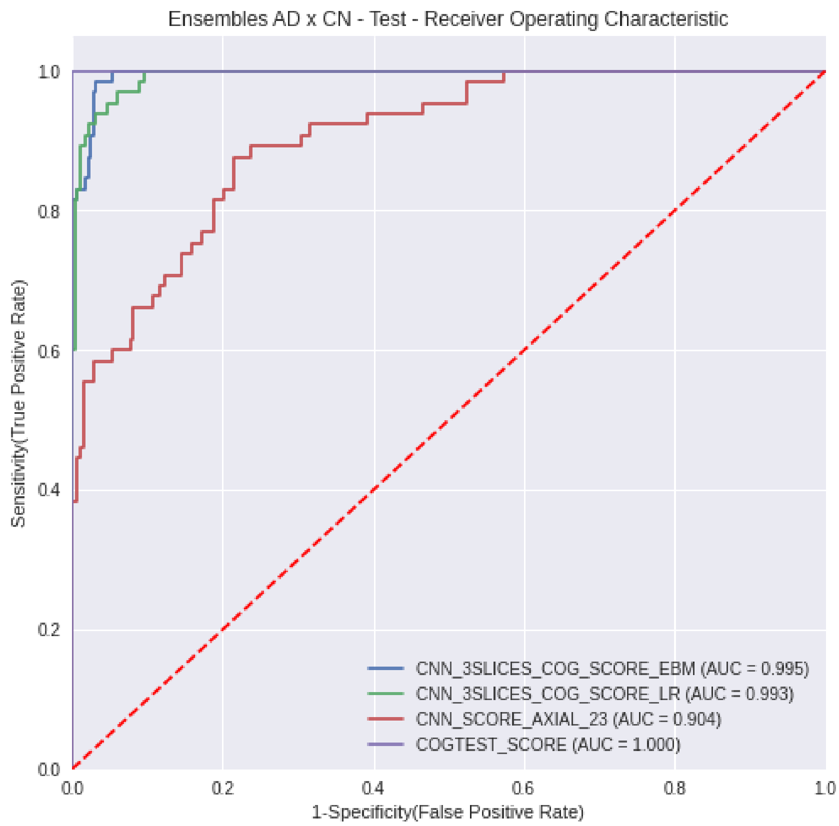

Figure 5.15: Ensemble of MRI CNNs + Cognitive Tests model ROC Curves for ADxCN case

periments. While for the ADxCN case we get an improvement on results by combining the 3 MRI CNNs into an ensemble, we see a similar performance between the single MRI CNNs and the 3 slices CNNs. We argue that since MCI patients are considered early stages or even a stage that precedes the pathology, they might not indicate much apparent structural changes in their brains when compared to patients in more advanced stages of the disease. That also explains why we see an increase in performance when taking into consideration the CDRSB as a feature, since this psychological exam can perceive changes in the behavior of the patient since very early stages of a cognitive impairment. Table 5.12 presents performance metrics for all experiments for the MCIxCN case, while Figure 5.20 presents a visual comparison between AUCs. We see that all models performed above 0.9 AUC, being the top-performing models the ones that used the CDRSB as feature. Also, when compared to the ADxCN case, we see that all models have inferior Precision, leading to a higher number of false positive cases.

In conclusion, it is clear that using the ensemble strategy to diagnose AD patients increases the performance when compared to single MRI models. However, when dealing with the MCI diagnosis, the ensembles showed similar

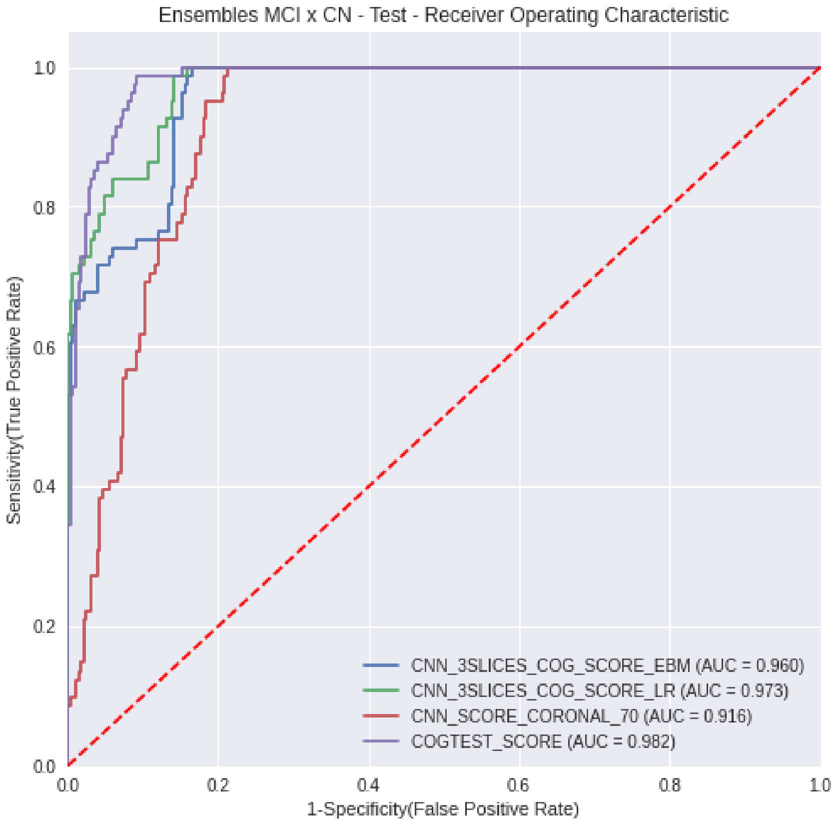

Figure 5.16: Ensemble of MRI CNNs + Cognitive Tests model ROC Curves for MCIxCN case

Model	AUC	F1	Acc	Prec	Recall	Pred_Threshold
SAGITTAL_50	0.904	0.732	0.843	0.591	0.963	0.265
CORONAL_70	0.916	0.688	0.835	0.595	0.815	0.466
AXIAL_8	0.915	0.708	0.843	0.605	0.852	0.322
COGTEST_SCORE	0.982	0.854	0.929	0.784	0.938	0.677
ENSEMBLE_1	0.917	0.705	0.846	0.615	0.827	0.306
ENSEMBLE_2	0.915	0.724	0.852	0.617	0.877	0.00006
ENSEMBLE_3	0.982	0.818	0.909	0.740	0.914	0.457
ENSEMBLE_4	0.960	0.759	0.871	0.649	0.914	0.00003
CDRSB	0.978	0.810	0.896	0.681	1.000	0.500

Table 5.12: All Experiments Performance on MCIxCN Test Set. ENSEMBLE_1 refers to the first ensemble, using only the 3 CNNs models; ENSEMBLE_2 to the second ensemble, using the 3CNNs plus demographics; ENSEMBLE_3 to the third ensemble, using the 3 CNNs models plus demographics and CDRSB; ENSEMBLE_4 the fourth ensemble, using the 3 CNNs models and the cognitive tests model.

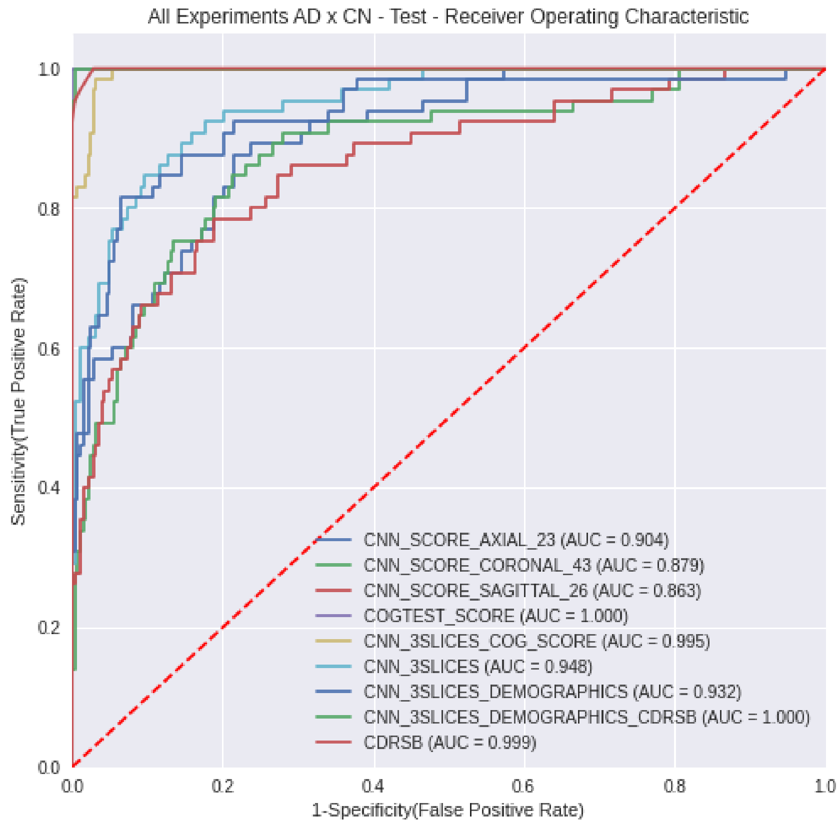

Figure 5.17: All models ROC Curves for ADxCN case

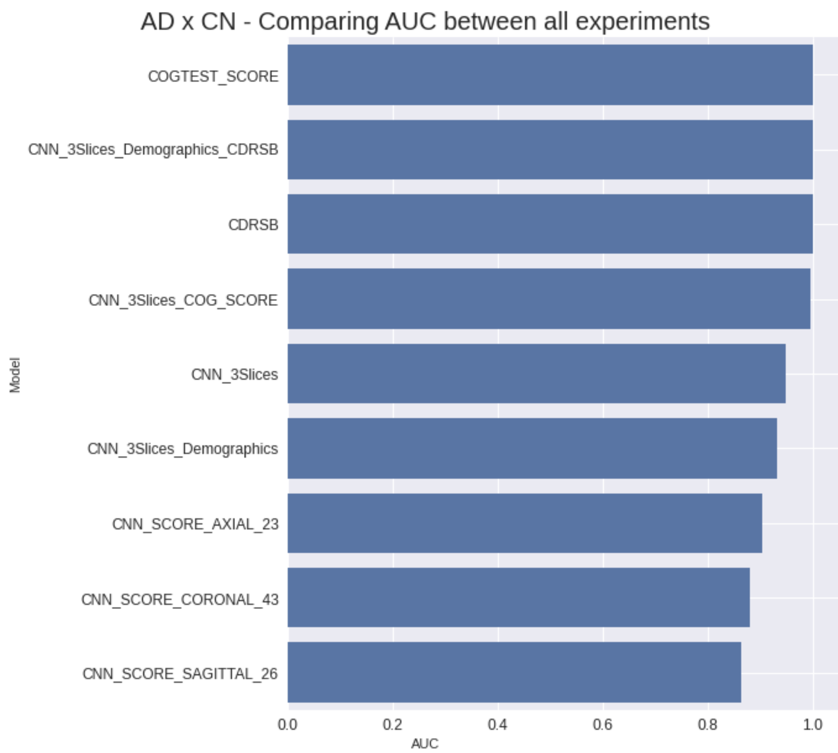

Figure 5.18: All Experiments AUCs for ADxCN case

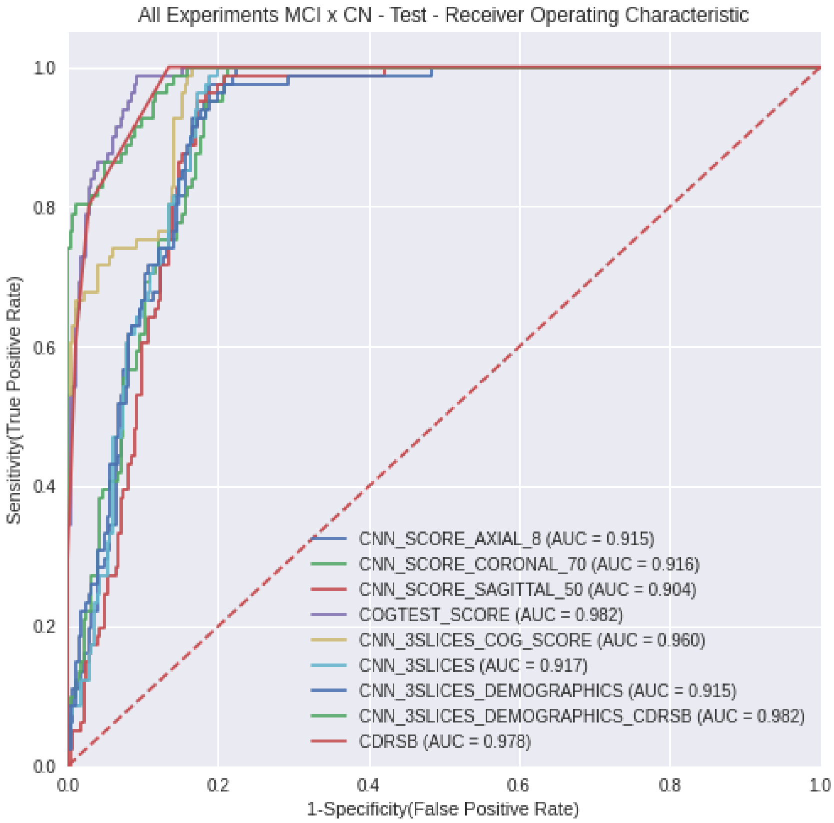

Figure 5.19: All Experiments ROC Curves for MCIxCN case

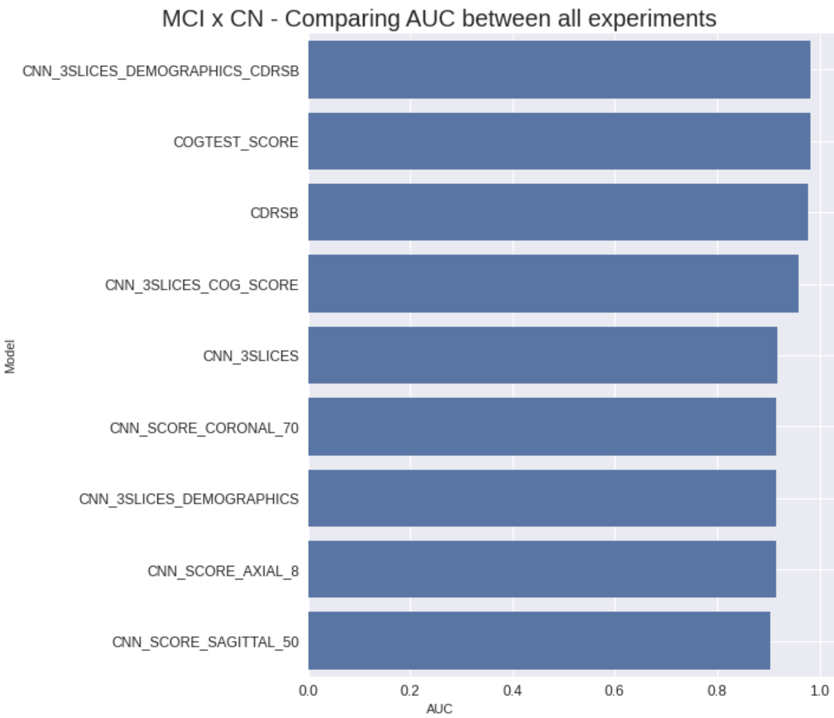

Figure 5.20: All Experiments AUCs for MCIxCN case

performance to the separate learning models. Additionally, the CDRSB feature offered an increase in performance for all cases. The cognitive tests model showed the highest performance for both cases, mostly due to combining several cognitive tests and demographic features of the patients.

That said, we propose using the second ensemble, which uses the three CNN slices models and some demographic features as a method of early passive trial, using the EBM model. The main advantages of this architecture would be: to extract relevant information with only passive cooperation of the patient; using information from different sources to execute the diagnosis in order to make it more robust; account for feature interaction; explaining the diagnosis over the different data sources. In a real-world scenario, our chosen architecture would be able to provide almost instant feedback about patients situation.

The next chapter presents the explanation of global and local predictions to elucidate how it is possible to interpret different outcomes, and also to assess situations where the models might fail.

6
Explanations of Predictions

As previously discussed in this work, global explainability can be seen as the overall behavior between the predictor variables and the interest variable, that are linked by a given machine learning model. In contrast, local explanations are inferences made on how a given model executed one single prediction. Therefore, both explanations complement each other: while global explanations might provide insights about feature importance as a whole and direct future studies and experiments, local explanations provide us insights about specific situations, helping to identify potential outliers or anomalous patterns.

First, we start discussing local explanations for the separate MRI learning experiments in section 6.1, followed by global explanations of the cognitive tests and ensemble models in sections 6.2 and 6.3, respectively. We provide a qualitative evaluation of the explanation methods, discussing some challenges faced during the process of obtaining reasonable explanations.

Afterward, local explanations is discussed for the final chosen ensemble in section 6.4, containing the three MRIs and demographics data as input. We also discuss how explanations might help uncover potentially misleading and biased predictions.

Lastly, we present a patient diagnosis simulation in section 6.4, to illustrate how the proposed architecture would provide a final diagnosis.

6.1
Local Explanations for the MRI CNN Models

For the Local Explanations, we have applied and compared two methods of deep learning explanations: Guided Grad-CAM (GC) and DeepLift (DLi - not to be confused with Deep Learning abbreviation DL). Although both approaches are backpropagation-based, GC tries to explain predictions based on the direct gradient calculation, while DLi propagates layer-wise contributions based on differences from a reference.

Before diving into arguing about true and false diagnoses and their predictions, let us discuss different configurations used to obtain meaningful explanations of the images.

6.1.1 Explanation Methods Configuration

Initially, we started using the DeepLift using as a reference an image containing only zeros, for it was its default parameter configuration. This led to feeble and visually meaningless explanations for all orientations in the ADxCN case. For this reason, we explored additional ways of generating good explanations, such as altering the reference image and also using the SmoothGrad strategy along with the DeepLift.

As explained earlier, the SmoothGrad (SG) algorithm was initially proposed as a standalone explanation method that averages several gradient explanations along with Gaussian noise - a noise that has a normal distribution with mean zero and standard deviation σ. However, this strategy can also be combined with any other explanations method, by applying Gaussian noise to each image, calculating individual explanations, and averaging the results.

Figure 6.1 shows several different configurations used to improve DeepLift visual explanations. The red color represents negative contributions to the diagnosis and the green areas represent positive contributions. In other words, green areas show potential areas of interest concerning the class we want to identify, Alzheimer's Disease. Red areas can be interpreted as regions that disturbed the AD diagnosis.

For comparison, we experimented the Guided Grad-CAM with the SmoothGrad, as shows Figure 6.2. We see that adding more noisy samples (items b and c of the figure) appears to saturate the important pixels and turns the explanation into an edge detector. Nonetheless, we consider the plain explanation of the GC as sufficient since it already provides clear explanations, as seen in item a) of the figure.

It is important to note that although we only show the coronal CNN here, for objectivity, the DeepLift behavior repeated itself over the sagittal and axial CNN models as well. An early conclusion of this behavior might be that, for the ADxCN case, although looking at different slices and orientations, the weights learned on all three CNN models share similar characteristics.

As said before, initial trials for the DeepLift used the reference as an image containing only zeros and obtained meaningless explanations, as shown in item a) of Figure 6.1: red and green pixels are mixed in a way that is not possible to distinguish any areas or textures of the brain that caught the attention of the model. Then, inspired by one of the DeepLift benchmark experiments [85] - using a blurred image as reference for the CIFAR-10 dataset classification - we opted to use as reference a blurred version of the input image itself, showing in item b) of Figure 6.1. In other words, instead of using an

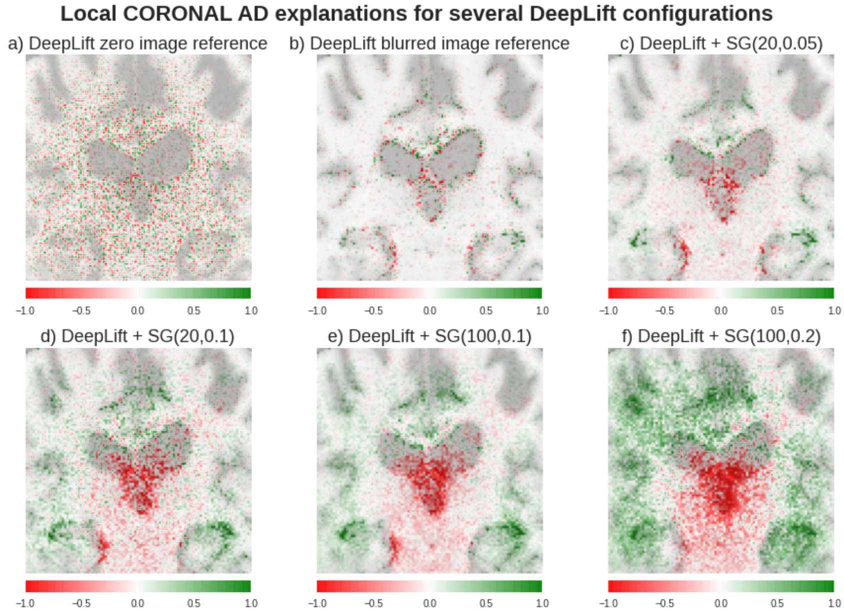

Figure 6.1: Different DeepLift Local Explanations for one AD patient using the CORONAL CNN. a) DeepLift using a zero image as reference; b) DeepLift using the blurred version of the input image as a reference; c) DeepLift with SmoothGrad using 20 samples and noise of 0.05 std, using the blurred image reference; d) DeepLift with SmoothGrad using 20 samples and noise of 0.1 std, using the blurred image reference; e) DeepLift with SmoothGrad using 100 samples and noise of 0.1 std, using the blurred image reference; f) DeepLift with SmoothGrad using 100 samples and noise of 0.2 std, using the blurred image reference;

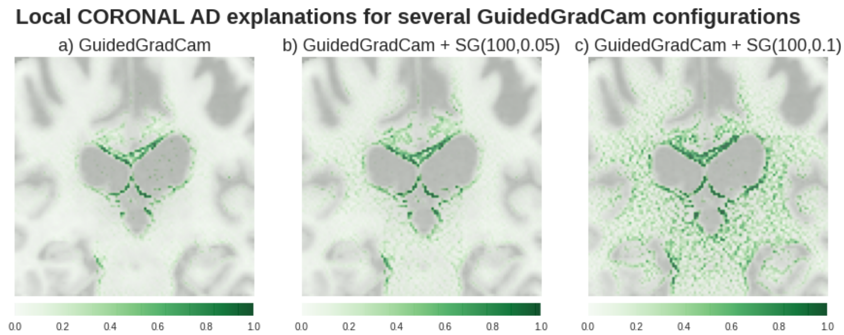

Figure 6.2: Different Guided Grad-CAM Local Explanations for one AD patient using the CORONAL CNN. a) Guided Grad-CAM; b) Guided Grad-CAM with SmoothGrad using 100 samples and noise of 0.05 std; c) Guided Grad-CAM with SmoothGrad using 100 samples and noise of 0.1 std;

image of zeros as reference to explain the image, we used a blurred version of the image to explain the image itself. This reference was obtained by applying a Gaussian Blur kernel of 5x5 with a standard deviation (std) of 1. Although some areas are still mixed, there is a good improvement compared to the first case.

Items c) to f) show the DeepLift with SmoothGrad (DeepLift+SG) with different parameters. The reference image for all these items is the blurred image. The values inside the parenthesis on each item subtitle is the number of averaged noisy samples and the standard deviation of the Gaussian noise added to the image, respectively. For example, item c) presents SG(20,0.05), meaning that the SmoothGrad is using 20 samples and gaussian noise of 0.05 std. All the configurations using SG show some visual improvement when compared to a) and b), such as red and green pixels not mixed anymore in the same region and consequently concentrations of red and green spots on specific areas, as it will be discussed further ahead.

Even the tiny addiction of noise, such as SG(20,0.05), eliminates most mixed contribution areas. When the standard deviation is 0.1, the important pixels start to merge, providing a sense of important regions on the lateral parts of the image. However, when std=0.2, it appears like the important regions are somehow saturated. Also, when comparing d) and e), we notice that adding more samples to the average actually smoothed the explanations. Therefore, we picked the DeepLift+SG(100,0.1) as our final DeepLift configuration, meaning: The DeepLift using the blurred image as reference input enhanced by the SmoothGrad using 100 samples with Gaussian noise of 0.1 std. For consistency, we will use the same configuration of the DeepLift both ADxCN and MCIxCN.

6.1.2
Local Explanations Discussion

For the following discussion, we opted to show 2 patients by class for each case, totalizing 8 cases. That way, it is possible to dive deeper into how each explanation per orientation/slice works and also how they complement each other for a potential diagnosis. The figures that will be shown ahead present three images per row. The first is the original image, the second is the GC explanation over the image, and the third is the DLi explanation. Also, the first row presents explanations for the sagittal orientation, the second one coronal and the third one axial. The number that follows the orientation name on the figures is the slice for each orientation that trained the CNNs. To orient the reader that is not familiarized with brain regions, we marked the most important brain regions on each image, followed by a corresponding number.

Every time a marked region is presented, its corresponding number is cited between parenthesis. The list follows ahead:

1. Hippocampus
2. Lateral Ventricles
3. Third Ventricle
4. Medial Temporal Lobe
5. Frontal Lobe
6. Cerebellum
7. Hypothalamus and Pituitary Gland
8. Sphenoid Sinus
9. Fourth Ventricle

Figures 6.3 and 6.4 present two different CN patients for the ADxCN case. The first conclusion that most likely attracts the attention of the reader is the absence of any Guided Grad-CAM explanation on all orientations of the second patient and on the axial orientation of the first patient. We argue that since the GC is a gradient-based algorithm, most likely, the activation functions of the layers were not high enough and the gradient might have vanished during its calculation. However, we can also think that the lack of any visual explanation might be an explanation itself. The models did not find anything that called their attention, hence the CN diagnosis and no positive pixels. On the other hand, DeepLift explanations are clear and very present over all figures, that already shows that this algorithm is somewhat more sensitive in comparison to the GC.

Moving on to the AD patients, explanations are shown on Figures 6.5 and 6.6. The GC got stronger activation values over all orientations than the CN patients, indicating the hippocampal region (1) on all orientations and the lateral ventricles (2) on the coronal orientation. The DLi also highlights the hippocampus (1) with positive contribution and with negative contribution the third and lateral ventricles (2,3), on coronal and sagittal orientations, respectively.

A key difference can be noticed on the coronal orientation explanations for the DeepLift, when comparing the CN and AD figures: there is a shift downwards of the central red region (negative), over the third ventricle (3), and also a green region appears right above this red region, over the lateral

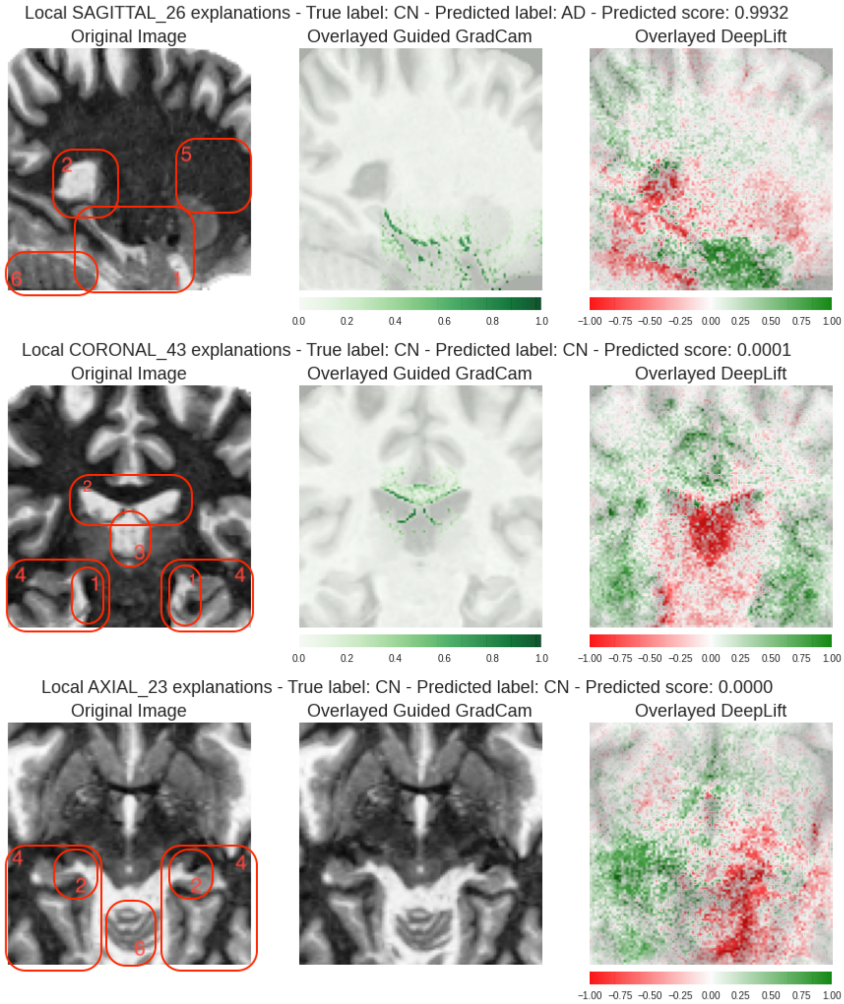

Figure 6.3: MRI Explanations for one CN patient for ADxCN case.

ventricles (2). This suggests that the CORONAL_43 CNN extracts relevant features from the lateral ventricles (2), as well as other areas indicated by the DLi, such as the hippocampus (1) and the temporal lobe (4), which is the region around the hippocampus on this particular view. In reality, early studies suggest a high correlation between the enlargement of ventricles and shrinking of the hippocampus, and the progression of Alzheimer's disease [35, 107, 108].

Curiously, on the axial orientation, there is an asymmetrical assignment of positive importance when comparing the two hemispheres of the brain. We argue that both explanation algorithms might have captured regions of the image that were easier for the CNN to learn in order to spot AD patterns located on the left side of the image, which is also the left side of the brain. While the GC did not assign any importance to the right side, the DLi assigned negative importance to it, suggesting that this area might have been confusing for the AXIAL_23 CNN to extract any relevant pattern.

It is possible to notice that the DeepLift explanations are much more

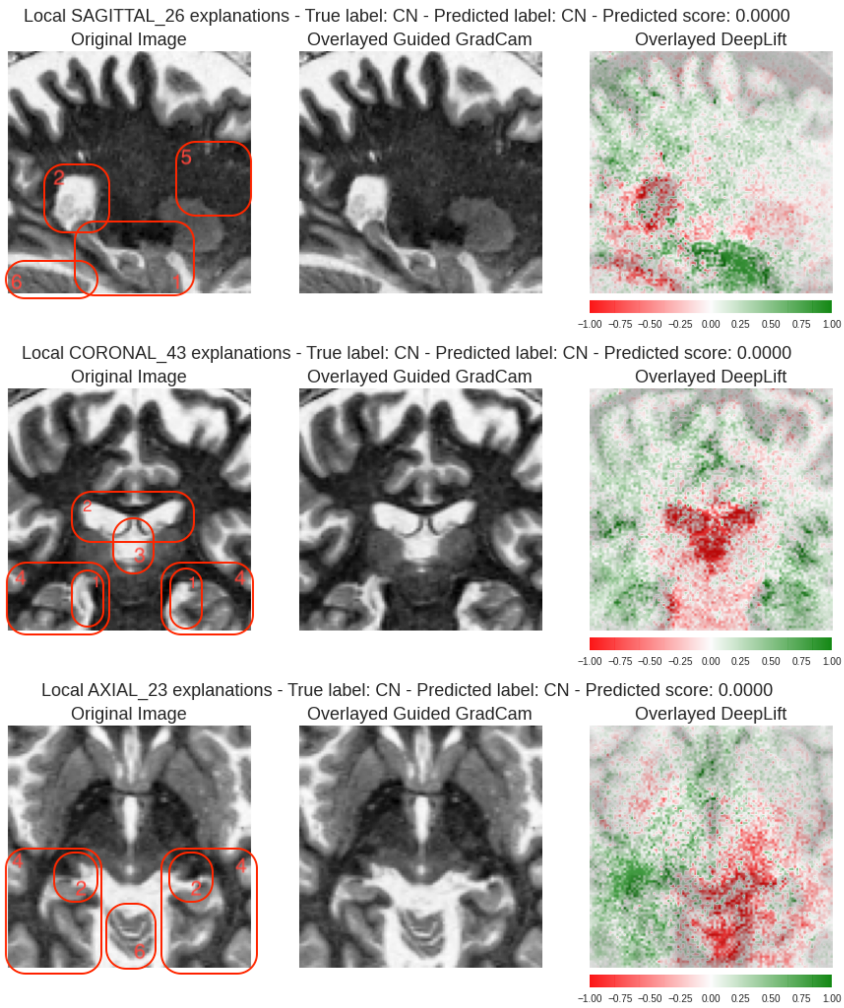

Figure 6.4: MRI Explanations for another CN patient for ADxCN case.

present and visually appealing when compared to the Guided Grad-CAM. We state that it was able to explain differences in textures and maybe even differences in the volume of gray and white matter of the brain when comparing healthy subjects with pathological subjects.

Moreover, the DLi algorithm indicates several regions with negative contributions for the AD diagnosis, such as the third ventricle (3) - on the central area of the coronal orientation - and the superior region of the cerebellum (6) and lateral ventricle (2) - lower-left corner and central areas of sagittal orientation. Since we modeled the problem as a binary classification, we can interpret negative contributions of the DeepLift as areas that contain slight or no difference between CN and AD patients. Perhaps, one could say that these negative highlighted areas might even be confusing and should be avoided when working on a proper diagnosis. Additionally, we can also conclude that non highlighted areas are less critical or indifferent for the diagnosis.

Figures 6.7, 6.8, 6.9 and 6.10 shows patients for the CNxMCI case. Here,

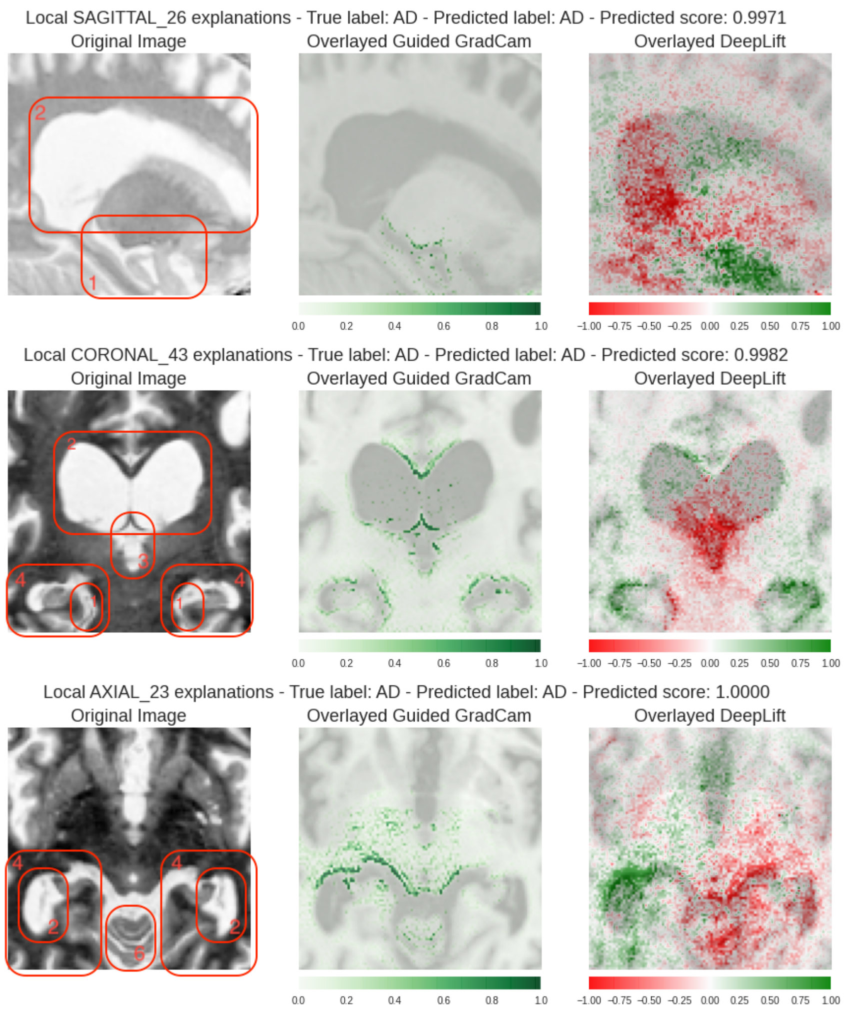

Figure 6.5: MRI Explanations for one AD patient for ADxCN case.

the lack of GC explanations for several images happens as well. In all cases with no explanation, the predicted score was bellow 0.45. This implies that the Guided Grad-CAM algorithm might need a stronger/higher predicted output in order to provide any explanation.

Figure 6.8, an explanation for a cognitive normal patient, shows the DeepLift with solid positive contributions on sagittal and coronal orientations. On the other hand, following the pattern mentioned earlier, GC provides no explanations at all. By looking at the images of this particular patient, it is possible to note some differences when compared to other patients from the MCIxCN case. First, on coronal orientation, a part of the skull appears on the upper part of the image. Second, on the sagittal orientation, we see a region on the lower right corner that does not belong to the brain, most likely a sinus region. Lastly, the contrast of the images of this figure looks different from the rest. Recalling that the DeepLift algorithm highly depends on a reference image to generate explanations, this might explain the apparent misleading

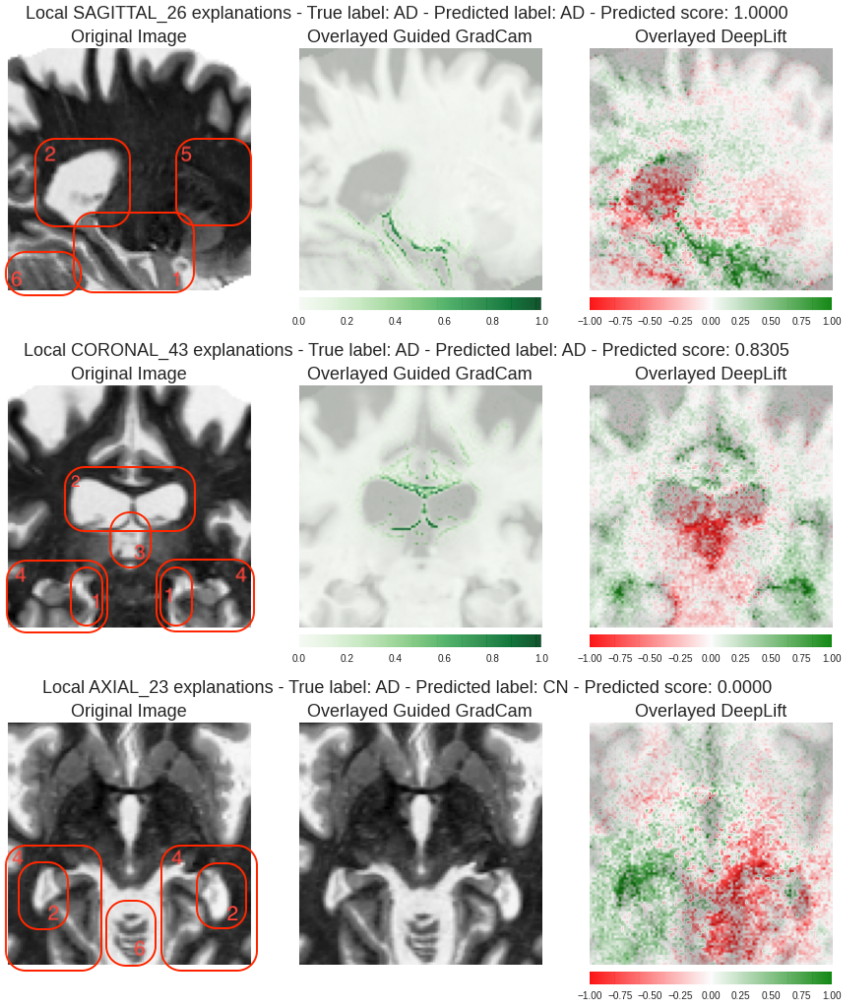

Figure 6.6: MRI Explanations for another AD patient for ADxCN case.

explanations for the DeepLift. Most likely, the preprocessing steps, such as brain registration and skull stripping, did not perform well for this particular MRI.

Both algorithms highlight the thalamus, hypothalamus, and pituitary gland regions (7) of the sagittal orientation over all CN and MCI patients, with no clear distinction. Disruptions in the hypothalamic-pituitary (7) region are associated with cognitive impairment and Alzheimer's Disease Pathology, with symptoms varying from disturbances during sleep to sex drive and drastic humor variations [109].

On the axial orientation, the region between the temporal lobes (4) and a cavity - missing part showing on Figures 6.7 and 6.9 - that seems to be the sphenoid sinus (8), gets highlighted. Most likely, this cavity was removed during the skull stripping process.

On the coronal orientation, GC still finds the lateral ventricles (2) as an important region, while DLi focus on the lower regions of the brain. Most of

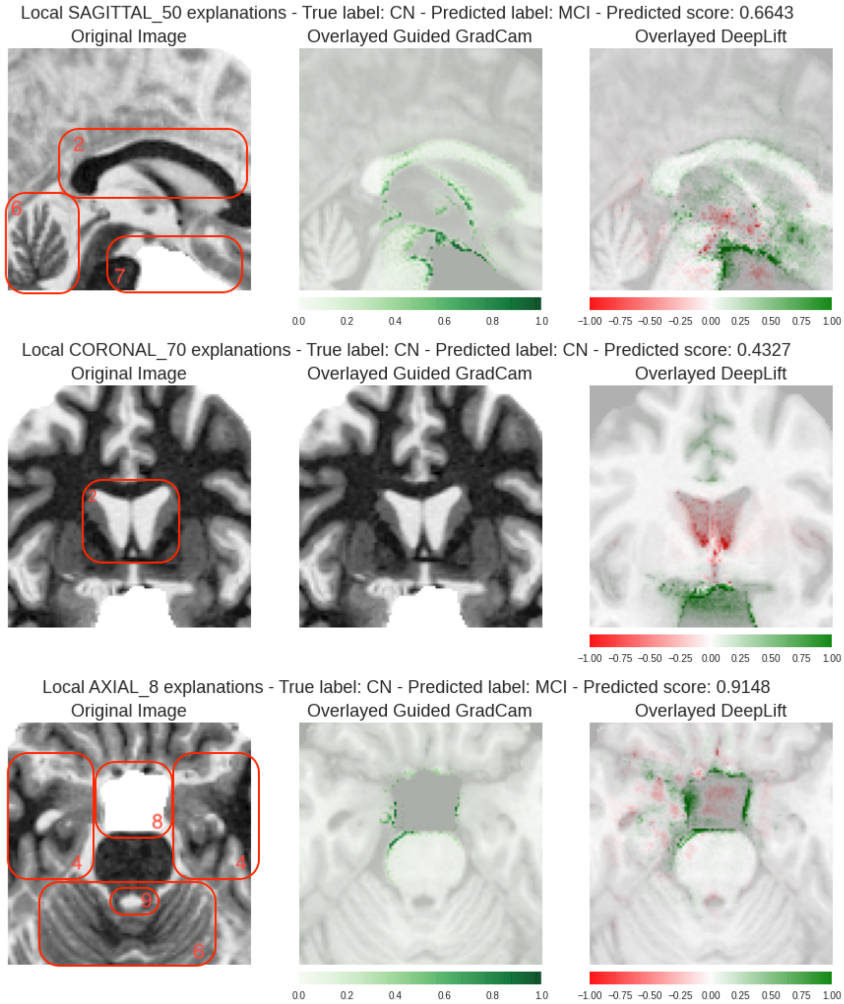

Figure 6.7: MRI Explanations for one CN patient for MCIxCN case.

the positive regions highlighted by the DeepLift on all orientations seem to be part of the inferior central region of the brain, near the temporal lobe (4).

By observing all figures presented so far, it is clear that the Guided Grad-CAM highlights more specific and sensitive areas of interest, especially edges of regions, for both cases. DeepLift also behaves similar to GC for the MCIxCN case, mainly focusing on edge regions. On the other hand, as discussed before, DeepLift appears to offer broader explanations for the AD case, highlighting diffused regions.

This also suggests that the trained models might perceive a wide variety of morphological changes of the brain when comparing AD and CN patients. These changes might be texture and volume of white and gray matter over the cerebral cortex (outermost layer of the brain, moving towards the border of the images), altering pixel intensity on the outer region of the images, as well as the atrophy of the hippocampus (1) and enlargement of ventricles (2,3), which affects the shape of well known inner structures of the brain. This is

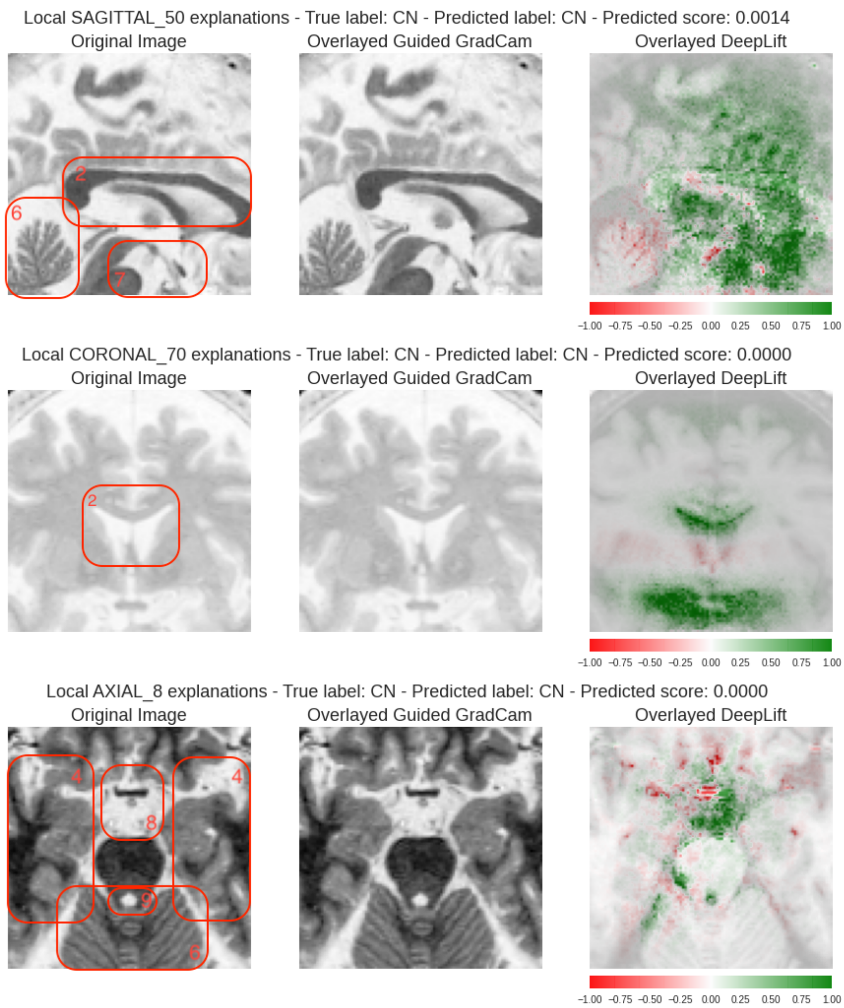

Figure 6.8: MRI Explanations for another CN patient for MCIxCN case.

in accordance with what [110] presented by measuring the direct volume of specific areas of the brain to compare CN, MCI, and AD patients. They also found significant changes in areas such as Amygdala, lateral ventricles, and variations on gray matter.

The difference in the behavior of the DeepLift could be explained by the fact that AD patients in general present clear changes in the morphology of the brain, due to their more advanced stage of the disease. Since MCI patients are considered on early or even pre-stages of the pathology, structural changes in the brain might manifest more subtly or not manifest at all, hence giving fewer 'clues' to the CNNs on where to look for patterns.

We conclude that both algorithms provided good explanations. While the Guided Grad-CAM showed itself more straightforward to use and also to analyze due to the presence of only positive contributions, the DeepLift showed more sensitive and broader explanations, suggesting that it can perceive changes that the Guided Grad-CAM cannot. We also want to remind the

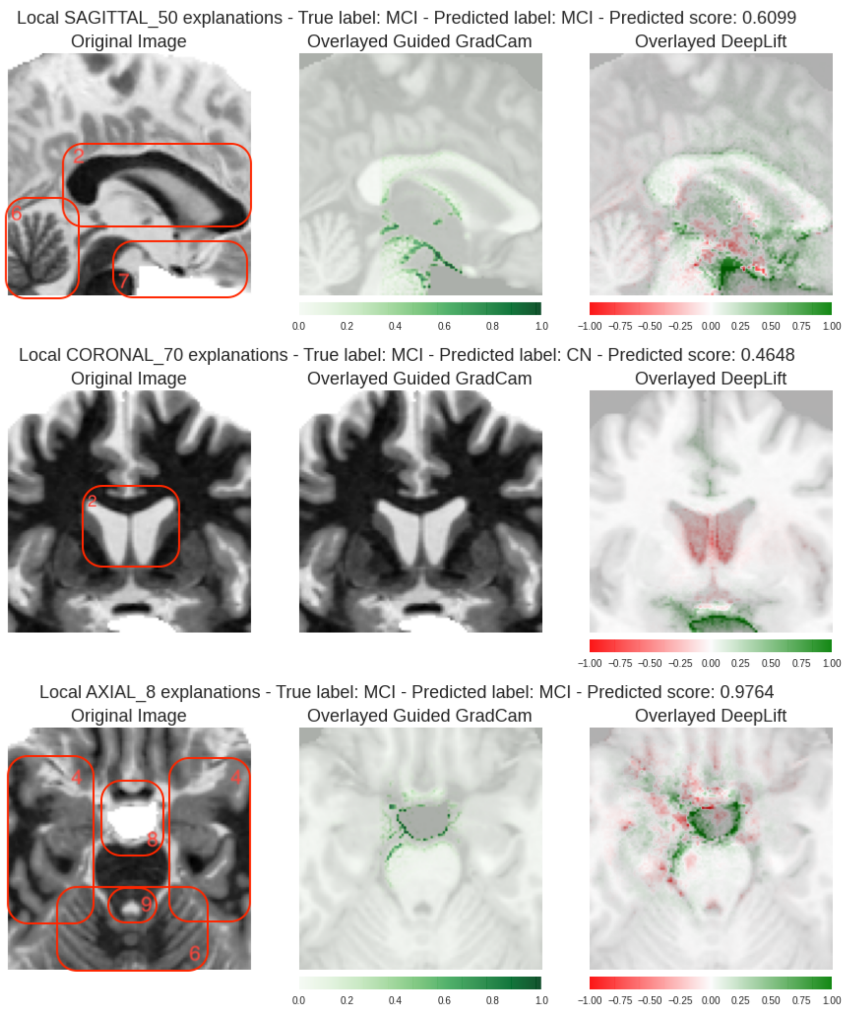

Figure 6.9: MRI Explanations for one MCI patient for MCIxCN case.

reader that, as stated by the DeepLift authors, the algorithm requires the choice of a reference input value, and it should be made using specific domain knowledge. This can greatly impact how the explanations are generated.

6.2 Global Explanations for the Cognitive Tests Model

Still following the Separate Learning Experiment, the global explanation for the tabular models is presented in the form of feature importance, ranked from the most important feature to the least important feature. Figure 6.11 presents global explanations for the cognitive test models. For simplicity, we only show the top 10 most important features.

For all cases, the CDRSB (Clinical Dementia Rating Sum of Box) is indicated as the most important feature. For the ADxCN case, one can see that the EBM also found the FAQ (Functional Activities Questionnaire) the second most relevant feature to the diagnosis, followed by several other cognitive tests

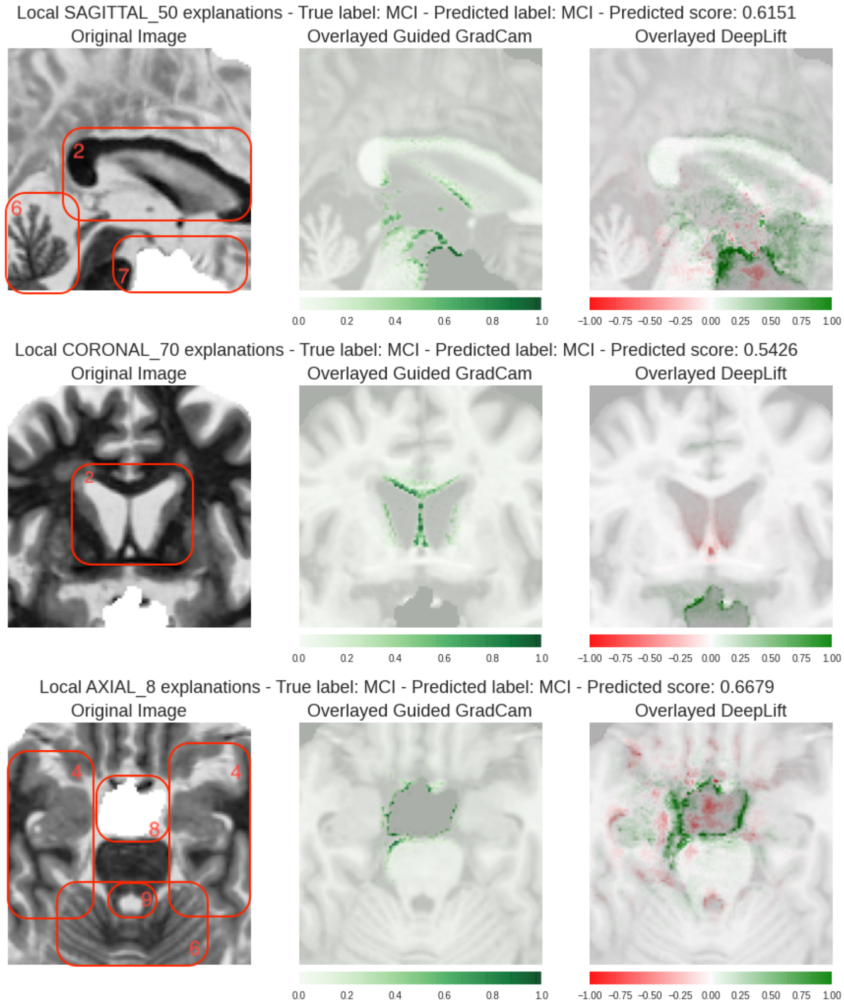

Figure 6.10: MRI Explanations for another MCI patient for MCIxCN case.

with less importance. Neither demographics nor feature interactions pairs are found in the top 10 features of the EBM. On the other hand, the linear model considers some demographic features as relevant, such as the race Asian, being widowed, race White and gender Male. Even though both models performed flawlessly on the test set, it is clear that they took different paths to find patterns in data. This difference might be explained by the non-linear and boosting nature of the EBM model compared to a more simplistic approach of the Logistic Regression.

For the MCIxCN case, we see the FAQ feature with less importance when compared to the AD diagnosis, for both models. Here, it seems like the CDRSB had more predictive power than in the ADxCN. Curiously, when looking at the LR model, the race Asian feature switches from a negative contribution to a positive contribution when comparing the AD and MCI diagnosis. The same switching contribution phenomenon is observed for other demographic features. Moreover, EBM considered age an important feature, suggesting that

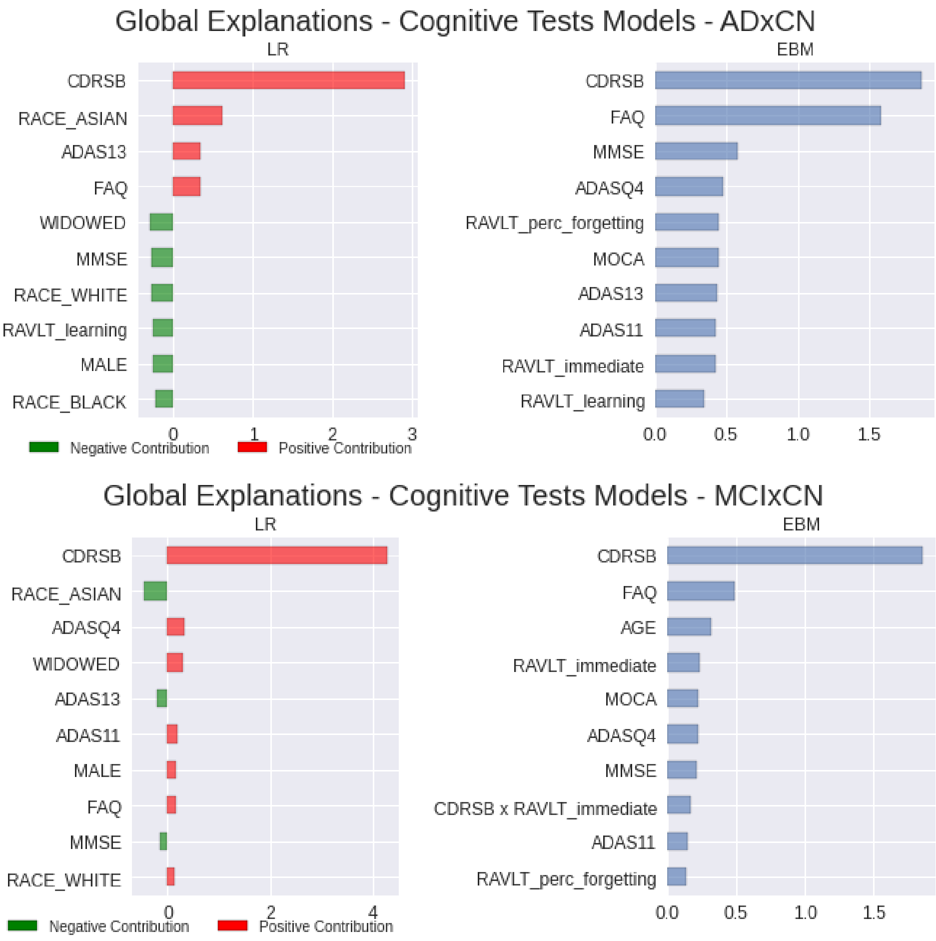

Figure 6.11: Global Explanations for the Cognitive Tests Model. Upper: ADxCN; Bottom:MCIxCN Left: Logistic Regression; Right: Explainable Boosting Machine

for the mild case diagnosis (MCI), the age might influence more when compared to the AD diagnosis. This is reasonable with what we know about Alzheimer's Disease and other dementia since most of these situations start showing with the aging of a person.

6.3 Global Explanations for the Ensemble Models

Figure 6.12 shows the feature importance for the first ensemble experiment, containing only the prediction score of the 3 CNNs (Coronal, Axial, and Sagittal). For the ADxCN, we see equal importance among all three orientations. For the MCIxCN case, there is a rank on each slice's importance, being the Axial orientation the most important one, followed by the Sagittal and then the Coronal. This behavior is consistent in both models. Another thing to notice on the EBM for the MCIxCN is that the interactions between the Axial and other slices have almost the same importance as the coronal

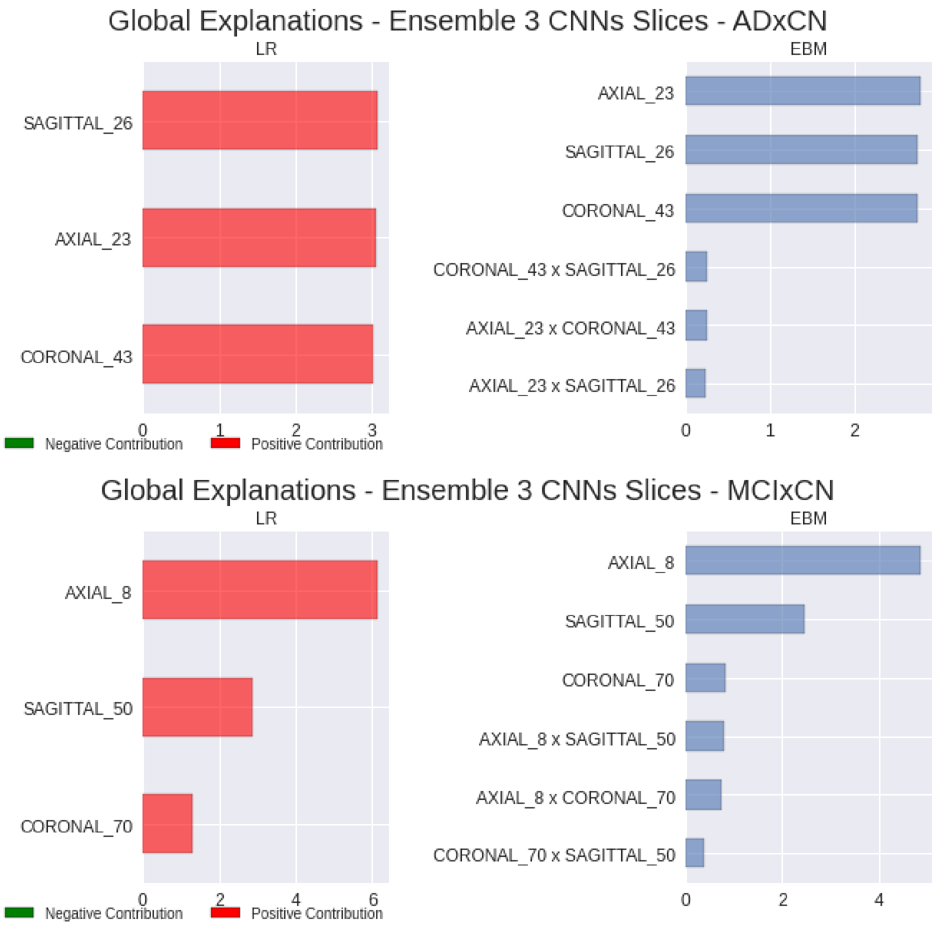

Figure 6.12: Global Explanations for the 3 CNNs Ensemble. Upper: ADxCN; Bottom:MCIxCN Left: Logistic Regression; Right: Explainable Boosting Machine

slice alone. Also, we recall that the CORONAL_70 and the AXIAL_8 have comparable performance for the separate learning experiments, with AUCs of 0.916 and 0.915, respectively.

Motivated by this odd behavior, we checked the linear correlation between the variables. We found out that there is a really high correlation between the CNN predictors for the MCIxCN, as shown in Table 6.1. Regularization applied during training for both models ended up ranking the Axial orientation as the most important feature, for it already offered sufficient information for the prediction. When we look at Table 6.2 for the ADxCN case, we also see a high correlation between the slices, but they are less correlated when compared to the MCIxCN.

As a matter of fact, this raises an important observation: looking at the brain over more angles (views, orientations) to identify mild cases against normal ones might not bring any relevant new evidence. However, when trying to identify more advanced AD cases, it is helpful to look at more

Features	Linear Correlation
CORONAL_70 x SAGITTAL_50	0.90
CORONAL_70 x AXIAL_8	0.79
AXIAL_8 x SAGITTAL_50	0.88

Table 6.1: Linear Correlation between CNN Models for MCIxCN

Features	Linear Correlation
CORONAL_43 x SAGITTAL_26	0.74
CORONAL_43 x AXIAL_23	0.78
AXIAL_23 x SAGITTAL_26	0.79

Table 6.2: Linear Correlation between CNN Model for ADxCN

orientations simultaneously than just one. This can be confirmed by analyzing the performance improvement between the 3 Slices CNNs Ensemble and single CNNs, as well as the equal feature importance distribution given to the ensemble features.

Looking at Figures 6.13, 6.14 and 6.15 we see the same pattern of similar importance to each CNN slice for the ADxCN and ranking importance for the MCIxCN case.

For the ensemble with the CNNs and demographics, shown in Figure 6.13, it is clear that all CNNs predictions have greater importance and that the EBM considered some interactions between demographics and AXIAL_8. We still see the demographics features playing a secondary role, just like in the Cognitive Tests Model.

When we consider the addition of the CDSRB, we see in Figure 6.14 that it ranks as one of the most important features for all situations. The ADxCN case distinctly ranks as the most important one for the Logistic Regression model and as the third rank for the EBM, having the same importance value as the second and first most important features. For the MCIxCN case, the AXIAL_8 still remains as the top 1 feature, even though we know that adding the CDRSB brought a good improvement in performance, as already discussed in the previous chapter.

Lastly, we observe that the global explanation for the ensemble using the CNN predictions and the Cognitive Tests Model prediction is similar to the ensembles of CNNs and CDRSB. The Cognitive Test prediction ranks equally among the CNNs predictions on the ADxCN case and it stays behind the SAGITTAL_50 on the MCIxCN.

After assessing the overall behavior of all experimented ensembles, we follow to local explanations in order to take a closer look at how models behave

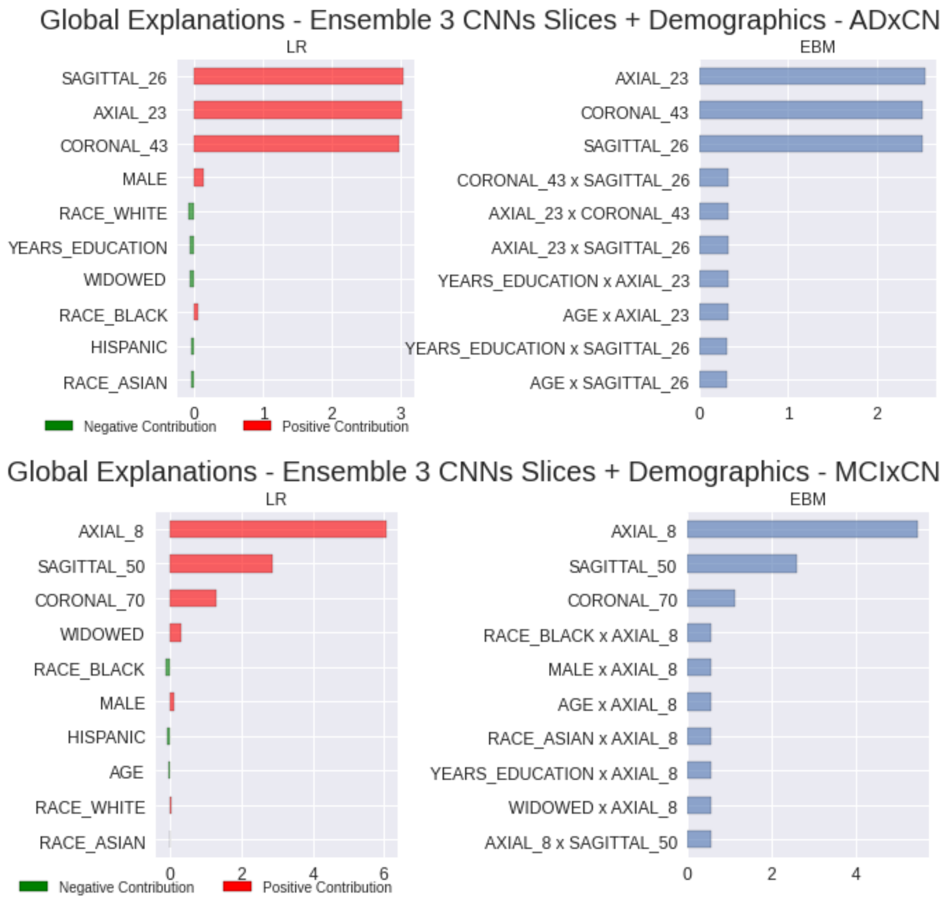

Figure 6.13: Global Explanations for the 3 CNNs + Demographics Ensemble. Upper: ADxCN; Bottom:MCIxCN Left: Logistic Regression; Right: Explainable Boosting Machine

under specific circumstances.

6.4 Local Explanations for the Final Ensemble Model

In this section, for objectivity, we only present local explanations of the final ensemble model, that is, the ensemble that considers the 3 CNNs prediction scores and demographic data about the patient. We discuss true and false predictions and how explanations can help address issues such as biased predictions. For each class on each case, we present two local explanations side by side, showing the top 10 most important features.

Figure 6.16 presents explanations for two true positives for the ADxCN case. Although it was earlier presented that the three CNNs prediction scores have the same global importance on Figure 6.13, there is a clear difference over MRI slices prediction importance at a local scale. Interestingly, on both patients the slices that got the least importance were the ones with really low scores, in comparison to the other slices with scores above 0.8.

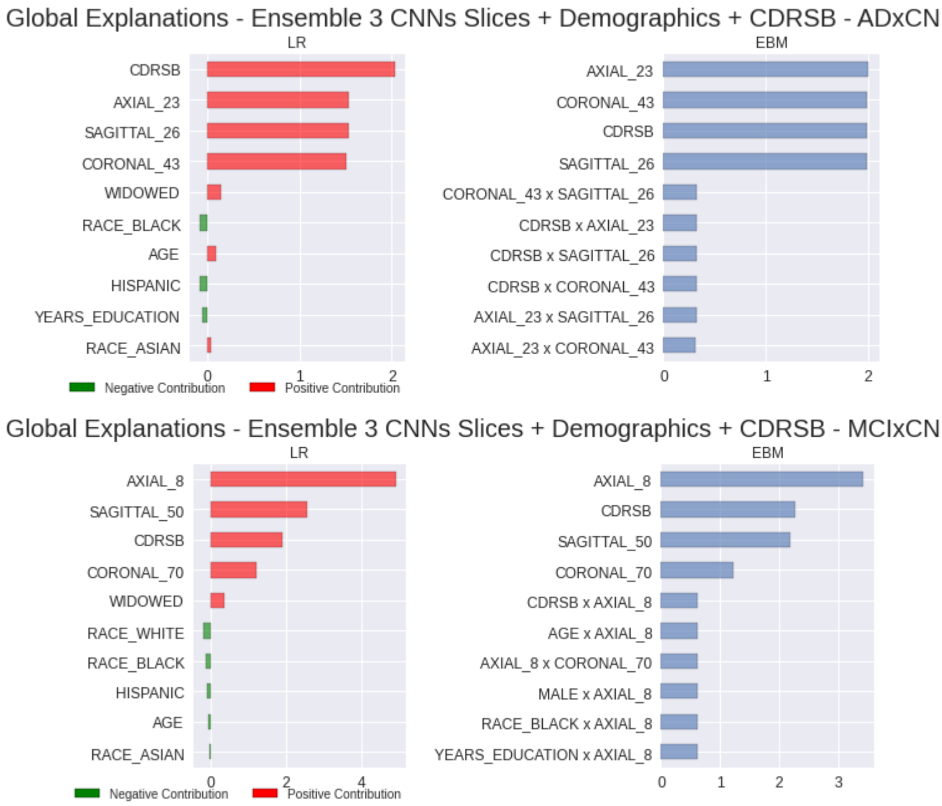

Figure 6.14: Global Explanations for the 3 CNNs + Demographics + CDRSB Ensemble. Upper: ADxCN; Bottom:MCIxCN Left: Logistic Regression; Right: Explainable Boosting Machine

On the left side patient, we see that a high level of education contributed to towards a negative diagnosis (the maximum number of years of education for these patients cohort is 20 - see Table 5.1). Furthermore, feature interactions are a mix of two MRI slice combinations and also one MRI slice with demographics data.

On the right side patient, there is a major contribution of the sagittal slice on the final prediction, for besides appearing as most important feature, it also appears on 6 feature interaction pairs. Still analyzing these interactions for this patient, we see that most interactions are actually combinations of demographics features with the sagittal slice.

These two local explanations suggest that the ensemble executed this diagnosis by combining morphological information of the brain with racial traits, age and educational level of the patient. In fact, this is comparable to what healthcare professionals would do.

Let us present two CN patients for the ADxCN case, shown in Figure 6.17. The sagittal slice appears as less important and opposite when compared to the other MRI slices. Axial and coronal slices play an equally important role in the negative diagnosis. Here we see that all three MRI slices were

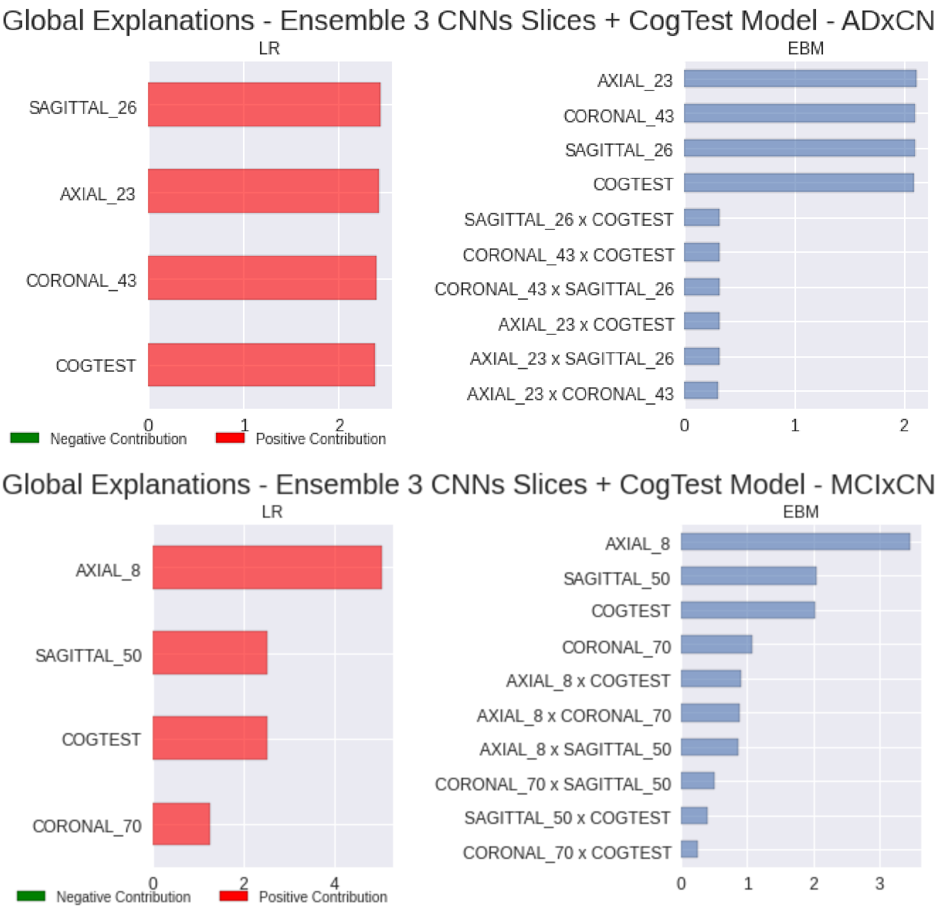

Figure 6.15: Global Explanations for the 3 CNNs + Cognitive Tests Model Ensemble. Upper: ADxCN; Bottom:MCIxCN Left: Logistic Regression; Right: Explainable Boosting Machine

combined on pair interactions. Additionally, it is possible to notice a strange use of the Hispanic ethnicity on the left side patient. This feature offers a negative contribution when combined with the axial slice and a positive contribution when combined with the sagittal. Perhaps this is due to the fact that the sagittal slice got a really high score of practically one, while the axial got a score of 0. Nonetheless, this is an attention point passive of further investigation.

Now, looking at false positive cases on Figure 6.18, it seems that even with weak evidence, such as only one MRI slice indicating the disease while all the other slices indicate a healthy subject with higher importance, the model still diagnoses a patient with Alzheimer's Disease. We remind the reader that this particular ensemble have low precision and high recall - see Table 5.11 - and that might explain this over sensitive behavior, where just few pieces of evidences are necessary for the model to get a positive diagnosis. It is also possible to see that the feature importances are closer in order of magnitude for the false positive cases when compared to true positive cases on Figure 6.16. Besides, for the right side patient, we see the features switching contribution

Figure 6.16: Local Ensemble Explanations for true AD (true positive).

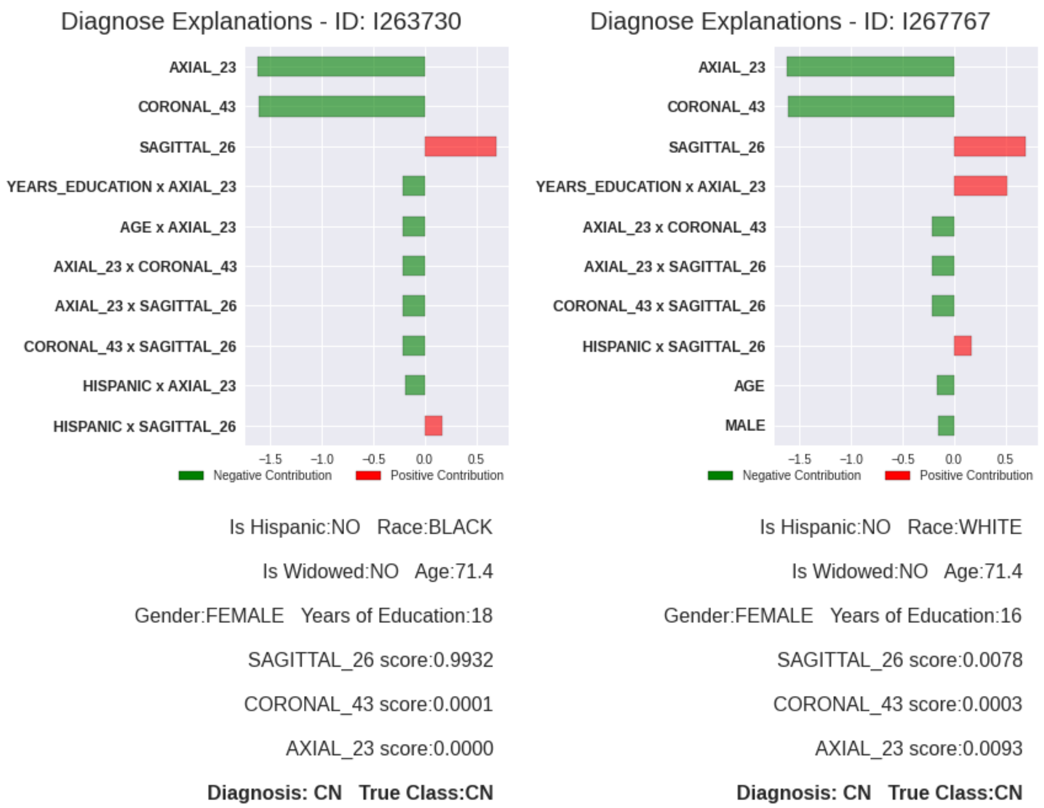

Figure 6.17: Local Ensemble Explanations for true CN (true negative of AD).

Figure 6.18: Local Ensemble Explanations for false AD (false positive).

sides across the top 10 rank, implying that the model might be confused about the diagnosis.

Although a false positive diagnosis could cause some stress for the patient, the doctors could request additional exams if they are not so sure about the model´s outcome. This scenario is preferred over a false negative diagnosis that could go untreated, leading to a loss of quality of life and other more severe outcomes.

Finally, the false AD negatives are presented in Figure 6.19. Here the race and ethnicity features appear with high negative contributions, along with the MRI slices. Since we chose to consider these features for our ensemble, we would like to call attention to any potential diagnostic bias that the cohort might have. Taking a quick look at the preprocessed data at Table 5.2, we see that the vast majority of people are white and non- hispanic, and a really small portion of data contains black people, Hispanics, Asians and other races. In other words, the minority races are underrepresented in this cohort.

Furthermore, for the right side patient of Figure 6.19, all MRI slices indicated a negative contribution towards the AD class, and the only positive feature is the years of education of the patient. Table 5.1 shows that the mean value is around 16 years. This might suggest that the ensemble captured an inverse relationship between the educational level and the likelihood of

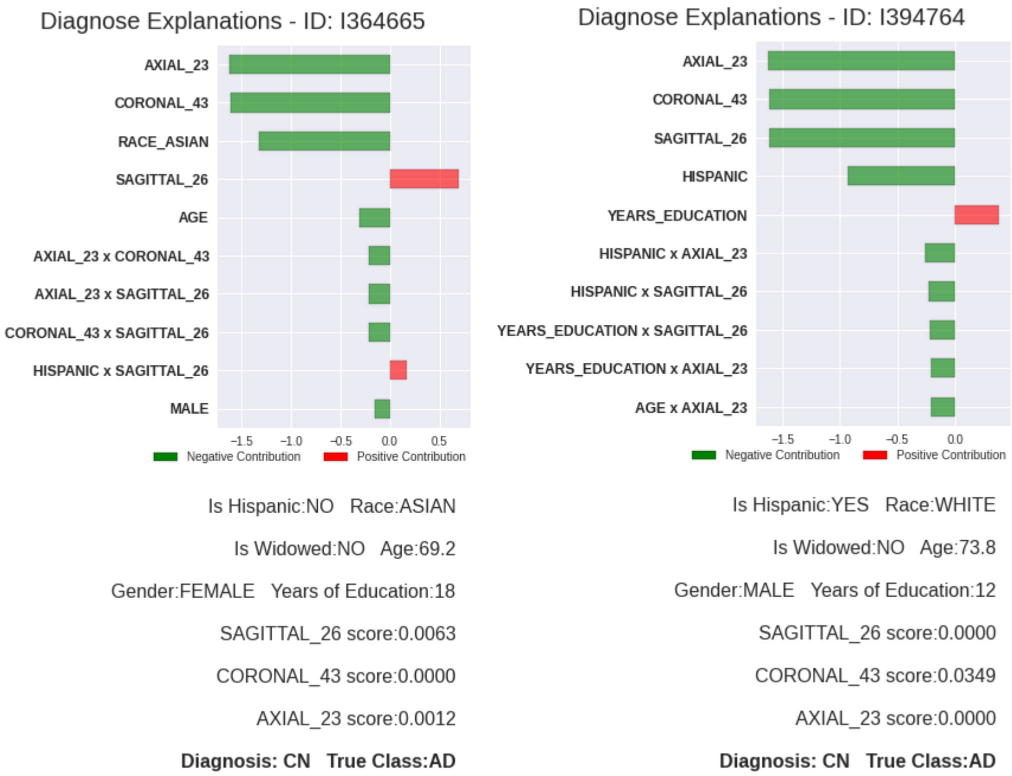

Figure 6.19: Local Ensemble Explanations for false CN (false negative of AD).

having Alzheimer's disease. In other words, the less educated the patient is, the higher the risk he has of having AD. As a matter of fact, an early study [111] suggests that lower educational levels are associated with Alzheimer's Disease progression.

As a bonus experiment, we removed all racial and ethnic features from the dataset and re-ran this ensemble experiment. It was possible to obtain an AUC of 0.942 and F1 of 0.683 for the test set, which is slightly higher than the original model - check Table 5.11. This result is shown here on purpose, to raise the attention to the importance of analyzing model explanations, rather than just its performance, in order to bring fairness and justice to AI-assisted healthcare applications. In fact, as already mentioned, the ensemble´s goal is to act as an early diagnosis tool based on a group of features relevant to this specific domain, and not necessarily propose a powerful black-box model that is obscure to healthcare professionals. Racial traits and other demographic features should be studied carefully before being embedded in automatic diagnostic tools.

Moving on to the CNxMCI case, let us start looking at true MCI patients (true positives), on Figure 6.20 and true CN patients (true negatives) on Figure 6.21. It is evident that the axial and sagittal MRI slices play an important role in the diagnosis of all patients. Also, all feature interactions ranked on the top 10 are interactions with the axial slice, most of which are combinations of demographics and MRI.

We observe more than one racial feature being combined with the axial slice, and this might be somewhat misleading since one person can only belong to one race, at least for this cohort and study. However, we would like a potential explanation to interpret this result: The fact that the right side MCI patient is not black, Asian, and is white, gets combined to the MRI diagnosis to offer a final explanation. In other words, the absence (zero value) of some features might also help the model on its decision. Furthermore, on the right side CN patient, the black race shows a small positive contribution, and we do not have any clear explanations.

Furthermore, on the left side patients of Figures 6.20 and 6.21, we see the age as a relevant positive factor. These patients are younger when compared to the right side ones. Table 5.1 shows that the age feature has a mean value of 72.45 and a minimum value of 55, so indeed, the age of these two patients is close to the lower boundary of this feature. This might imply that an inverse relationship between age and likelihood of a mild cognitive impairment or maybe that this feature is more relevant on younger patients than on older patients.

Figures 6.22 and 6.23 show false MCI and false CN diagnosis, respectively. The false MCI patients show a similar profile to the true MCI patients discussed earlier. A subtle difference between these false positive patients and the true positives is that the sagittal slice has lower scores and does not contribute positively. However, this was not enough evidence to make the model output a CN diagnosis.

The false negative patient on the left looks similar to the true negative patient on the right side of Figure 6.21, having all MRI slices as negative contribution and the black race as a positive contribution. On the other hand, the right side false negative patient presents a confusing diagnosis between the MRI slices. Sagittal and axial slices are similar in magnitude but opposite in contribution. Maybe these features end up canceling each other. Since the coronal slice does not have such high importance compared to the rest of the features, the canceling effect might remain, and the model might not have enough evidence to support a MCI diagnosis.

In conclusion, we saw presented and discussed a myriad of different situations, in order to assess the ensembles behavior. Over the topics discussed so far in this section, it is obvious that the explainability of the predictions can support better diagnostic procedures, just by the simple fact that is possible to question how the machine learning model 'thinks'.

Figure 6.20: Local Ensemble Explanations for true MCI.

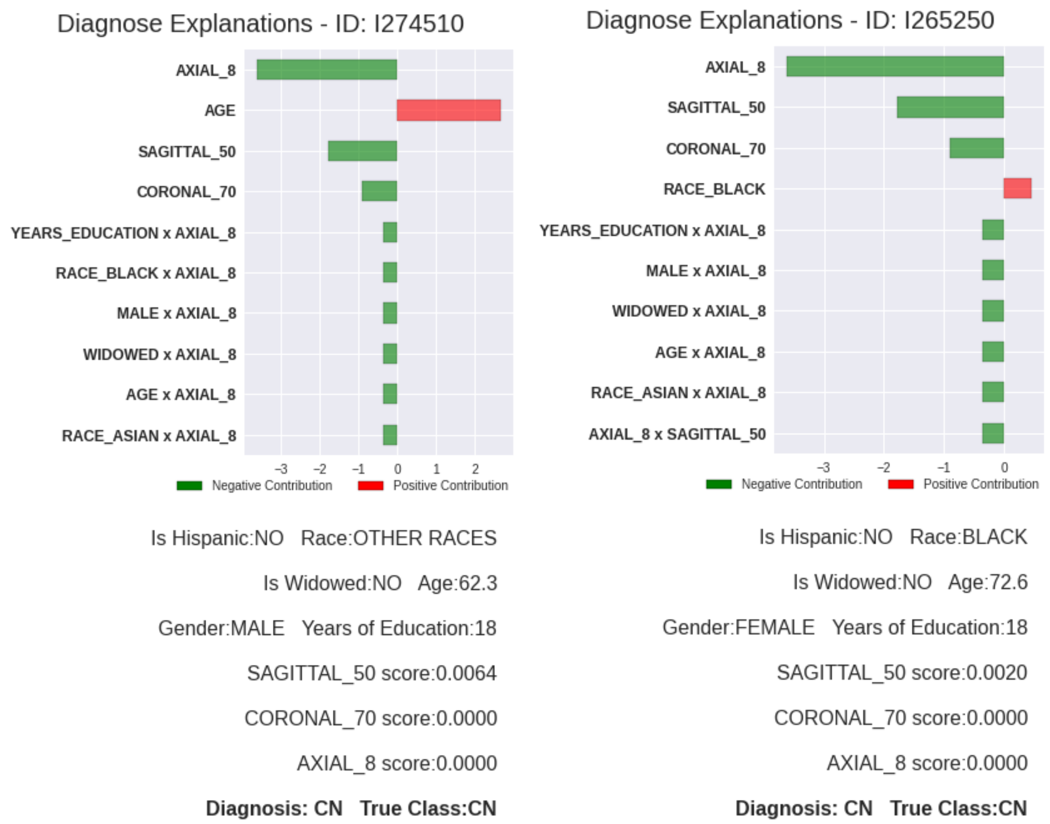

Figure 6.21: Local Ensemble Explanations for true CN (true negative of MCI).

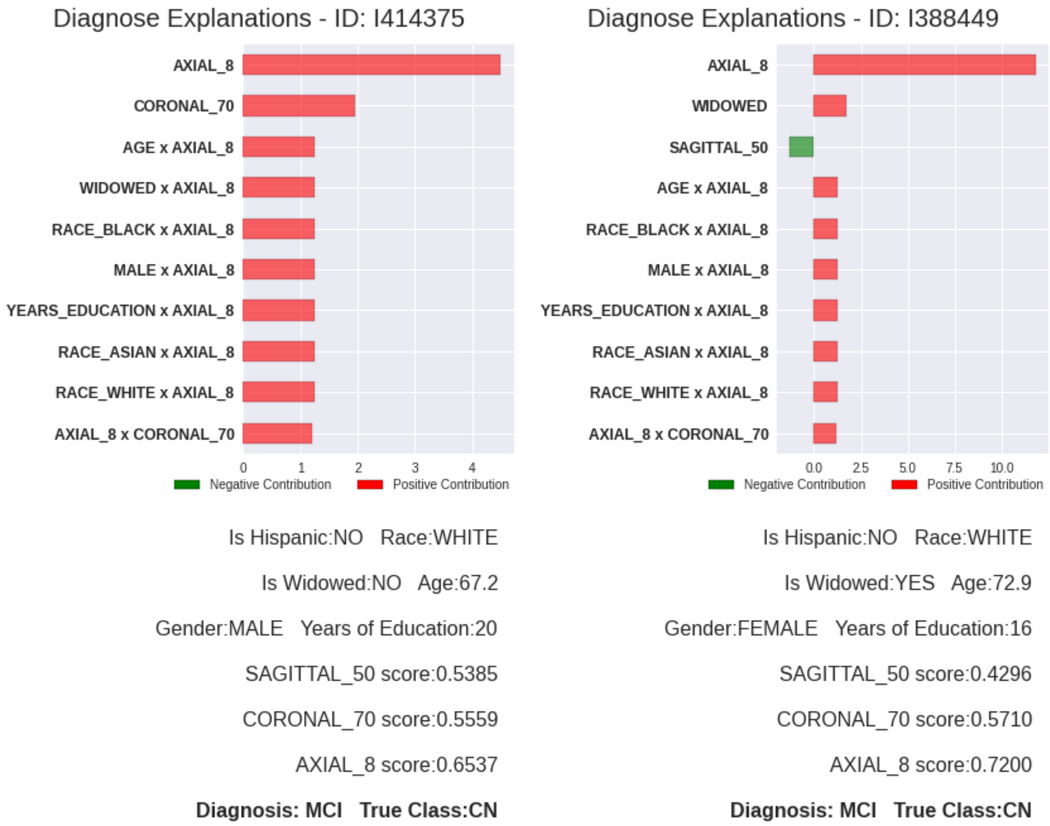

Figure 6.22: Local Ensemble Explanations for false MCI.

Figure 6.23: Local Ensemble Explanations for false CN (false negative of MCI).

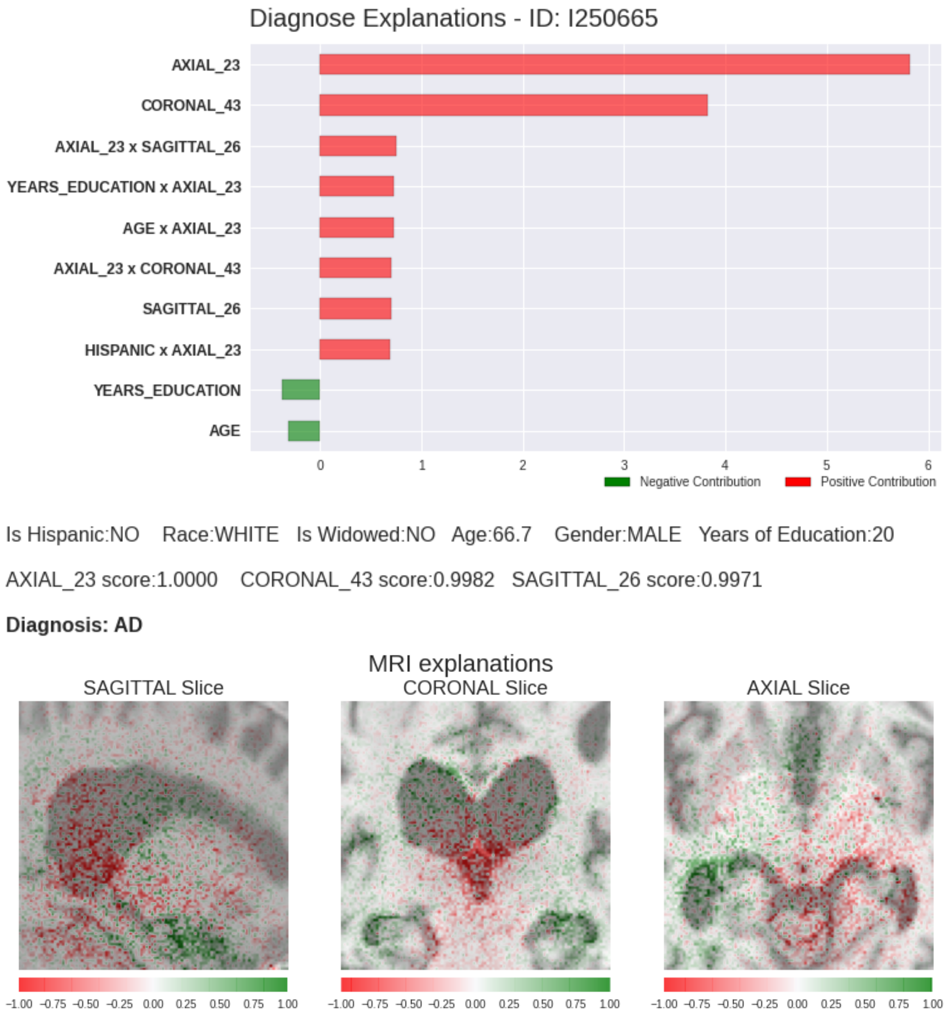

Figure 6.24: Diagnosis Explanation for an AD patient.

6.5 Patient Diagnosis Assessment

In this final section, we would like to present a simulation of a patient diagnosis by making a joint visualization of the ensemble explanations and CNNs explanations. Since we extensively discussed several algorithms and models under different circumstances and from different points of view, we just quickly present a patient diagnosis simulation that AI assists.

Figure 6.24 shows the diagnosis of a patient with Alzheimer's Disease and Figure 6.25 a patient with Mild Cognitive Impairment. By having MRI and ensemble explanations presented together, professionals can be guided by explanations on where and how to look for evidence to execute a final diagnosis. In other words, the machine will provide the diagnosis, showing the most relevant attributes that led to the current decision, shining some light on a black-box diagnostic.

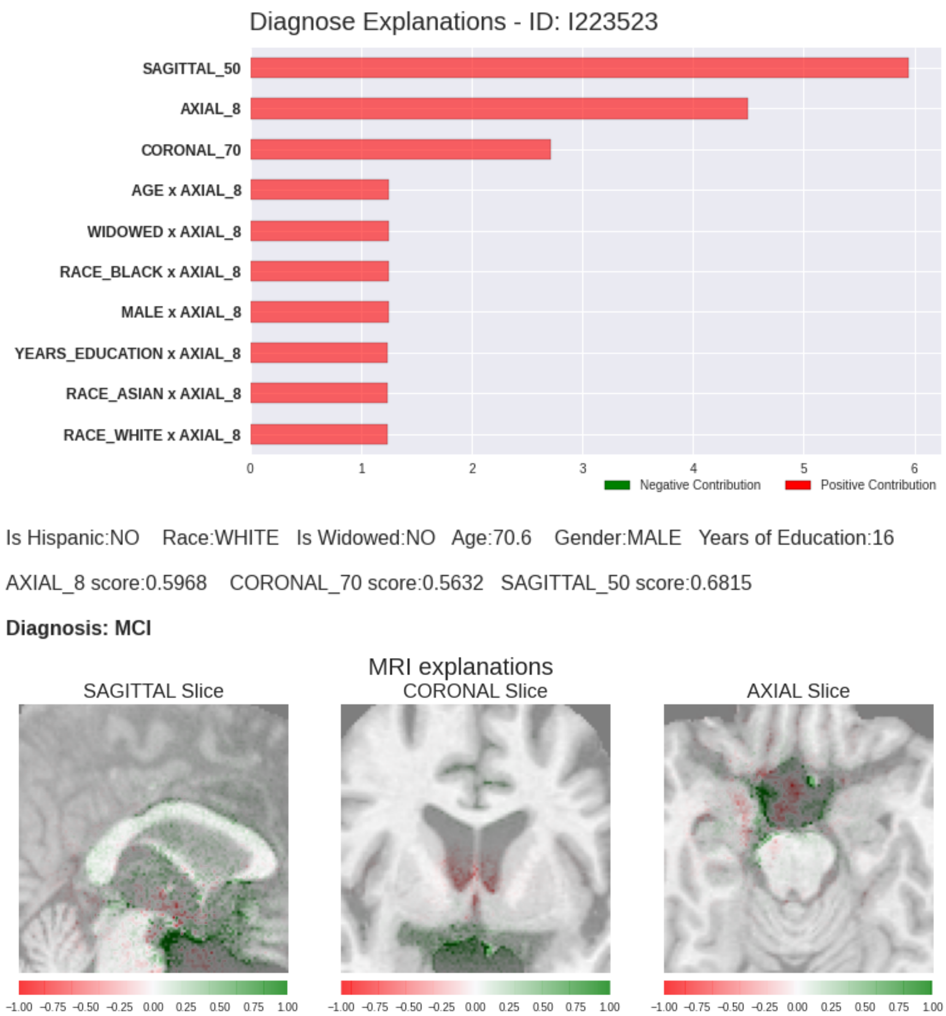

Figure 6.25: Diagnosis Explanation for an MCI patient.

7 Conclusion

This work presented an explainable artificial intelligence approach to support the Alzheimer's Disease and Mild Cognitive Impairment diagnosis. Besides processing different types of information, in order to imitate how a doctor would actually execute a patient diagnosis, we were also preoccupied on explaining the diagnosis through the analysis of most important features.

All the data used was obtained through the Alzheimer's Disease Neuroimaging Initiative (ADNI) and consisted of results of several cognitive tests, demographics and also magnetic resonance imaging (MRI). Each data type was preprocessed separately, and later aligned in order to make possible the ensemble experimentation with different data types. The MRI preprocessing mainly consisted of a registration against a reference atlas, as well as skull stripping and cropping process.

The baseline experiments consisted of three 2D deep convolutional neural networks (CNNs) that processed one orientation of the 3D MRI - sagittal, coronal an axial - as well as a more traditional machine learning model, that used demographics along with cognitive tests data. Afterwards, we analysed a few different ensemble based learning architectures, that mainly consisted on the training of another model that used the prediction scores of the baseline models, along with demographics and cognitive tests features. The ensemble learning training strategy allowed more data to be ingested by each separate learning module, increasing the potential of generalization of the models.

We found out that when classifying AD patients against CN, just by combining the 3 CNNs prediction scores, it was possible to increase the performance considerably, from 0.904 to 0.948. Unfortunately, the same did not hold true when classifying MCI patients, for the CNNs ensemble performed the same of the best single orientation CNN. Moreover, we realized that the Clinical Dementia Rating Scale Sum of Boxes (CDRSB) test alone had a really good predictive power for the chosen cohort of patients. Hence, the models that used the CDRSB as a feature had superior performance on both AD and MCI classification.

We decided to choose as our final ensemble the one using the 3 CNNs prediction scores along with some demographics such as race, age and years of education. Even thought this was not the most powerful presented ensemble, it offered excellent performance and a reasonable amount of necessary attributes. We argue that this ensemble is close to a real doctor diagnosis, by analysing different data sources and providing a consolidated diagnosis based on a group of observations, instead of the observation upon one single factor.

We opted for a qualitative evaluation of all explanations, since most quantitative evaluations are still under development and discussion in the community. For the proposed explanations, we started exploring two different image explanation algorithms: Guided Grad-CAM and DeepLift. Several steps were necessary in order to obtain a reasonable explanation for the MRIs with the DeepLift. However, this effort was compensated, for DeepLift algorithm seemed more sensitive to more subtle changes in the brain structure than the Guided Grad-CAM and ended being chosen to compose the final patient diagnosis.

Since we used two intrinsically interpretable machine learning models for the ensembles, no additional explanation algorithm was need. By analysing global explanations of the ensembles we realized that the CDRSB and the MRI prediction scores have a great importance on the overall behavior of the models. As expected, demographic features played a complementary role on explanations.

Moreover, the discussion of local explanations for the chosen ensemble, i.e. discussing model behavior for specific predictions, raised an important discussion about prediction bias, fairness and also misleading diagnosis. Here we saw most of the demographics features playing a more relevant role, when compared to the global explanation visualizations. The 3 CNNs prediction scores appeared combined, as feature interactions, with a considerable frequency, showing the importance of analysis the brain from different points of view, not only focusing on single areas, such as made on previous works.

Since we structured the proposed methodology for the current working diagnosis of the disease, we see as a limitation of our work the lack of information about the progression of mild and advanced cases. Therefore for future works, we would like to evolve the present work to predict the risk of Alzheimer's Disease and Mild Cognitive Impairment progression, explaining the most important factors that influentiate on the progression.

Also, since there is a limitation on using MRI scans to detect MCI patients, due to the fact that some of these still do not manifest structural changes in the brain, it would be interesting to build an ensemble processing functional scans, such as the Positron Emission Tomography (PET), along with structural MRIs and demographics. Moreover, since we saw the major contribution of CDRSB scores in predictions, it might also be interesting to build a proxy model of the CDRSB, based on MRIs, PETs and other data collected through non invasive procedures.

Additionally, it would be interesting to explore more thoroughly how subcategories and partial results of the neuropsychological tests correlate specific brain regions and functioning aspects concerning Alzheimer's Disease and its progression.

In conclusion, we argue that it is important to evaluate models under the lens of predictive performance and also both global and local explanations. Therefore, is possible to assess potential bias, misleading predictions and avoid common black-box modelling pitfalls. We urge to the community to pay more attention to these topics, instead of just focusing on increasing models performance on specific tasks. We lay out some of the most relevant benefits of adding explainability analysis to Machine Learning strategies for Alzheimer's Disease diagnosis and any other healthcare domain:

- Help clear obscure diagnosis;
- Assist healthcare professionals to avoid misleading diagnosis;
- Guide professionals towards new researches and new findings about Alzheimer's disease and other dementia related features;
- Can uncover potentially biased predictions.

Furthermore, it is important to note that any type of AI assisted diagnosis tool should be taken into further investigation considering specific race, gender and age cohorts in order to assess both models performance and model explainability. It is vastly know that, minorities in general have less access to education and healthcare around the world and specially in our country.

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